1 Jan 7: Elementary number theory

The goal of this lecture is to prove Fermat's little theorem.

Theorem 1.1. *Let p be a prime, and let a be any integer. Then $a^p - a$ is divisible by p.*

1.1 Modular arithmetic [2.7, 2.9]

An *equivalence relation* on a set S is a relation $\sim$ between certain pairs of elements of S. We write $a \sim b$ if a and b are *equivalent*. An equivalence relation is required to be

- *transitive*: if $a \sim b$ and $b \sim c$, then $a \sim c$.
- *symmetric*: if $a \sim b$, then $b \sim a$.
- *reflexive*: for all a, $a \sim a$.

An equivalence relation $\sim$ partitions S into *equivalence classes*.

Definition 1.2. Let n be a positive integer. For integers a, b, we write

$$a \equiv b \pmod{n}$$

if $a - b$ is divisible by n, i.e., $a - b = nk$ for some integer k.

Lemma 1.3 (Addition and multiplication modulo n)**.** *If $a' \equiv a \pmod{n}$ and $b' \equiv b \pmod{n}$, then $a' + b' \equiv a + b \pmod{n}$ and $a'b' \equiv ab \pmod{n}$.*

Proof. Suppose $a' = a + nk$ and $b' = b + n\ell$. Then

$$a' + b' = (a + b) + n(k + \ell),$$

and

$$a'b' = ab + n(a\ell + bk + k\ell). \qquad \square$$

Definition 1.4. Let $\mathbb{Z}/n\mathbb{Z}$ denote the set of equivalence classes of $\mathbb{Z}$ with respect to the equivalence relation $\equiv$. These equivalence classes are also referred to as *congruence classes* modulo n.

By the lemma above, addition and multiplication of congruence classes modulo n is well-defined. If we write $\overline{a}$ to denote the congruence class of a, then

$$\overline{a} + \overline{b} = \overline{a + b},$$

and similarly

$$\overline{a}\overline{b} = \overline{ab}.$$

The associative, commutative, and distributive laws carry over for addition and multiplication of elements of $\mathbb{Z}/n\mathbb{Z}$.

Example 1.5. $\mathbb{Z}/6\mathbb{Z}$ has 6 elements. The elements $\overline{2}$ and $\overline{8}$ are the same element since $2 \equiv 8 \pmod{6}$.

We have $\overline{2} \cdot \overline{5} = \overline{10}$, and $\overline{8} \cdot \overline{5} = \overline{40}$. Fortunately, $\overline{10} = \overline{40}$ since $10 \equiv 40 \pmod{6}$. We usually take the remainder when divided by 6 and say $\overline{2} \cdot \overline{5} = \overline{4}$.

1.2 Bezout's lemma [2.3]

We recall division with remainder: let n be an integer, and let a be a positive integer. Then there exists an integer q and an integer $0 \leq r < a$ such that

$$n = aq + r.$$

Definition 1.6. Let a and b be integers, not both zero. The *greatest common divisor* of a and b, denoted $\gcd(a, b)$, is the largest integer which divides both a and b. If $\gcd(a, b) = 1$, we say that a and b are *coprime* or *relatively prime*.

The GCD satisfies the property that

$$\gcd(a, b) = \gcd(a + bk, b)$$

for any integer k. Indeed, if d divides both a and b, then d divides both $a + bk$ and b, and conversely.

As such, we can compute GCD's using the *Euclidean algorithm*, which works by repeated division with remainder.

Example 1.7. For example, for $a = 314$, $b = 136$, since

$$314 = 2 \cdot 136 + 42, \quad 136 = 3 \cdot 42 + 10, \quad 42 = 4 \cdot 10 + 2,$$

we have

$$\gcd(314, 136) = \gcd(42, 136) = \gcd(42, 10) = \gcd(2, 10) = 2.$$

Proposition 1.8 (Bezout's lemma)**.** *For any integers a and b, not both zero, there exist integers r and s such that*

$$\gcd(a, b) = ra + sb.$$

Proof. Let $d = \gcd(a, b)$ Let ℓ be the smallest positive integer that can be expressed as

$$\ell = ra + sb$$

for some r and s.

We claim that $\ell | a$. Use division with remainder to write

$$a = \ell q + m$$

for $0 \leq m < \ell$. Then m can also be expressed in the form $ra + sb$:

$$m = a - \ell q = a - q(ra + sb) = (1 - qr)a - (qs)b.$$

Since ℓ was assumed to be minimal, $m = 0$, so $\ell | a$.

Similarly, $\ell | b$, so ℓ divides both a and b. Since d is the greatest common divisor,

$$\ell \leq d.$$

On the other hand, d divides both ra and sb, so d also divides ℓ, so

$$d \leq \ell.$$

Thus, $\ell = d$. $\square$

Corollary 1.9. *Let e be an integer which divides both a and b. Then e divides* $\gcd(a, b)$.

Proof. Let

$$\gcd(a, b) = ra + sb.$$

Since e divides both terms on the right hand side, it also divides $\gcd(a, b)$. □

Corollary 1.10. *Let p be a prime, and let a and b be integers. If $p|ab$, then $p|a$ or $p|b$.*

Proof. Suppose that p divides ab, but p does not divide a.

Since p is prime, $\gcd(a, p) = 1$, so by Bezout's lemma there exist $r, s \in \mathbb{Z}$ such that

$$1 = ra + sp.$$

Multiplying both sides by b,

$$b = rab + spb.$$

Both terms on the right are multiples of p by the assumption $p|ab$, so $p|b$. □

Corollary 1.11 ($\mathbb{Z}/p\mathbb{Z}$ has inverses)**.** *Let p be a prime, and let a be an integer which is not divisible by p. There exists an integer b such that $ab \equiv 1 \pmod{p}$.*

Proof. As in the proof above, there exist $r, s \in \mathbb{Z}$ such that

$$1 = ra + sp.$$

So $ra \equiv 1 \pmod{p}$. Clearly, we can take $b = r$. □

1.3 Proof of Fermat's little theorem

Proof. If a is divisible by p, then it is apparent that $a^p - a$ is divisible by p. Assume $p \nmid a$.

1. Consider the set

   $$\{\overline{1}, \overline{2}, \ldots, \overline{p-1}\}$$

   of nonzero congruence classes modulo p. Then consider the set

   $$\{\overline{a}, \overline{2a}, \ldots, \overline{(p-1)a}\}$$

   of congruence classes modulo p.

2. We claim that they're the same set. Indeed, since both sets have $p - 1$ elements, we just need to show that $\overline{j}$ appears in the second set for every $j \in \{1, \ldots, p-1\}$.

   In other words, we want $ka \equiv j \pmod{p}$ for some $k \not\equiv 0 \pmod{p}$. Let b be such that $ab \equiv 1 \pmod{p}$, and let $k = jb$. Then

   $$ka \equiv jba \equiv j \pmod{p}.$$

   Obviously $k \not\equiv 0 \pmod{p}$ since $j \not\equiv 0 \pmod{p}$.

3. Then

$$\begin{aligned} 1 \cdot 2 \cdots (p-1) &\equiv a \cdot (2a) \cdots (p-1)a \\ &\equiv 1 \cdot 2 \cdots (p-1) \cdot a^{p-1} \pmod{p}. \end{aligned}$$

Multiplying both sides by an inverse of $(p-1)!$ gives

$$a^{p-1} \equiv 1 \pmod{p}.$$

□

1.4 $(\mathbb{Z}/n\mathbb{Z})^\times$

Corollaries 1.10 and 1.11 are not true if p is not prime. For example, $4|2 \cdot 2$ but 4 does not divide 2, and there is no integer b such that $2b \equiv 1 \pmod{4}$, because $2b$ cannot be odd.

Here are some generalizations of them to general n.

Lemma 1.12. *Suppose n be a positive integer. If $n|ab$, then b is a multiple of $n/\gcd(a,n)$.*

Proof. Let $d = \gcd(a,n)$. Suppose

$$d = ra + sn.$$

Then $db = rab + snb$ is a multiple of n, so b is a multiple of n/d. □

Lemma 1.13. *Let n be a positive integer, and a be an integer such that $\gcd(a,n) = 1$. There exists an integer b such that $ab \equiv 1 \pmod{n}$.*

Proof. Since $\gcd(a,n) = 1$, there exist $r, s \in \mathbb{Z}$ such that

$$1 = ra + sn.$$

So $ra \equiv 1 \pmod{p}$, and we can take $b = r$. □

Definition 1.14. Let $(\mathbb{Z}/n\mathbb{Z})^\times$ denote the set of congruence classes $\overline{a}$ modulo n such that $\gcd(a,n) = 1$. Note that this does not depend on the choice of a, only on $a \pmod{n}$, since $\gcd(a+nk,n) = \gcd(a,n)$ as mentioned previously.

Definition 1.15. In the special case when $n = p$ is a prime, $(\mathbb{Z}/p\mathbb{Z})^\times$ is just all of the elements of $\mathbb{Z}/p\mathbb{Z}$ other than $\overline{0}$.

1.5 Least common multiple

Definition 1.16. Let a and b be integers, both not zero. The *least common multiple* of a and b, denoted $\operatorname{lcm}(a,b)$ is the smallest positive integer which is a multiple of both a and b.

Proposition 1.17. *Let a and b be positive integers. If $d = \gcd(a, b)$ and $m = \operatorname{lcm}(a, b)$, then $ab = dm$.*

Proof. Suppose $m = ak$. Since $b|m$, by Lemma 1.12, $k \geq b/d$, so $m \geq ab/d$. On the other hand, it is clear that ab/d is a multiple of both a and b, so $m \leq ab/d$. $\square$

2 Jan 12: Basic group theory definitions

2.1 Groups, subgroups, and product groups [2.1–2.3, 2.11]

Definition 2.1 (Law of composition)**.** A *law of composition* on a set S is a map

$$S \times S \to S.$$

For example, addition and multiplication of integers.

Example 2.2. Let T be a set, and let S denote the set of all functions $g\colon T \to T$. Function composition

$$(g, f) \mapsto g \circ f$$

is a law of composition on S, where

$$g \circ f\colon T \xrightarrow{f} T \xrightarrow{g} T,$$

i.e., $g \circ f$ is the function $t \mapsto g(f(t))$.

Definition 2.3 (Group axioms)**.** A *group* is a set G with a law of composition such that

1. the law of composition is **associative**: $a(bc) = (ab)c$ for all $a, b, c \in G$.
2. G contains an **identity** element $e \in G$ such that $ea = ae = a$ for all $a \in G$.
3. every element $a \in G$ has an **inverse**, an element b such that $ab = ba = e$.

Proposition 2.4. *In a group,*

1. *the identity is unique. We often denote it by* 1 *or* 0*.*
2. *the inverse of an element* a *is unique. We usually denote it by* a^{-1}*.*
3. $(ab)^{-1} = b^{-1}a^{-1}$*.*
4. *the* ***cancellation law*** *holds: if* $ab = ac$*, then* $b = c$*.*

Proof.

1. If e and e' are both identities, then

$$e = ee' = e'.$$

4. Multiplying both sides of $ab = ac$ by a^{-1} on the left gives $b = c$.

□

Example 2.5.

1. The set $\mathbb{Z}/n\mathbb{Z}$ equipped with addition is a group. The identity is the congruence class $\overline{0}$.

2. For $n > 1$, the set $\mathbb{Z}/n\mathbb{Z}$ equipped with multiplication is *not* a group. The identity would have to be $\overline{1}$, but $\overline{0}$ does not have a multiplicative inverse.

3. Let p be a prime. Recall that $(\mathbb{Z}/p\mathbb{Z})^\times$ denote the set of *nonzero* elements of $\mathbb{Z}/p\mathbb{Z}$. Then $(\mathbb{Z}/p\mathbb{Z})^\times$ is a group under multiplication.

4. In general, $(\mathbb{Z}/n\mathbb{Z})^\times$ is also a group under multiplication. Recall that this is the set of congruence classes $\overline{a}$ where a is relatively prime to n.

Definition 2.6. A group G is called *commutative* or *abelian* if $ab = ba$ for all $a, b \in G$.

Example 2.7. The examples above are abelian. An example of a nonabelian group is

$$\mathrm{GL}_n(\mathbb{R}) \coloneqq \{n \times n \text{ real matrices with nonzero determinant}\}.$$

The *order* of a group G is the number of elements of G, and denoted $|G|$. It could be infinite.

Definition 2.8. A *subgroup* of a group G is a subset H satisfying

1. the identity is contained in H.

2. if $a, b \in H$, then $ab \in H$. This property is referred to as *closure*.

3. if $a \in H$, then $a^{-1} \in H$.

The subgroup is called *proper* if it is not equal to G or $\{1\}$.

Example 2.9. The special linear group

$$\mathrm{SL}_n(\mathbb{R}) = \{A \in \mathrm{GL}_n(\mathbb{R}) : \det A = 1\}$$

is a subgroup of $\mathrm{GL}_n(\mathbb{R})$.

Definition 2.10. Let G and G' be groups. The *product group* consists of the set of pairs

$$G \times G' = \{(a, a') : a \in G, a \in G'\},$$

and the law of composition is given by

$$(a, a') \cdot (b, b') = (ab, a'b').$$

The identity of $G \times G'$ is $(1_G, 1_{G'})$.

2.2 Permutations [1.5]

Definition 2.11. A *permutation* (of length n) is a bijective map $\sigma\colon \{1, 2, \ldots, n\} \to \{1, 2, \ldots, n\}$.

Here is an example of a permutation of length 6.

n	1	2	3	4	5	6
$\sigma(n)$	3	5	4	1	2	6

We express permutations using *cycle notation* which works like this.

- Pick an arbitrary index, for example 1.
- We see where σ sends 1. In this example, $\sigma(1) = 3$.
- We see where σ sends 3. In this example, $\sigma(3) = 4$.
- We see where σ sends 4. In this example, $\sigma(4) = 1$.
- We are back where we started. We indicate the cycle $\sigma\colon 1 \to 3 \to 4 \to 1$ using the notation
$$(134).$$
- We collect all cycles, and usually ignore 1-cycles, The σ above is
$$(134)(25)(6), \text{ or } (134)(25).$$

Note: the cycle notation is not unique. We can also express (134) as

$$(341) \text{ or } (413)$$

by choosing a different starting index.

Example 2.12. In cycle notation,

$$(1452) \circ (134)(25) = (135).$$

In general, *bijective* functions from a set T to itself form a group under composition. The identity is the function $\mathrm{id}(t) = t$, and inverses exist by the requirement that the functions are bijective.

Definition 2.13. The group of permutations of the set $\{1, 2, \ldots, n\}$ is called the *symmetric group* and denoted S_n. It has order $n!$

Example 2.14. The group S_3 has 6 elements. Let $x = (123)$ and $y = (12)$. Since x is a 3-cycle and y is a 2-cycle,

$$x^3 = 1, \quad y^2 = 1. \tag{$\heartsuit$}$$

One can verify without computation that the six elements

$$1, x, x^2, y, xy, x^2y$$

are distinct, using the cancellation law.

So S_3 consists of these 6 elements. Observe that

$$yx = (12) \circ (123) = (23) = (132) \circ (12) = x^2 y. \qquad (\diamondsuit)$$

This rule lets us move all occurrences of y to the right. For example,

$$x^{-1}y^3x^2y = x^2yx^2y = x^2(yx)xy = x^2(x^2y)xy = x(yx)y = x(x^2y)y = 1.$$

The elements x and y and the equations $(\heartsuit)$ and $(\diamondsuit)$ are called a set of *generators and relations* for S_3, and we write

$$S_3 = \langle x, y \mid x^3 = 1, y^2 = 1, yx = x^2y \rangle.$$

This is called a *presentation* of the group S_3.

2.3 Orders [2.4]

For any $x \in G$, the *cyclic subgroup* generated by x consists of the elements

$$\ldots, x^{-2}, x^{-1}, 1, x, x^2, \ldots$$

and is denoted $\langle x \rangle$.

Definition 2.15. Let x be an element of a group G. The *order* of x is the smallest positive integer n such that $x^n = 1$.

If no such integer exists, then x has *infinite order*.

Proposition 2.16. *Let x be an element of G of order n. Let k and j be integers.*

1. *If $x^k = 1$, then $k = nq$ for some integer q.*
2. *If $x^k = x^j$, then $k - j = nq$ for some integer q.*

Proof.

1. Let $k = nq + r$ for $0 \leq r < n$. Then if $x^k = 1$, since $x^n = 1$, we have

$$1 = x^k = x^{nq+r} = (x^n)^q x^r = x^r.$$

   By minimality of n, we must have $r = 0$.

2. Follows from 1. □

Example 2.17. Some applications of the above properties of orders:

1. If x has order n, then $\langle x \rangle$ is a finite subgroup of order n, consisting of the elements

$$1, x, x^2, \ldots, x^n.$$

2. Let $G = (\mathbb{Z}/p\mathbb{Z})^\times$. Fermat's little theorem is the statement that for any $a \in G$,
$$a^{p-1} = 1.$$
Thus, the order of every element of G divides $p-1$.

The formulation of Fermat's little theorem in 2. above generalizes to any finite group.

Theorem 2.18 (Lagrange's theorem)**.** *Let G be a finite group. Then for any $a \in G$,*
$$a^{|G|} = 1.$$

Proof for abelian groups. The proof is similar to the proof of Fermat's little theorem we saw in Lecture 1.

Let $G = \{g_1, \ldots, g_n\}$, where $n = |G|$. Then $G = \{ag_1, \ldots, ag_n\}$ is the same set because the (left) multiplication by a map $G \to G$ is bijective; it has inverse (left) multiplication by a^{-1}.

Taking the product of all elements in G,
$$g_1 \cdots g_n = a^n (g_1 \cdots g_n).$$
This calculation requires G to be abelian. By cancellation, $a^n = 1$. $\square$

Corollary 2.19. *In a finite group G, the order of every element divides $|G|$.*

3 Jan 14: Homomorphisms and isomorphisms

3.1 Dihedral group

Let $A_1A_2\cdots A_n$ be a regular n-gon, with center O. The *dihedral group* D_n consists of the symmetries of the regular n-gon. It has order

$$|D_n| = 2n.$$

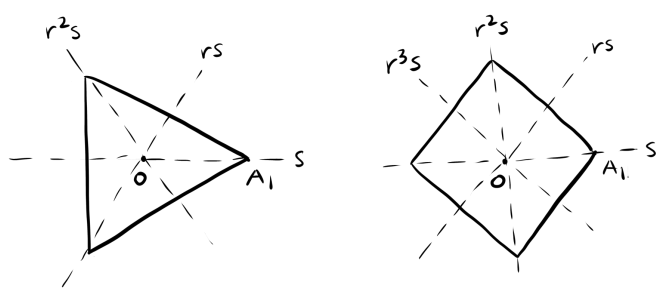
- There are n rotations in D_n. Let r denote rotation by $2\pi/n$ around O. It satisfies

  $$r^n = 1.$$

  The other rotations are

  $$1, r, r^2, \ldots, r^{n-1}.$$

- There are n reflections in D_n.



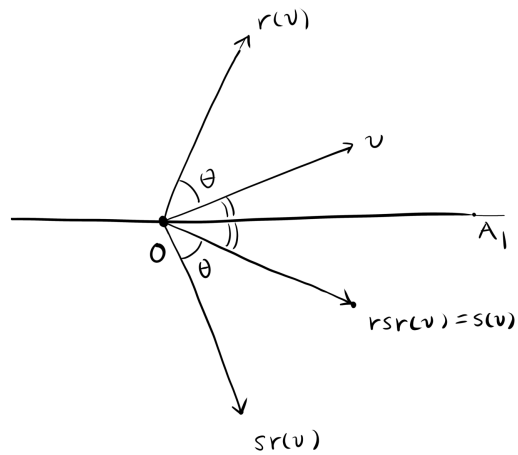
  Let s denote reflection across OA_1. It satisfies

  $$s^2 = 1.$$

  The other reflections are

  $$s, rs, r^2s, \ldots, r^{n-1}s.$$

The transformations r and s satisfy $rsr = s$, since



The dihedral group has the presentation

$$D_n = \langle r, s \mid r^n = 1, s^2 = 1, rsr = s \rangle.$$

The third relation can equivalently be written

$$rsr = s, \quad sr = r^{n-1}s.$$

We also have $r^k s r^k = s$ and $sr^k = r^{n-k}s$.

For example,

$$D_3 = \langle r, s \mid r^3 = 1, s^2 = 1, sr = r^2 s \rangle.$$

3.2 Definitions [2.5, 2.6]

Definition 3.1. Let G and G' be groups. A *homomorpmism* is a map $\varphi\colon G \to G'$ such that

$$\varphi(ab) = \varphi(a)\varphi(b)$$

for all $a, b \in G$. The product ab is taken in G, and the product $\varphi(a)\varphi(b)$ is taken in G'.

In other words, a homomorphism is a map which is compatible with the laws of composition on G and G'.

Proposition 3.2. *Let $\varphi\colon G \to G'$ be a homomorphism.*

1. *It maps the identity to the identity: $\varphi(1_G) = \varphi(1_{G'})$.*
2. *It maps inverses to inverses: $\varphi(a^{-1}) = \varphi(a)^{-1}$.*

Proof. Since φ is a homomorphism,

$$\varphi(1_G)\varphi(1_G) = \varphi(1_G),$$

so $\varphi(1_G) = 1_{G'}$. In addition,

$$\varphi(a^{-1})\varphi(a) = \varphi(1_G) = 1_{G'},$$

so $\varphi(a^{-1}) = \varphi(a)^{-1}$. $\square$

Definition 3.3. A homomorphism $\varphi\colon G \to G'$ is an *isomorphism* if it is bijective.

We say that two groups G and G' are *isomorphic* if there exists an isomorphism $\varphi\colon G \to G'$.

Example 3.4. The map $\varphi\colon \mathbb{Z}/6\mathbb{Z} \to (\mathbb{Z}/7\mathbb{Z})^\times$ given by

$\overline{a}$	0	1	2	3	4	5	(mod 6)
$\varphi(\overline{a})$	1	3	2	6	4	5	(mod 7)

is an isomorphism. It is bijective by the chart above. It is a homomorphism because it is actually given by

$$\varphi(\overline{a}) \equiv 3^a \pmod 7,$$

so

$$\varphi(\overline{a} + \overline{b}) = \overline{3^{a+b}} = \overline{3^a} \cdot \overline{3^b} = \varphi(\overline{a})\varphi(\overline{b}).$$

It is well defined by Fermat's little theorem, because

$$3^{a+6k} \equiv 3^a \pmod 7$$

since $3^6 \equiv 1 \pmod 7$, so $3^a \pmod 7$ only depends on $a \pmod 6$.

3.3 Sign of permutations [1.5]

A *permutation matrix* is an $n \times n$ matrix with entries in $\{0, 1\}$, which has exactly one 1 in each row and column. Every permutation matrix has determinant equal to ± 1.

Definition 3.5. Given a permutation σ, the associated permutation matrix is the matrix with

$$P_{\sigma(i),i} = 1$$

for $1 \le i \le n$, and 0 in all other entries. This is the unique matrix with the property

$$P \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x_{\sigma^{-1}(1)} \\ \vdots \\ x_{\sigma^{-1}(n)} \end{bmatrix}.$$

Example 3.6. If $\sigma = (123)$, then the associated permutation matrix P is below and satisfies

$$PX = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3 \\ x_1 \\ x_2 \end{bmatrix}.$$

Definition 3.7. The *sign* of σ is the determinant $\det P = \pm 1$ of its permutation matrix.

Proposition 3.8. *If σ and τ are permutations with associated permutation matrices P and Q, then the permutation matrix of $\sigma\tau$ is PQ.*

Corollary 3.9. *The map* $\operatorname{sign}\colon S_n \to \{\pm 1\}$ *sending σ to its sign is a homomorphism of groups. (Here $\{\pm 1\}$ is a group under multiplication.)*

3.4 Kernel and image [2.5]

Definition 3.10. Let $\varphi\colon G \to G'$ be a homomorphism of groups. The *image* of φ is

$$\operatorname{im}(\varphi) = \{x \in G' : x = \varphi(a) \text{ for some } a \in G\}.$$

The *kernel* of φ is

$$\ker(\varphi) = \{a \in G : \varphi(a) = 1_{G'}\}.$$

Lemma 3.11. *The kernel and image of a homomorphism $\varphi\colon G \to G'$ are subgroups of G and G', respectively.*

Proof. We verify closure in each case and omit the verification of the other axioms.

1. Suppose $x, y \in \operatorname{im}(\varphi)$. Then $x = \varphi(a)$, $y = \varphi(b)$ for some $a, b \in G$. So $xy = \varphi(ab)$ is also in $\operatorname{im}(\varphi)$.

2. Suppose $a, b \in \ker(\varphi)$. Then $\varphi(ab) = \varphi(a)\varphi(b) = 1_{G'} \cdot 1_{G'} = 1_{G'}$, so $ab \in \ker(\varphi)$. □

Lemma 3.12. *A homomorphism $\varphi\colon G \to G'$ is injective if and only if* $\ker(\varphi)$ *is the trivial subgroup $\{1_G\}$.*

Proof. First suppose φ is injective. Since $\varphi(1_G) = 1_{G'}$, this means that if $\varphi(a) = 1_{G'}$, then $a = 1_G$, so $\ker(\varphi)$ is the trivial subgroup.

Now, suppose $\ker(\varphi) = \{1_G\}$. If $\varphi(a) = \varphi(b)$ for some $a, b \in G$, then

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = 1_{G'}.$$

Thus, $ab^{-1} = 1_G$, so $a = b$. □

Example 3.13. Consider the map

$$\varphi\colon \mathbb{Z}/15\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z},$$

which sends

$$\overline{a} \mapsto (\overline{a}, \overline{a}).$$

One can check that this is well-defined and a homomorphism.

The kernel of φ consists of all congruence classes $\overline{a}$ such that

$$a \equiv 0 \pmod 3 \text{ and } a \equiv 0 \pmod 5.$$

Since 3 and 5 are relatively prime, this implies that $a \equiv 0 \pmod{15}$, so $\ker(\varphi)$ is trivial.

The lemma then tells us that φ is injective. Both the target and the source have 15 elements, so φ is bijective, and thus it is an isomorphism.

3.5 Normal subgroups

However, not every subgroup of G can be the kernel of some homomorphism! The kernel always has the following property.

Definition 3.14. Let $a, g \in G$. The element gag^{-1} is called the *conjugate* of a by g. We say that two elements a and a' are *conjugate* if there exists $g \in G$ such that $a' = gag^{-1}$.

Definition 3.15 (Normal subgroup)**.** A subgroup N of G is *normal* if for all $a \in N$ and all $g \in G$, the conjugate gag^{-1} is also in N.

Proposition 3.16. *The kernel of a homomorphism $\varphi\colon G \to G'$ is normal.*

Proof. Suppose $a \in \ker(\varphi)$. For any $g \in G$, we have

$$\varphi(gag^{-1}) = \varphi(g) \cdot 1_{G'} \cdot \varphi(g)^{-1} = 1_{G'}. \qquad \square$$

Example 3.17.

1. If G is abelian, then every subgroup is normal, because $gag^{-1} = a$ for all a, g.
2. In general, the *center* of a group G is

$$\{z \in G : zg = gz \quad \forall g \in G\}.$$

   It is always a normal subgroup of G.
3. $\mathrm{SL}_n(\mathbb{R})$ is a normal subgroup of $\mathrm{GL}_n(\mathbb{R})$ since it is the kernel of the homomorphism $\det\colon \mathrm{GL}_n(\mathbb{R}) \to \mathbb{R}^\times$. (Here, $\mathbb{R}^\times$ denotes the group of nonzero real numbers under multiplication.)

Example 3.18. Recall our usual presentation

$$S_3 = \langle x, y | x^3 = 1, y^2 = 1, yx = x^2 y \rangle.$$

The cyclic subgroup generated by y, which consists of the elements $\{1, y\}$, is not normal. This is because

$$xyx^{-1} = xyx^2 = x(x^2y)x = xx^2(x^2y) = x^2y,$$

which is not in $\{1, y\}$.

3.6 Isomorphism classes [2.5]

Lemma 3.19. *If $\varphi\colon G \to G'$ is an isomorphism, then its inverse $\varphi^{-1}\colon G' \to G$ is also an isomorphism.*

As mentioned earlier, we say that G and G' are isomorphic if there exists an isomorphism $\varphi\colon G \to G'$. The *isomorphism class* of G consists of all groups isomorphic to G.

Example 3.20. Suppose $x \in G$ is an element of order n. The cyclic subgroup $\langle x \rangle$ is isomorphic to $\mathbb{Z}/n\mathbb{Z}$. The map

$$\varphi(\overline{a}) = x^a$$

is an isomorphism $\varphi\colon \mathbb{Z}/n\mathbb{Z} \to \langle x \rangle$.

Example 3.21. The groups S_3 and D_3 are isomorphic since they both have the presentation

$$\langle x, y | x^3 = 1, y^2 = 1, yx = x^2 y \rangle.$$

4 Jan 21: Cosets

4.1 Cosets [2.6, 2.8]

Definition 4.1 (Left cosets)**.** Let H be a subgroup of a group G. Let a be any element of G. We denote by aH the set

$$aH \coloneqq \{g \in G : g = ah \text{ for some } h \in H\}.$$

In other words, $aH = \{ah : h \in H\}$. This set is called a *left coset* of H.

Example 4.2. Let G be the additive group $\mathbb{Z}$, and let $H = 100\mathbb{Z}$ be the set of all multiples of 100, i.e.,

$$H = \{\dots, -200, -100, 0, 100, 200, \dots\}.$$

H is a subgroup of $\mathbb{Z}$.

Recall that elements of $\mathbb{Z}/100\mathbb{Z}$ are congruence classes modulo 100. The element we denote $\overline{a}$ is the set of all integers which are congruent to $a \pmod{100}$, so

$$\overline{3} = \{\dots, -197, -97, 3, 103, 203, \dots\} = \{3 + h : h \in 100\mathbb{Z}\}.$$

Thus $\overline{3} = 3 + 100\mathbb{Z}$ is a left coset of H. Note also that

$$3 + 100\mathbb{Z} = 103 + 100\mathbb{Z} = 12403 + 100\mathbb{Z},$$

etc.

Let $\varphi\colon G \to G'$ be a homomorphism. Let $K = \ker(\varphi)$. We know that K is the set of all elements of G which map to $1_{G'}$. In general, for $g' \in G$, the set of all elements of G which map to g' is called the *fiber* over g'.

Proposition 4.3. *Let $\varphi\colon G \to G'$ be a homomorphism. Let $a \in G$ be any element. The set of all elements $x \in G$ such that $\varphi(x) = \varphi(a)$ is the left coset aK.*

Proof. Suppose $\varphi(x) = \varphi(a)$. Then

$$\varphi(a^{-1}x) = \varphi(a)^{-1}\varphi(x) = 1_{G'},$$

so $a^{-1}x \in K$, which implies $x = a(a^{-1}x) \in aK$.

On the other hand, if $x \in aK$, then $x = ak$ for some $k \in K$, so

$$\varphi(x) = \varphi(a)\varphi(k) = \varphi(a). \qquad \square$$

4.2 Counting formula [2.8]

We can also view left cosets of H as equivalence classes for the following equivalence relation.

Let H be a subgroup of G. We define

$$a \sim b \text{ if } b = ah \text{ for some } h \in H.$$

We check that it is indeed an equivalence relation.

- (Transitivity) If $a \sim b$ and $b \sim c$, then $b = ah_1$ and $c = bh_2$, so $c = ah_1h_2$, so $a \sim c$.
- (Symmetry) If $a \sim b$, then $b = ah$ so $a = bh^{-1}$, so $b \sim a$.
- (Reflexivity) We have $a = a \cdot 1_G$, so $a \sim a$.

Corollary 4.4. *The left cosets of H partition G.*

Example 4.5. Recall again that

$$S_3 = \langle x, y | x^3 = 1, y^2 = 1, yx = x^2y \rangle$$

has the 6 elements

$$1, x, x^2, y, xy, x^2y.$$

Let $H = \langle y \rangle$ be the subgroup generated by y. We calculate the 6 sets aH:

$$H = \{1, y\}, \quad xH = \{x, xy\}, \quad x^2H = \{x^2, x^2y\},$$

and

$$yH = \{y, 1\}, \quad xyH = \{xy, x\}, \quad x^2yH = \{x^2y, x^2\}.$$

Note that these are the same as the three above. So there are 3 distinct left cosets of H

$$H = \{1, y\} = yH, \quad xH = \{x, xy\} = xyH, \quad x^2H = \{x^2, x^2y\} = x^2yH,$$

and these three sets partition S_3.

Definition 4.6. The number of left cosets of H is called the *index* of H in G and denoted $[G : H]$. If $|G|$ is infinite, it could be infinite.

Lemma 4.7. *All left cosets aH of H have the same number of elements. (It could be infinite.)*

Proof. We have a bijection $H \to aH$ given by $h \mapsto ah$, with inverse $g \mapsto a^{-1}g$. Thus aH has $|H|$ elements. □

Theorem 4.8 (Counting formula)**.** *For any subgroup H of G,*

$$|G| = |H|[G : H].$$

Proof. This is because G is partitioned into $[G : H]$ equivalence classes, each of which has $|H|$ elements. □

4.3 Lagrange's theorem [2.8]

Theorem 4.9. *Let H be a subgroup of a finite group G. Then $|H|$ divides $|G|$.*

Proof. This follows directly from the above theorem. □

Remark 4.10. Let G be a finite group, and let $a \in G$ be any element Let n be the order of G. Then Lagrange's theorem tells us that n divides $|G|$ because the cyclic subgroup

$$H = \langle a \rangle = \{1, a, \ldots, a^{n-1}\}$$

satisfies $|H| = n$. From this we also get

$$a^{|G|} = 1.$$

Corollary 4.11. *Let p be a prime. Any group G of order p is isomorphic to $\mathbb{Z}/p\mathbb{Z}$.*

Proof. Let a be any element of G other than the identity. Then a has order p, so the subgroup $\langle a \rangle$ has p elements. So $G = \langle a \rangle$, which is isomorphic to $\mathbb{Z}/p\mathbb{Z}$. □

Proposition 4.12 (Groups of order 4)**.** *Let G be a group of order 4. Then G is isomorphic to either $\mathbb{Z}/4\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.*

The dihedral group D_2 is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. It is also called the Klein Four Group.

Proof. By the corollary of Lagrange's theorem mentioned last time, the order of every element divides $|G| = 4$.

Case 1: G has an element x of order 4. Then $G = \langle x \rangle$ so it's isomorphic to $\mathbb{Z}/4\mathbb{Z}$.

Case 2: Every element of G other than the identity has order 2. Then for any $x, y \in G$, we have

$$xyxy = 1,$$

which (using $x^2 = y^2 = 1$), implies $yx = xy$. So G is abelian.

We can check directly that the map

$$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \to G$$

given by

$$(0,0) \mapsto 1_G, \quad (1,0) \mapsto x, \quad (0,1) \mapsto y, \quad (1,1) \mapsto xy$$

is an isomorphism. □

4.4 More on the counting formula [2.8]

Corollary 4.13. *For any homomorphism $\varphi\colon G \to G'$,*

$$|G| = |\ker(\varphi)||\operatorname{im}(\varphi)|.$$

Proof. By Proposition 4.3, the cosets of $\ker(\varphi)$ are the nonempty fibers of φ, which are in bijection with $\operatorname{im}(\varphi)$. Thus, G is partitioned into $|\operatorname{im}(\varphi)|$ cosets of $\ker(\varphi)$, from which the formula follows. □

Proposition 4.14. *If $G \supseteq H \supseteq K$ is a chain of subgroups of a group G, then*

$$[G : K] = [G : H][H : K].$$

Proof. Suppose $[G : H] = n$, and $[H : K] = m$. Then we have partitions

$$G = a_1 H \cup \cdots \cup a_n H,$$

and

$$H = b_1 K \cup \cdots \cup b_m K.$$

The second line lets us note that each $a_j H$, for $1 \leq j \leq n$, is partitioned into m cosets of K

$$a_j H = a_j b_1 K \cup \cdots \cup a_j b_m K.$$

So G is partitioned into mn cosets of K, so $[G : K] = mn$.

The cases where one of $[G : H]$ or $[H : K]$ is infinite are similar. □

4.5 Right cosets [2.8]

Definition 4.15. Let H be a subgroup of G. A *right coset* of H is a set

$$Ha \coloneqq \{ha : h \in H\}.$$

Proposition 4.16. *If H is a normal subgroup of G, then $gH = Hg$ for all $g \in G$.*

Proof. For any $h \in H$, we have

$$gh = ghg^{-1}g \in Hg,$$

where we have used the assumption $ghg^{-1} \in H$ since H is normal. Thus, $gh \subseteq Hg$. Similarly, $Hg \subseteq gH$. □

Example 4.17. We return to the subgroup $H = \langle y \rangle$ of S_3, which is not normal. Earlier, we calculated the left cosets

$$\{1, y\}, \quad \{x, xy\}, \quad \{x^2, x^2 y\}.$$

We similarly calculate the right cosets

$$H = \{1, y\}, \quad Hx = \{x, yx\} = \{x, x^2 y\}, \quad Hx^2 = \{x^2, yx^2\} = \{x^2, xy\},$$

and

$$Hy = \{y, 1\}, \quad Hxy = \{xy, yxy\} = \{xy, x^2\}, \quad Hx^2y = \{x^2y, yx^2y\} = \{x^2y, x\}.$$

Thus, there are also three distinct right cosets

$$\{1, y\}, \quad \{x, x^2y\}, \quad \{x^2, xy\},$$

which also partition G, but they are different from the left cosets.

5 Jan 26: Quotients and correspondence theorem

5.1 Quotients [2.12]

In this subsection, we define, for a **normal** subgroup N of G, the *quotient group* G/N. Note that the quotient group construction only applies when N is normal.

Recall that we had an equivalence relation on G given by

$$a \sim b \text{ if } b = an \text{ for some } n \in N.$$

The equivalence classes are the left cosets of N.

Definition 5.1. The elements of G/N are the left cosets aN. The group operation $\cdot$ on these elements is given by

$$(aN) \cdot (bN) = (ab)N.$$

Example 5.2. For $G = \mathbb{Z}$ and $H = n\mathbb{Z}$, the cosets are $\overline{a} = a + n\mathbb{Z}$, the congruence classes modulo n. The quotient group given by the above definition is the same as the group which we have already been denoting $\mathbb{Z}/n\mathbb{Z}$.

Before we go on, we need to check that the group operation in Definition 5.1 is actually well-defined, since a coset can be written as aN for different choices of a. In fact, recall that

$$aN = a'N \iff a' \in aN.$$

Lemma 5.3. *If $aN = a'N$ and $bN = b'N$ for $a, b, a', b' \in G$, then*

$$(ab)N = (a'b')N.$$

Proof. The assumptions tell us that $a' = an_1$ and $b' = bn_2$ for some $n_1, n_2 \in N$. Then

$$a'b' = an_1bn_2 = ab(b^{-1}n_1b)n_2 \in abN,$$

where we have used the fact that $b^{-1}n_1b \in N$ since N is normal. Thus, $(a'b')N = (ab)N$. □

Thus, we have shown that the group operation defined on G/N actually makes sense. The group axioms can easily be checked. The identity of G/N is the coset $1 \cdot N = N$.

Remark 5.4.

1. Note that if G is a finite group, then by the counting formula,

$$|G/N| = |G|/|N|.$$

2. The group operation on G/N can equivalently be defined by

$$(aN) \cdot (bN) = \{xy : x \in aN, y \in bN\},$$

i.e., the product of cosets is the literal set of products of the elements in the cosets. This is the same as our previous definition by the same kind of calculation as in the proof of Lemma 5.3: if $x = an_1$ and $y = bn_2$, then

$$xy = an_1bn_2 = ab(b^{-1}n_1b)n_2 \in abN.$$

Example 5.5. Let $G = \mathbb{Z}/6\mathbb{Z}$, and let $N = \{0, 3\}$. Then G/N consists of the cosets

$$N = \{0, 3\} = 3 + N, \quad 1 + N = \{1, 4\} = 4 + N, \quad 2 + N = \{2, 5\} = 5 + N.$$

The coset $\{0, 3\}$ is the identity, and the group operation is commutative and given by

$$\{1, 4\} + \{1, 4\} = \{2, 5\}, \quad \{1, 4\} + \{2, 5\} = \{0, 3\}, \quad \{2, 5\} + \{2, 5\} = \{1, 4\}.$$

The quotient group G/N is isomophic to $\mathbb{Z}/3\mathbb{Z}$.

Example 5.6. Let G and H be groups, and let N be the subgroup of $G \times H$ given by

$$N = \{(g, 1_H) : g \in G\} \simeq G.$$

Then

$$G \times H/N \simeq H,$$

via the isomorphism $(g, h)N = (1_G, h)N \mapsto h$.

Example 5.7. Let $G = \mathbb{Z}/4\mathbb{Z}$, and let $N = \{0, 2\}$, which is isomorphic to $\mathbb{Z}/2\mathbb{Z}$. Then G/N also has two elements so it is isomorphic to $\mathbb{Z}/2\mathbb{Z}$. In this case,

$$G \ncong N \times G/N$$

since $\mathbb{Z}/4\mathbb{Z}$ is not isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

5.2 First isomorphism theorem [2.12]

Theorem 5.8 (First isomorphism theorem)**.** *Let $\varphi\colon G \to G'$ be a homomorphism of groups with kernel K. Then G/K is isomorphic to* $\mathrm{im}(\varphi)$.

Proof. The isomorphism is given by

$$\overline{\varphi}(aK) = \varphi(a).$$

This is well-defined and injective because K is the kernel. It is surjective by definition of the image of φ. It is a group homomorphism because

$$\overline{\varphi}((aK) \cdot (bK)) = \overline{\varphi}((ab)K) = \varphi(a)\varphi(b) = \overline{\varphi}(aK)\overline{\varphi}(bK). \qquad \square$$

Example 5.9. The kernel of the homomorphism $\varphi\colon \mathbb{R}^\times \to \mathbb{R}^\times$ given by

$$\varphi(x) = x^2$$

is $\{\pm 1\}$, and the image is $\mathbb{R}_{>0}$, the subgroup of positive real numbers. Thus,

$$\mathbb{R}^\times/\{\pm 1\} \simeq \mathbb{R}_{>0}.$$

Example 5.10. Consider the homomorphism $\varphi\colon D_4 \to \mathbb{Z}/4\mathbb{Z}$ given by

g	1	r	r^2	r^3	s	rs	r^2s	r^3s
$\varphi(g)$	0	2	0	2	2	0	2	0

.

We see that $\ker(\varphi) = \{1, r^2, rs, r^3s\}$. The two cosets of $\ker(\varphi)$ are

$$\{1, r^2, rs, r^3s\} \quad \text{and} \quad \{r, r^3, s, r^2s\}.$$

The quotient $D_4/\ker(\varphi)$ is isomorphic to $\mathbb{Z}/2\mathbb{Z}$, which is also isomorphic to the image $\{0, 2\} \subseteq \mathbb{Z}/4\mathbb{Z}$.

Example 5.11. (Artin, Exercise 2.8.6) Suppose $\varphi\colon G \to G'$ is a nontrivial homomorphism of groups with $|G| = 18$ and $|G'| = 15$. What is the order of its kernel?

Solution. We have $G/\ker(\varphi) \simeq \operatorname{im}(\varphi)$, so

$$18/|\ker(\varphi)| = |\operatorname{im}(\varphi)|.$$

Since $\operatorname{im}(\varphi)$ is a subgroup of G', the order of $\operatorname{im}(\varphi)$ is a divisor of $|G'| = 15$. From the above displayed equation, $|\operatorname{im}(\varphi)|$ also divides 18.

Thus, $|\operatorname{im}(\varphi)|$ divides $\gcd(15, 18) = 3$, and by assumption $|\operatorname{im}(\varphi)| \neq 1$, so $|\operatorname{im}(\varphi) = 3|$. This gives us $|\ker(\varphi)| = 6$.

5.3 Quotient map [2.12]

Proposition 5.12. *Let N be a normal subgroup of G. There is a canonical surjective homomorphism*

$$\pi\colon G \to G/N$$

given by $\pi(g) = gN$. The kernel of π is N.

5.4 Correspondence theorem [2.10]

Let $\varphi\colon G \to G'$ be a homomorphism of groups. We study the relationship between

$$\{\text{subgroups of } G\} \quad \text{and} \quad \{\text{subgroups of } G'\}.$$

We first observe the following way to go between them.

Proposition 5.13. *Let $\varphi\colon G \to G'$ be a homomorphism with kernel K.*

1. *If H is a subgroup of G, then $\varphi(H) = \{\varphi(h) : h \in H\}$ is a subgroup of G'.*

2. *If H' is a subgroup of G', then*

$$\varphi^{-1}(H') \coloneqq \{h \in G : \varphi(h) \in H'\}$$

is a subgroup of G which contains K.

Proof. The first statement is the fact that the image of a homomorphism is a subgroup, which we have seen before.

For the second part, we check closure, and the other group axioms are similar. Suppose $h_1, h_2 \in \varphi^{-1}(H)$. Then

$$\varphi(h_1 h_2) = \varphi(h_1)\varphi(h_2) \in H',$$

where we have used $\varphi(h_1), \varphi(h_2) \in H'$ and the closure condition for H'.

In addition, we see that

$$K \subseteq \varphi^{-1}(H')$$

because $\varphi(a) = 1_{G'} \in H'$ for all $a \in K$. □

Remark 5.14.

1. Note that the method in 2. above only lets us obtain subgroups of G containing K.

2. The method in 1. does not in general give us all subgroups of G' either.

   For example, if $\varphi\colon \mathbb{R}^\times \to \mathbb{R}^\times$ is given by

$$\varphi(x) = x^2,$$

   the entire group $\mathbb{R}^\times$ is not equal to $\varphi(H)$ for any subgroup H of $\mathbb{R}^\times$, since the image of φ consists of positive real numbers.

   However, if φ is *surjective*, then we have the following theorem.

Theorem 5.15 (Correspondence theorem)**.** *If $\varphi\colon G \to G'$ is a surjective homomorphism with kernel K, then the map*

$$\begin{aligned} \{\text{subgroups of } G \text{ containing } K\} &\to \{\text{subgroups of } G'\} \\ H &\mapsto \varphi(H) \end{aligned}$$

is a bijection with inverse

$$\varphi^{-1}(H') \leftarrow\!\shortmid H'.$$

Example 5.16. Consider the homomorphism $\varphi\colon \mathbb{C}^\times \to \mathbb{C}^\times$ given by

$$\varphi(z) = z^2.$$

Here $K = \{\pm 1\}$. The subgroup $\mathbb{R}^\times$ of $\mathbb{C}^\times$ contains K, and it corresponds to the subgroup $\mathbb{R}_{>0}$ of $\mathbb{C}^\times$ consisting of positive real numbers.

Corollary 5.17. *For a normal subgroup N of G, the subgroups of G/N correspond to the subgroups of G which contain N.*

Proof of correspondence theorem.

1. For H' a subgroup of G, we show that $\varphi(\varphi^{-1}(H')) = H'$. By definition of $\varphi^{-1}(H')$, we have
$$\varphi(\varphi^{-1}(H)) \subseteq H'.$$
Since φ is surjective, for each $h' \in H'$, there exists $h \in G$ such that $\varphi(h) = h'$. By definition again, $h \in \varphi^{-1}(H)$. Thus,
$$h' = \varphi(h) \in \varphi(\varphi^{-1}(H)),$$
so $H' \subseteq \varphi(\varphi^{-1}(H))$.

2. For H a subgroup of G containing K, we show that $\varphi^{-1}(\varphi(H)) = H$. By definition, we have
$$H \subseteq \varphi^{-1}(\varphi(H)).$$
Now suppose $g \in \varphi^{-1}(\varphi(H))$. This means that $\varphi(g) \in \varphi(H)$, which by definition means that $\varphi(g) = \varphi(h)$ for some $h \in H$. This means that $g \in hK$, but since $K \subseteq H$, this implies
$$g \in hK \subseteq H.$$
Thus $\varphi^{-1}(\varphi(H)) \subseteq H$. $\square$

6 Jan 28: Group actions

6.1 Correspondence theorem [2.10]

The correspondence between subgroups also gives us some information about normal subgroups.

Theorem 6.1 (Correspondence theorem)**.** *If $\varphi\colon G \to G'$ is a surjective homomorphism with kernel K, then the map*

$$\begin{aligned}\{\text{subgroups of } G \text{ containing } K\} &\to \{\text{subgroups of } G'\}\\ H &\mapsto \varphi(H)\end{aligned}$$

is a bijection with inverse

$$\varphi^{-1}(H') \leftarrow\!\shortmid H'.$$

Theorem 6.2 (Correspondence theorem, cont'd)**.** *If H and H' are corresponding subgroups, then H is normal if and only if H' is normal.*

Proof.

1. Suppose H' is normal. Then for any $h \in \varphi^{-1}(H')$ and $g \in G$, we have

   $$\varphi(ghg^{-1}) = \varphi(g)\varphi(h)\varphi(g)^{-1} \in H',$$

   where we have used $\varphi(h) \in H'$ and the fact that H' is normal. (This implication does not require φ to be surjective.)

2. Suppose H is normal. Then for any $g' \in G'$ and $h \in H$, we have

   $$g'\varphi(h)g'^{-1} = \varphi(ghg^{-1}) \in \varphi(H),$$

   where $g \in G$ is some element such that $\varphi(g) = g'$. Here we have used surjectivity of φ. □

6.2 Symmetry

Example 6.3. Recall that S_n is the set of bijective maps from $\{1, 2, \ldots, n\}$ to itself.

Example 6.4. Recall that

$$\begin{aligned}\mathrm{GL}_n(\mathbb{R}) &= \{\text{invertible } n \times n \text{ matrices } A \text{ with real entries}\}\\ &= \{\text{bijective linear maps } f\colon \mathbb{R}^n \to \mathbb{R}^n\}.\end{aligned}$$

A linear map is a function which preserves the vector space structure on $\mathbb{R}^n$, i.e.,

$$f(au + bv) = af(u) + bf(v).$$

The matrix A corresponds to the linear map $f(v) = Av$.

Example 6.5. The dihedral group D_n is the group of symmetries of a regular n-gon $A_1A_2\cdots A_n$. What we really mean is

$$D_n = \{\text{isometries from } A_1A_2\ldots A_n \text{ to itself}\}.$$

Here, an *isometry* of the plane is a map $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ which preserves distance, i.e.,

$$\|f(u) - f(v)\| = \|u - v\|$$

for all $u, v \in \mathbb{R}^2$.

Not all isometries are linear maps (and not all linear maps are isometries), but all of the elements of the dihedral group are linear maps.

Here are the elements of D_n in matrix form. Suppose that $A_1A_2\ldots A_n$ is centered at $(0,0)$, and $A_1 = (1,0)$.

The element r^k is a rotation by $\theta = 2\pi k/n$ around the origin, so

$$r^k = \begin{bmatrix} \cos(2\pi k/n) & -\sin(2\pi k/n) \\ \sin(2\pi k/n) & \cos(2\pi k/n) \end{bmatrix}.$$

The element s is reflection across the x-axis, so it sends (x, y) to $(x, -y)$, so

$$s = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

The other reflections are

$$r^k s = \begin{bmatrix} \cos(2\pi k/n) & \sin(2\pi k/n) \\ \sin(2\pi k/n) & -\cos(2\pi k/n) \end{bmatrix}.$$

6.3 Group actions [6.7]

In the previous subsection, we saw that many groups we've seen consist of bijective maps from some set to itself, which preserve some additional structure on the set.

Another example of this is the set of automorphisms of a group G, which are the bijective maps from G to itself which preserve the group structure.

Definition 6.6. Let S be a set and let G be a group. An action of G on S is a map $*\colon G \times S \to S$ satisfying

1. $1 * s = s$ for all $s \in S$.
2. (associativity) $(gh) * s = g * (h * s)$ for all $g, h \in G$ and $s \in S$.

Given an action of G ond S, every element $g \in G$ can be viewed as a bijective map

$$g\colon S \to S, \quad s \mapsto g * s.$$

It is bijective because the axioms imply that $g^{-1}\colon S \to S$ is its inverse:

$$g * (g^{-1} * s) = 1 * s = s \quad \forall s \in S.$$

Example 6.7. The map

$$*\colon G \times G \to G, \quad g * a = gag^{-1}$$

is an action of G on G itself. This is G acting on itself by conjugation. We check that it satisfies the group action axioms.

1. $1 * a = 1a1 = a$ for all $a \in G$.
2. $(gh) * a = (gh)a(gh)^{-1} = ghah^{-1}g^{-1} = g * (hah^{-1}) = g * (h * a)$.

In this case, every element g acts by an automorphism of G.

Example 6.8. The map

$$*\colon G \times G \to G, \quad g * a = ga$$

is also a group action. We check associativity:

$$(gh) * a = gha = g * (ha) = g * (h * a).$$

In this case, the elements g don't act by automorphisms, only bijections.

6.4 Orbits and stabilizers [6.7]

Let G be a group acting on a set S.

Definition 6.9. Let $s \in S$. The *orbit* of s is

$$O_s = \{s' \in S : s' = gs \text{ for some } g \in G\}.$$

We can define an equivalence relation on S by

$$s \sim s' \text{ if } s' = sg \text{ for some } g \in G.$$

Then the orbits O_s are the equivalence classes of S under this equivalence relation. The orbits of the action of G on S partition S.

Example 6.10.

1. Consider the action of $\mathbb{R}^\times$ on the set $\mathbb{R}$ given by $c * x = cx$. The orbit of $x = 0$ is $\{0\}$, and the orbit of any nonzero x is $\mathbb{R} - \{0\}$.
2. Consider the action of $\mathbb{R}^\times$ on $\mathbb{R}^2$ given by $c * (x, y) = (cx, c^{-1}y)$. The orbits are

   $$\{(0,0)\}, \quad \{(x,0) : x \neq 0\}, \quad \{(0,y) : y \neq 0\}, \quad \{(x,y) : xy = a\}$$

   for all nonzero real numbers a.

Definition 6.11. A group action is *transitive* if there is only one orbit, i.e., for any two elements $s, s' \in S$, there exists $g \in G$ such that $s' = gs$.

Example 6.12. The action of S_n on $\{1, 2, \ldots, n\}$ is transitive.

Definition 6.13. Let $s \in S$. The *stabilizer* is

$$G_s = \{g \in G : gs = s\}.$$

It is the set of elements of G which fix s. The group action axioms imply that G_s is a actually a subgroup of G.

Example 6.14. Consider the action of the dihedral group $G = D_6$ acting on the set of vertices $S = \{A_1, A_2, \ldots, A_6\}$ of a regular hexagon. This action is transitive, so the orbit of A_1 is the entire set

$$O_{A_1} = \{A_1, A_2, \ldots, A_6\}.$$

The only elements of D_6 which fixes A_1 are the identity and reflection across OA_1. Thus

$$G_{A_1} = \{1, s\} = \langle s \rangle.$$

The stabilizers of the other vertices are also groups of order 2.

Lemma 6.15. *Let H be a subgroup of a group G. For any element $a \in G$, the subset*

$$aHa^{-1} = \{g \in G : g = aha^{-1} \text{ for some } h \in H\}$$

is also a subgroup of G.

Proposition 6.16. *Let G be a group acting on a set S, and let $s \in S$. Then for any $a \in G$, the stabilizer of $s' = as$ is the conjugate subgroup aG_sa^{-1}.*

Proof. Let $g \in G$. Note that

$$gs' = s' \iff gas = as \iff a^{-1}gas = s \iff a^{-1}ga \in G_s \iff g \in aG_sa^{-1}.$$

□

6.5 Operation on cosets [6.8]

Let G be a group, and let H by any subgroup of G. Let G/H denote the set of left cosets of H.

Following Artin, we write $[C]$ to denote a coset C when viewed as an element of the set G/H. We have an action of G on G/H defined by

$$g[C] = [gC]$$

where $gC = \{gc : c \in C\}$.

This action can equivalently be defined by $g[aH] = [gaH]$.

Proposition 6.17. *The action of G on G/H is transitive and the stabilizer of the coset $[H]$ is the subgroup H.*

Example 6.18. Let $G = S_3$, $H = \langle y \rangle = \{1, y\}$. Then

$$G/H = \{\{1, y\}, \{x, xy\}, \{x^2, x^2y\}\}.$$

The element x acts by a 3-cycle

$$[\{1, y\}] \to [\{x, xy\}] \to [\{x^2, x^2y\}] \to [\{1, y\}].$$

The element y fixes $[H] = [\{1, y\}]$, and swaps

$$[\{x, xy\}] \leftrightarrow [\{x^2, x^2y\}].$$

The stabilizer of the *element* $[\{1, y\}] \in G/H$ is the *subgroup* $H = \{1, y\} \subseteq G$.

Note that left multiplication by y does not fix the elements of $\{1, y\}$, rather, it fixes the entire coset $[\{1, y\}]$.

7 Feb 2: Orbit-stabilizer theorem, Burnside's lemma, permutation representations

7.1 Orbit-stabilizer theorem [6.8, 6.9]

Theorem 7.1. *Let S be a finite set on which a group G acts. Let $s \in S$. The elements of O_s are in bijection with the left cosets of G_s. In particular,*

$$|G| = |G_s| \cdot |O_s|.$$

Proof. Note that

$$O_s = \{gs : g \in G\}.$$

However, sometimes $gs = g's$ for different $g, g' \in G$. This happens exactly when $g^{-1}g's = s$, i.e.,

$$g^{-1}g' \in G_s,$$

or $g' \in gG_s$.

Thus, we have a bijection $G/G_s \to O_s$ given by $[gG_s] \mapsto gs$. This is well-defined by definition of G_s, surjective by definition of O_s, and injective by the discussion above.

Thus, $|O_s| = [G : G_s]$, the number of left cosets of G_s, and the formula in the theorem statement follows from the counting formula

$$|G| = |G_s|[G : G_s].$$

□

7.2 Orbits partition S [6.9]

Note that if S is partitioned into k orbits $O_1, O_2, \ldots, O_k$, then we also have the formula

$$|S| = |O_1| + \cdots + |O_k|.$$

7.3 Burnside's lemma

Definition 7.2 (Fixed points)**.** Let G be a group acting on a set S. For any $g \in G$, the set of *fixed points* of g is the set

$$S^g := \{s \in S : gs = s\}.$$

Theorem 7.3 (Burnside's lemma)**.** *Let G be a finite group acting on a finite set S. Then*

$$\text{number of orbits of } G \text{ acting on } S = \frac{1}{|G|} \sum_{g \in G} |S^g|.$$

Proof. Suppose S is partitioned into k orbits $O_1, \ldots, O_k$.

1. Let $s \in S$. The orbit stabilizer theorem gives

$$\frac{1}{|O_s|} = \frac{|G_s|}{|G|}.$$

We take the sum over $s \in S$ of both sides, to get

$$\sum_{s \in S} \frac{1}{|O_s|} = \frac{1}{|G|} \sum_{s \in S} |G_s|. \tag{$\heartsuit$}$$

2. However note that the left hand side of ($\heartsuit$) is equal to

$$\sum_{s \in S} \frac{1}{|O_s|} = \sum_{i=1}^{k} \sum_{s \in O_i} \frac{1}{|O_i|} = \sum_{i=1}^{k} 1 = k,$$

the number of orbits! So in fact,

$$k = \frac{1}{|G|} \sum_{s \in S} |G_s|.$$

3. On the other hand,

$$\sum_{s \in S} |G_s| = \sum_{g \in G} |S^g|.$$

This equation comes from counting the number of elements in the set

$$X = \{(g, s) \in G \times S : gs = s\}$$

in two different ways: for fixed $s_0 \in S$, the number of pairs (g, s_0) in X is $|G_{s_0}|$. Then we sum over s_0. Alternatively, for each $g_0 \in G$, the number of pairs (g_0, s) in X is $|S^{g_0}|$, and we sum over g_0.

□

Example 7.4. Let p be a prime. Find the number of different ways to color the edges of a regular p-gon using a colors, where two colorings are considered the same if one can be obtained from the other by rotating the p-gon.

Solution. If we don't consider rotations, the number of colorings is clearly a^p; for each of the p edges, we choose one of the a colors. Let S be the set of these colorings.

Let $G = \{1, r, r^2, \ldots, r^{p-1}\}$ be the group of rotations of the regular p-gon. We have an action of G on S. Recall that for any group action, we have an equivalence relation on S defined by

$$s' \sim s \text{ if } s' = gs \text{ for some } g,$$

whose equivalence classes are the orbits of the action.

Thus, the problem is exactly asking for the number of orbits of G acting on the set S of size a^p. By Burnside's lemma, we just need to calculate

$$\frac{1}{p} \cdot \sum_{g \in G} |S^g|.$$

Note that G is naturally isomorphic to $\mathbb{Z}/p\mathbb{Z}$. It is easier to think about G this way. There are two cases.

1. $g = 1$. Every element of S is fixed by the identity so,

$$|S^g| = |S| = a^p. \tag{$\spadesuit$}$$

2. $g = r^j$ for $j = 1, \ldots, p-1$. Note that s is fixed by g if and only if s is fixed by the entire subgroup $\langle g \rangle$ generated by g. In this case, since p is prime, $\langle g \rangle$ is the entire group G.

   The only colorings which are fixed by any rotation are the ones where every edge is the same color. There are a of these, so $|S^g| = a$

Thus, the sum in ($\spadesuit$) has one term equal to a^p and $(p-1)$ terms equal to a, and the final answer is

$$\boxed{\frac{a^p + (p-1)a}{p}.}$$

Example 7.5. Do the same as above but for a regular 25-gon.

Solution. The setup is the same. The set S has size a^{25}, and the group G which is isomorphic to $\mathbb{Z}/25\mathbb{Z}$ acts on S. We want to find the number of orbits, which Burnside's lemma tells us is equal to the sum

$$\frac{1}{25} \sum_{g \in G} |S^g|.$$

Since 25 is not prime, there are three cases now. The cases correspond to subgroups of $\mathbb{Z}/25\mathbb{Z}$, which also correspond to the divisors of 25.

1. $g = \overline{0}$. As before, $|S^g| = |S| = a^{25}$.

2. $\langle g \rangle = \{\overline{0}, \overline{5}, \overline{10}, \overline{15}, \overline{20}\}$. If we label the edges of the 25-gon $e_1, e_2, \ldots, e_{25}$, the colorings which are fixed by r^5 are determined by the colors of any 5 consecutive edges, e.g., $e_1, e_2, \ldots, e_5$. Then e_6 has the same color as e_1 because $r^5(e_1) = e_6$, etc.

   Thus, there are a^5 colorings which are fixed by $\{\overline{0}, \overline{5}, \overline{10}, \overline{15}, \overline{20}\}$, so for any generator of this subgroup, $|S^g| = a^5$.

3. $\langle g \rangle = \mathbb{Z}/25\mathbb{Z}$. Same as before, the only colorings fixed by the entire group G are the ones where all edges are the same color. So $|S^g| = a$.

Now, there are four elements $g \in \mathbb{Z}/25\mathbb{Z}$ such that $\langle g \rangle = \{\overline{0}, \overline{5}, \overline{10}, \overline{15}, \overline{20}\}$. There are 20 elements such that $\langle g \rangle = \mathbb{Z}/25\mathbb{Z}$; they correspond to the integers from $1, 2, \ldots, 25$ which are coprime to 25. Thus in the sum we want to compute, there is one term equal to a^{25}, four terms equal to a^5, and 20 terms equal to a. The final answer is

$$\boxed{\frac{a^{25} + 4a^5 + 20a}{25}.}$$

7.4 Permutation repsentations [6.11]

If a group G acts on a finite set of n elements, this gives us a homomorphism $G \to S_n$.

Proposition 7.6. *Let G be a group. Let $S = \{1, 2, \ldots, n\}$. There is a bijection*

$$\{\text{actions of } G \text{ on } S\} \leftrightarrow \{\text{homomorphisms } G \to S_n\}.$$

Proof. The proof is trivial but this is how it works. Suppose we are given an action of G on S. We want to define a homomorphism $\varphi\colon G \to S_n$.

Recall that elements of S_n are bijective maps from S to itself, and the group operation is function composition. We simply define

$$\varphi(g) = m_g,$$

where $m_g\colon S \to S$ is the map $m_g(s) = g * s$. Note that $m_g \in S_n$.

We check it satisfies $\varphi(gh) = \varphi(g) \circ \varphi(h)$. This is equivalent to $m_{gh} = m_g \circ m_h$, which is true since

$$m_{gh}(s) = (gh) * s = g * (h * s) = m_g(m_h(s)).$$

The map in the other direction is defined the same way. Given φ, the action of G on S is defined by $g * s = \varphi(g)(s)$. □

Definition 7.7 (Faithful group action)**.** An action of G on S is *faithful* if the only element $g \in G$ which fixes every element of S is the identity element of G, i.e.,

$$gs = s \quad \forall s \in S \implies g = 1.$$

If $S = \{1, 2, \ldots, n\}$, the action of G on S being faithful is equivalent to the corresponding homomorphism $G \to S_n$ being injective.

Example 7.8.

1. For any group G and any set S, there is an action of G on S defined by $gs = s$ for all $g \in G$ and $s \in S$.

   It is not faithful (unless G has only one element). If $S = \{1, 2, \ldots, n\}$, it corresponds to the trivial homomorphism $G \to S_n$.

2. We saw on problem set 2 that there's an injective homomorphim $D_5 \to S_5$, because D_5 acts faithfully on the vertices of a regular pentagon.

3. There is no injective homomorphism $D_5 \to S_4$, because $|D_5| = 10$ and $|S_4| = 24$. However, 10 does not divide 24, so S_4 does not have a subgroup isomorphic to D_5, by Lagrange's theorem.

8 Feb 4: Class equation

8.1 Cayley's theorem [7.1]

Recall that we have an action of G on itself by left multiplication

$$G \times G \to G, \quad (g, x) \mapsto gx.$$

For each $g \in G$, the (left) multiplication-by-g map m_g is bijective, and moreover, it does not fix all (or any) $x \in G$ unless $g = 1$. Thus this action is faithful.

Theorem 8.1. *A finite group G of order n is isomorphic to a subgroup of S_n.*

Proof. From last time, the above group action gives us a homomorphism φ from G to

$$\mathrm{Perm}(G) := \text{ the group of permutations of the set of elements of } G,$$

which is isomorphic to S_n, since $|G| = n$. Specifically, the homomorphism is given by $\varphi(g) = m_g$.

Since the left multiplication action is faithful, φ is an injective homomorphism from $G \to S_n$, so G is isomorphic to the subgroup $\mathrm{im}(\varphi)$ of S_n. $\square$

8.2 Class equation [7.2]

Also recall that we have another action of G on itself by *conjugation*. It is the map

$$G \times G \to G, \quad (g, x) \mapsto gxg^{-1}.$$

Question 8.2. What are the orbits of the action of G on itself by left multiplication?

Definition 8.3. A *conjugacy class* in a group G is an orbit of the conjugation action of G on itself. The conjugacy class of an element $a \in G$ is

$$C(x) = \{gag^{-1} : g \in G\}.$$

Example 8.4. The conjugacy class of the identity element is always just

$$C(1) = \{1\}.$$

Example 8.5. In $\mathrm{GL}_n(\mathbb{R})$, two matrices A and B are conjugate if

$$B = XAX^{-1}$$

for some invertible $n \times n$ matrix X. So conjugate elements of $\mathrm{GL}_n(\mathbb{R})$ have the same characteristic polynomial.

The converse is not quite true. For example

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

both have characteristic polynomial $(t-1)^2$, but they are not conjugate.

Definition 8.6. For any $x \in G$, the *centralizer* of x in G is the stabilizer of x under the conjugation action. We denote it

$$Z(x) = \{g \in G : gxg^{-1} = x\} = \{g \in G : gx = xg\}.$$

Definition 8.7. The *center* of a group G is the set

$$Z = \{z \in G : zg = gz \text{ for all } g \in G\}$$

of elements which commute with all elements of G.

Example 8.8.

1. The centralizer of the identity element is always the whole group.
2. If G is abelian, $Z(x) = Z = G$ for any $x \in G$.
3. The centralizer of the matrix $x = \begin{bmatrix} 2 & \\ & 1 \end{bmatrix} \in \mathrm{GL}_2(\mathbb{R})$ consists of all matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ such that

$$\begin{bmatrix} 2a & b \\ 2c & d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 2 & \\ & 1 \end{bmatrix} = \begin{bmatrix} 2 & \\ & 1 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 2a & 2b \\ c & d \end{bmatrix}.$$

   Solving this equation gives $c = d = 0$, so

$$C(x) = \left\{ \begin{bmatrix} a & \\ & d \end{bmatrix} : a, d \in \mathbb{R}^\times \right\}.$$

4. The center of $\mathrm{GL}_n(\mathbb{R})$ consists of the scalar matrices $c \cdot I_n$ for $c \in \mathbb{R}^\times$.

Proposition 8.9. *Let G be a group and let $x \in G$.*

1. *The centralizer $Z(x)$ contains x, and it contains Z.*
2. *The following are equivalent.*

   (i) $Z(x) = G$. (ii) $x \in Z$. (iii) $C(x) = \{x\}$.

The *class equation* is the formula

$$|G| = \sum_{\substack{\text{conjugacy} \\ \text{classes } C}} |C|.$$

This equation is trivial, but it is useful because there are several restrictions on the numbers that appear on the right hand side.

- At least one of the terms in the sum on the right is equal to 1, it corresponds to the conjugacy class $C(1)$.

- Moreover, every 1 that appears on the right hand side corresponds to an element of the center Z.
- Every term on the right divides $|G|$, since by the orbit-stabilizer theorem,

$$|G| = |Z(x)| \cdot |C(x)|.$$

Example 8.10. We compute the class equation for S_3, by first determining the centralizers, and then using the orbit-stabilizer theorem.

1. The centralizer $Z(x)$ contains x, which has order 3, so $Z(x)$ is a subgroup of S_3 whose order is a multiple of 3, so $|Z(x)| = 3$ or 6.

   It is not the whole group because x is not in the center of S_3, since

   $$yx = x^2y \neq xy,$$

   so $|Z(x)| = 3$, and $|C(x)| = 2$.

2. The centralizer $Z(y)$ contains y, so $Z(y)$ is a subgroup of S_3 whose order is even, so $|Z(y)| = 2$ or 6.

   Again, y is not in the center of S_3, so $|Z(y)| = 2$, and $|C(y)| = 3$.

So we have found a conjugacy class of size 3 and a conjugacy class of size 2, and together with $\{1\}$, these partition S_3. Thus, the class equation is

$$6 = 1 + 2 + 3.$$

8.3 Application of class equation [7.3]

In this subsection, let p be a prime. We use the class equation to classify groups of order p^2.

Definition 8.11. A *p-group* is a group G whose order is a positive power of p.

Proposition 8.12. *The center of a p-group is not trivial.*

Proof. Suppose $|G| = p^e$, with $e \geq 1$. The class equation reads

$$p^e = \sum \text{divisor of } p^e,$$

and 1's on the right hand side correspond exactly to the elements of the center. All terms which are not equal to 1 are divisible by p, so the number of 1's in the sum is a multiple of p.

There is at least one 1 in the sum, corresponding to the identity, so there must actually be at least p 1's, so the center is not trivial. □

Proposition 8.13. *A group G of order p^2 is abelian.*

Proof. By the above proposition, Z has order either p or p^2. In fact, we show now that a group of order p^2 cannot have center of order p. This would imply $Z = G$, so G is abelian.

Suppose to the contrary that $|Z| = p$. Let $x \notin Z$. The fact we use is that the centralizer $Z(x)$ is a subgroup of G so its order divides p^2. It contains Z and x, so it has order greater than p, so $|Z(x)| = p^2$. Thus $Z(x) = G$, but $Z(x) = G$ is equivalent to $x \in Z$, a contradiction. $\square$

9 Feb 11: Conjugation in S_n

9.1 Conjugation in the symmetric group [7.5]

The main observation here is that **conjugation in the symmetric group is basically renaming the indices**.

For example, let $p = (1\ 3\ 4)\ (2\ 5)$, and let $q = (1\ 4\ 5\ 2)$. We compute qpq^{-1}. We forget about the cycle decomposition for a second. The permutations p and q are the functions

i	1	2	3	4	5
$p(i)$	3	5	4	1	2

i	1	2	3	4	5
$q(i)$	4	1	3	5	2

Then

i	$q(1)$	$q(2)$	$q(3)$	$q(4)$	$q(5)$
$qpq^{-1}(i)$	$q(3)$	$q(5)$	$q(4)$	$q(1)$	$q(2)$

This is because $qpq^{-1}(q(i)) = q(p(i))$ for any i, so we just added a $q(\cdot)$ to every entry of the first table.

The function q tells us how to "rename" the indices. So in this case, the cycle decomposition of qpq^{-1} is

$$(q(1)\ q(3)\ q(4))\ (q(2)\ q(5)) = (4\ 3\ 5)\ (1\ 2).$$

If we the cycle decomposition for qpq^{-1} under p, then we can read off the table for q.

$$\frac{(1\ 3\ 4)\ (2\ 5)}{(4\ 3\ 5)\ (1\ 2)}$$

Theorem 9.1. *Two elements of S_n are in the same conjugacy class if and only if they have the same cycle type.*

A couple more: $(1\ 2)\ (3\ 4)$ is conjugate to $(1\ 3)\ (2\ 4)$, and $(1\ 2)\ (3\ 4\ 5)\ (6\ 7)$ is conjugate to $(1\ 5)\ (2\ 4)\ (3\ 7\ 6)$.

Thus, the conjugacy classes in S_n correspond to *partitions* of n.

9.2 Class equations of S_4 and S_5 [7.5]

Example 9.2. The partitions of 4 are

$$4 = 4, \quad 4 = 3 + 1, \quad 4 = 2 + 2, \quad 4 = 2 + 1 + 1, \quad 4 = 1 + 1 + 1 + 1.$$

The corresponding conjugacy classes of S_4 and their sizes are

partition	$4 = 4$	$4 = 3 + 1$	$4 = 2 + 2$	$4 = 2 + 1 + 1$
shape	$(-\ -\ -\ -)$	$(-\ -\ -)$	$(-\ -)\ (-\ -)$	$(-\ -)$
$\lvert C\rvert$	6	8	3	6

in addition to the conjugacy class of the identity. The class equation is

$$24 = 1 + 3 + 6 + 6 + 8.$$

Example 9.3. The partitions of 5 are

$$5,\ 4+1,\ 3+2,\ 3+1+1,\ 2+2+1,\ 2+1+1+1,\ 1+1+1+1+1.$$

There are 24 elements of the form $(-\ -\ -\ -\ -)$, 30 $(-\ -\ -\ -)$'s, 20 $(-\ -\ -)\ (-\ -)$'s, 20 $(-\ -\ -)$'s, 15 $(-\ -)\ (-\ -)$'s, 10 $(-\ -)$'s, and the identity. The class equation is

$$120 = 1 + 10 + 15 + 20 + 20 + 30 + 24.$$

9.3 Signs of permutations

Recall that we defined the sign homomorphism

$$\operatorname{sign}\colon S_n \to \{\pm 1\},$$

by $\operatorname{sign}(\sigma) = \det P_\sigma$, where P_σ is the permutation matrix associated to the permutation σ. Recall that this is a group homomorphism, meaning that we have

$$\operatorname{sign}(\sigma\tau) = \operatorname{sign}(\sigma)\operatorname{sign}(\tau).$$

Definition 9.4. A *transposition* is a permutation of the form $(i\ j)$, i.e., it is a permutation which just swaps two indices i and j.

Example 9.5. The permutation matrix associated to the transposition $(1\ 2)$ is

$$\begin{pmatrix} & 1 & & & \\ 1 & & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix},$$

which is one row-swap away from the identity matrix, so it has determinant 1. Thus the sign of $(1\ 2)$ is -1. A similar argument shows that $\operatorname{sign}(i\ j) = -1$ for any transposition $(i\ j)$.

Proposition 9.6. *Every element of S_n can be expressed as a product of transpositions.*

Example 9.7. Let $\sigma \in S_5$ be the permutation

i	1	2	3	4	5
$\sigma(i)$	3	5	4	1	2

Working backwards,

i	1	2	3	4	5
$(1\ 3) \circ \sigma(i)$	1	5	4	3	2

i	1	2	3	4	5
$(2\ 5) \circ (3\ 4) \circ (1\ 3) \circ \sigma(i)$	1	2	3	4	5

Proof of Proposition 9.6. We note that

$$\begin{aligned}(i_1\ i_2\ \cdots\ i_k) &= (i_1\ i_k) \circ (i_1\ i_2\ \cdots\ i_{k-1}) \\ &= (i_1\ i_k) \circ (i_1\ i_{k-1}) \circ \cdots \circ (i_1\ i_2).\end{aligned}$$

Since every permutation can be decomposed into cycles, every permutation can be written as a product of transpositions. □

We call permutations with sign $+1$ *even*. Permutations with sign -1 are called *odd*.

Example 9.8. Transpositions are odd. By the decomposition in the proof of Proposition 9.6, a k-cycle is even if and only if k is odd. The permutation

$$(1\ 2)\ (3\ 4\ 5)\ (6\ 7\ 8\ 9)$$

has sign $(-1) \cdot 1 \cdot (-1) = 1$.

9.4 Alternating group [7.5]

Definition 9.9. The *alternating group* A_n is the kernel of the sign homomorphism, i.e., it is the subgroup of S_n of even permutations.

Example 9.10. The elements of S_3 are

$$1, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2).$$

The elements of A_3 are

$$1, (1\ 2\ 3), (1\ 3\ 2).$$

Question 9.11. What is the order of A_5?

Note that conjugate permutations have the same sign. In S_5, the conjugacy classes which contain even permutations are the ones consisting of permutations with the cycle types, in addition to the identity.

shape	$(-\ -\ -\ -\ -)$	$(-\ -\ -)$	$(-\ -)\ (-\ -)$
number	24	20	15

However, the class equation of A_5 is **not** $60 = 1 + 15 + 20 + 24$.

- Two permutations p and p' may be conjugate in S_5 but not in A_5, i.e., there could exist $q \in S_5$ such that

  $$p' = qpq^{-1},$$

  but there may not exist q in the smaller group A_5 which makes the same equation hold.

- If two elements are conjugate in A_5, then they are clearly also conjugate in S_5.
- Thus, some of the even conjugacy classes of S_5 remain a class in A_5, and some split up in A_5.

We will see shortly that the conjugacy class of 5-cycles splits into two conjugacy classes of size 12 in A_5, and the class equation of A_5 is

$$60 = 1 + 15 + 20 + 12 + 12.$$

9.5 Simple groups [7.4]

Definition 9.12. A group G is *simple* if it contains normal subgroup other than $\{1\}$ and G.

Theorem 9.13. *The alternating group A_5 is simple.*

Proof. Suppose N is a normal subgroup of A_5. By definition of normal, it must be a union of conjugacy classes, including the conjugacy class $\{1\}$. By Lagrange's theorem, the order of N must divide 60.

There is no divisor of 60 other than 1 and 60 which can be written as a sum of some of the terms 1, 15, 20, 12, 12, if the term 1 must be included. □

The group A_4 contains the proper normal subgroup

$$\{1, (1\ 2)\ (3\ 4), (1\ 3)\ (2\ 4), (1\ 4)\ (2\ 3)\},$$

so it is not simple.

Theorem 9.14. *The alternating group A_n is simple for $n \geq 5$*

9.6 Class equation of A_5

Lemma 9.15. *If p and p' are conjugate in S_n and the centralizer $Z(p)$ contains an odd permutation, then p and p' are also conjugate in A_n.*

Proof. Suppose σ has sign -1, and $\sigma p \sigma^{-1} = p$.

Since p and p' are conjugate, there exists $q \in S_n$ such that $qpq^{-1} = p'$. If q is even, then p and p' are conjugate in A_n.

If q is odd, note that

$$(q\sigma)p(q\sigma)^{-1} = q\sigma p\sigma^{-1}q^{-1} = p'$$

as well, and $q\sigma \in A_n$, so p and p' are conjugate in A_n. □

Lemma 9.16. *Let $p \in S_n$. If the centralizer $Z(p)$ contains only even permutations, then the conjugacy class $C(p)$ splits into two conjugacy classes (of equal size) in A_n.*

Proof. We temporarily denote by $Z_G(p)$ and $C_G(p)$ the centralizer and conjugacy class of p in the group G, respectively.

The assumption is that $Z_{S_n}(p) = Z_{A_n}(p)$. By the orbit-stabilizer theorem

$$|A_n| = |Z_{A_n}(p)| \cdot |C_{A_n}(p)|,$$

and

$$|S_n| = |Z_{S_n}(p)| \cdot |C_{S_n}(p)|.$$

Since $|S_n| = 2|A_n|$, and $|Z_{S_n}(p)| = |Z_{A_n}(p)|$, the above two equations give us

$$|C_{S_n}(p)| = 2 \cdot |C_{A_n}(p)|.$$

Thus, $C(p)$ splits into two conjugacy classes in A_n, with half the size. $\square$

In A_5, there are 3 cases:

1. The centralizer of the 3-cycle $(1\ 2\ 3) = (1\ 2\ 3)\ (4)\ (5)$ contains the odd permutation $(4\ 5)$. Thus, the set of all 3-cycles forms a conjugacy class in A_5. Recall that there are 20 of these.

2. The centralizer of $(1\ 2)\ (3\ 4)$ contains the odd permutation $(1\ 2)$. Thus, the set of all permutations with cycle type $(-\ -)\ (-\ -)$ forms a conjugacy class in A_5 as well as in S_5. There are 15 of these.

3. Let $p = (1\ 2\ 3\ 4\ 5)$. This is an even permutation. The permutations q such that $qpq^{-1} = p$ are (the identity or) given by the tables

$$\frac{(1\ 2\ 3\ 4\ 5)}{(2\ 3\ 4\ 5\ 1)}, \frac{(1\ 2\ 3\ 4\ 5)}{(3\ 4\ 5\ 1\ 2)}, \frac{(1\ 2\ 3\ 4\ 5)}{(4\ 5\ 1\ 2\ 3)}, \frac{(1\ 2\ 3\ 4\ 5)}{(5\ 1\ 2\ 3\ 4)},$$

   i.e., $(1\ 2\ 3\ 4\ 5)$, $(1\ 3\ 5\ 2\ 4)$, $(1\ 4\ 2\ 5\ 3)$, $(1\ 5\ 4\ 3\ 2)$, and the identity.

   These are the powers of p, so they are all even. Thus the set of 5-cycles forms two conjugacy classes in A_5, with 12 elements each.

   Indeed, the 5-cycles $(1\ 2\ 3\ 4\ 5)$ and $(1\ 3\ 5\ 2\ 4)$ are not conjugate in A_5.

Thus, the class equation of A_5 is

$$60 = 1 + 15 + 20 + 12 + 12.$$

10 Feb 16: Sylow theorems

10.1 Sylow p-subgroups [7.7]

Let p be a prime. Recall that we call a group of order p^e for $e > 0$ a *p-group*.

Definition 10.1. Let G be a group of order $p^e m$, where p does not divide m. A *Sylow p-subgroup* of G is a subgroup of G of order p^e.

The Sylow theorems are useful for the classification of finite groups.

10.2 First Sylow Theorem [7.7]

Theorem 10.2. *Every finite group G whose order is divisible by p contains a Sylow p-subgroup.*

Corollary 10.3 (Partial converse to Lagrange's theorem)**.** *A finite group G whose order is divisible by p contains an element of order p.*

Proof. By the First Sylow Theorem, G contains a subgroup H of order p^e, for some $e > 0$. Orders of elements of H must divide p^e by Lagrange's theorem, so H contains an element x of order p^k for some $k > 0$. Then $x^{p^{k-1}}$ has order p. □

10.3 Second Sylow Theorem [7.7]

Theorem 10.4. *Let G be a group whose order is divisible by p.*

(a) Any two Sylow p-subgroups of G are conjugate subgroups.

(b) Every subgroup of G whose order is a power of p is contained in a Sylow p-subgroup.

Corollary 10.5. *A group G contains only one Sylow p-subgroup H if and only if H is normal.*

Proof. If H is normal, then $gHg^{-1} = H$ for all $g \in G$. By the Second Sylow Theorem, every Sylow p-subgroup is of the form gHg^{-1}, so H is the only Sylow p-subgroup.

The other direction follows from the fact that if H is a Sylow p-subgroup, then any conjugate gHg^{-1} is also a Sylow p-subgroup. So if H is the only Sylow p-subgroup, then $gHg^{-1} = H$ for all g. □

10.4 Third Sylow Theorem [7.7]

Theorem 10.6. *Let G be a group of order $p^e m$, with m not divisible by p. Let n_p be the number of Sylow p-subgroups of G. Then $n_p | m$ and $n_p \equiv 1 \pmod{p}$.*

10.5 Simple groups

We first see an application of the Sylow theorems to showing that groups of certain orders cannot be simple.

Recall that a simple group is a group which has no proper normal subgroup. In a sense which we will not make precise, every finite group can be "broken down" into simple groups (although a group is **not** necessarily a product of simple groups).

There is a complete classification of finite simple groups.

Example 10.7. Let p and q be distinct primes. If G is a group of order pq, then G is not simple.

Proof. Assume to the contrary that G is simple. By Corollary 10.5, this means that G has more than one Sylow p-subgroup. Since $n_p|q$, we must have $n_p = q$. Since $n_p \equiv 1 \pmod{p}$, we must have

$$q \equiv 1 \pmod{p}.$$

In particular, this implies $q > p$. The same argument shows that we must have $p \equiv 1 \pmod{q}$, so $p > q$. This is clearly impossible. □

The following lemma lets us do more.

Lemma 10.8. *Let p and q be distinct primes, and let G be a group whose order is divisible by both p and q.*

(a) The intersection of a Sylow p-subgroup and a Sylow q-subgroup is $\{1\}$.

(b) The intersection of two different subgroups of order p is $\{1\}$.

Proof.

(a) Let H be a subgroup of order p^e and let K be a subgroup of order q^f. By Lagrange's theorem, $|H \cap K|$ divides both $|H| = p^e$ and $|K| = q^f$, which are coprime, so $|H \cap K| = 1$.

(b) Let H_1 and H_2 be two distinct subgroups of order p. Then $|H_1 \cap H_2|$ divides p, and it is not equal to p because $H_1 \neq H_2$, so $|H_1 \cap H_2| = 1$.

□

10.6 Classification of groups [2.11, 7.7]

In the examples below, we classify groups of some small orders. We will use the following facts about product groups in these examples.

Proposition 10.9 (Artin Proposition 2.11.4)**.** *Let H and K be subgroups of a group G. Consider the function $f\colon H \times K \to G$ given by.*

$$f(h, k) = hk.$$

Its image is $HK \coloneqq \{hk : h \in H,\ k \in K\}$.

(a) f is injective if and only if $H \cap K = \{1\}$.

(b) f is a homomorphism if and only if $hk = kh$ for all $h \in H$ and $k \in K$.

(c) If H is normal, then HK is a subgroup of G.

(d) f is an isomorphism from the product group $H \times K$ to G if and only if $H \cap K = \{1\}$, $HK = G$, and H and K are normal.

Example 10.10. Every group of order 15 is isomorphic to $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$. (This product is isomorphic to $\mathbb{Z}/15\mathbb{Z}$.)

Proof.

- The number of Sylow 3-subgroups divides 5 and is congruent to $1 \pmod 3$. Thus $n_3 = 1$. By Corollary 10.5, the unique Sylow 3-subgroup H is normal.
- The number of Sylow 5-subgroups divides 3 and is congruent to $1 \pmod 5$. Thus $n_5 = 1$. By Corollary 10.5, the unique Sylow 5-subgroup K is normal.
- By Lemma 10.8, $H \cap K = \{1\}$, so the multiplication map $f\colon H \times K \to G$ is injective. Furthermore, $|H \times K| = 15 = |G|$, so it is surjective, i.e., $HK = G$. Finally, since H and K are normal, the proposition tells us that
$$G \cong H \times K \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}.$$ □

Example 10.11. There are two isomorphism classes of groups of order 21: the class of $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$, and the class of the group with presentation

$$\langle x, y | x^7 = 1, y^3 = 1, yx = x^2 y \rangle.$$

Proof. We first analyze the number of Sylow p-subgroups again.

- We have $n_7 \mid 3$ and $n_7 \equiv 1 \pmod 7$, so there is a unique Sylow 7-subgroup, call it K, and it is normal.
- On the other hand, the Third Sylow Theorem tells us that $n_3 \mid 7$ and $n_3 \equiv 1 \pmod 3$, so $n_3 = 1$ or 7. Let H be a Sylow 3-subgroup. It might not be normal.

Let x be a generator of K, and let y be a generator of H. By the same reasoning as in the previous example, $KH = G$. The elements of G are

$$\begin{array}{ccccccc} 1, & x, & x^2, & x^3, & x^4, & x^5, & x^6, \\ y, & xy, & x^2y, & x^3y, & x^4y, & x^5y, & x^6y, \\ y^2, & xy^2, & x^2y^2, & x^3y^2, & x^4y^2, & x^5y^2, & x^6y^2. \end{array}$$

We have $x^7 = y^3 = 1$. Furthermore, since K is normal, we have $yxy^{-1} \in K$, so $yxy^{-1} = x^i$ for some $1 \leq i \leq 6$. The relations

$$x^7 = 1, \quad y^3 = 1, \quad yx = x^i y$$

would determine the multiplication table for the 21 elements of G listed above, because they let us move occurrences of y to the right.

Since $y^3 = 1$, we must have

$$x = y^3 x y^{-3} = y^2 x^i y^{-2} = y x^{i^2} y^{-1} = x^{i^3},$$

where we have used the fact that $yxy^{-1} = x^i$ implies $yx^k y^{-1} = x^{ki}$ for any k. Since x has order 7, this means i satisfies $i^3 \equiv 1 \pmod 7$, so $i = 1, 2, 4$.

- If $i = 1$, then $yx = xy$, and G is abelian. So both H and K are normal, and $G \cong H \times K \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$.
- If $i = 4$, then $yxy^{-1} = x^4$. A straightforward calculation tells us that

  $$y^2 x y^{-2} = y x^4 y^{-1} = x^{16} = x^2.$$

  Since we can choose y to be any generator of H, replacing y by y^2 reduces us to the case $i = 2$, where $yxy^{-1} = x^2$.
- If $i = 2$, then G has the generators and relations $x^7 = y^3 = 1$, $yx = x^2 y$. However, we need to check that a group of order 21 with these generators and relations actually exists. This can be done by explicitly exhibiting one.

  Here is the group: the group generated by the matrices

  $$x = \begin{bmatrix} 1 & 1 \\ & 1 \end{bmatrix}, \quad y = \begin{bmatrix} 2 & \\ & 1 \end{bmatrix},$$

  except we treat the entries as elements of $\mathbb{Z}/7\mathbb{Z}$ when we do matrix multiplication! So they do indeed satisfy $x^7 = y^3 = 1$ and $yx = x^2 y$.

  People always refer to $\mathbb{Z}/7\mathbb{Z}$ as $\mathbb{F}_7$ in this context, because it is a field. We will get to this later. $\square$

Remark 10.12. This last part is important because sometimes generators and relations cause the group to collapse. For example, we showed that in

$$\langle x, y | x^7 = y^3 = 1, yx = x^3 y \rangle,$$

the equation $x^{3^3} = x$ holds, so $x^{26} = 1$. But combined with $x^7 = 1$, this implies $x = 1$. The group with the above presentation has only 3 elements. It is the same as $\langle y | y^3 = 1 \rangle$, the cyclic group of order 3.

The same kind of argument can be used to classify groups of order 6.

Example 10.13. A group G of order 6 is isomorphic to either $\mathbb{Z}/6\mathbb{Z}$ or S_3.

Proof. By similar reasoning as before, G has exactly one Sylow 3-subgroup K, which is normal. Let x be a generator of K.

There is also at least one a Sylow 2-subgroup H, although it need not be normal. Let y be a generator of K. Then

$$x^3 = 1, \quad y^2 = 1.$$

Since K is normal, $yxy^{-1} = x^i$ for $i = 1$ or 2, so $yx = xy$ or $yx = x^2y$.

The case $i = 1$ corresponds to the abelian group $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, and the case $i = 2$ corresponds to S_3, which we can see using the standard presentation of S_3. $\square$

11 Feb 18: Proof of Sylow theorems

Answers in Section 11.6.

11.1 Group action on subsets [6.10]

Let G be a group acting on a set S, and let U be a subset of S of order r. Then

$$gU \coloneqq \{gu : u \in U\}$$

is also a subset of S of order r. This defines an action of G on the set of subsets of S of order r.

Example 11.1.

1. The group S_3 acts on $\{1, 2, 3\}$. It also acts on the set of 2-element subsets $\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$. For example,

$$(1\ 2\ 3) * \{1, 2\} = \{2, 3\}, \quad (2\ 3) * \{1, 3\} = \{1, 2\}.$$

2. The dihedral group D_4 acts on the 4 vertices of a square, so it also acts on the set of pairs of vertices. There are $\binom{4}{2} = 6$ pairs.

   The 4 pairs of adjacent vertices form an orbit, and the 2 pairs of diagonally opposite vertices form an orbit.

We went over the exam problem 2(e) on group actions (exam solutions are in Canvas).

11.2 Proof of First Sylow Theorem [7.7]

Theorem 11.2. *Every finite group G whose order is divisible by p contains a Sylow p-subgroup.*

Lemma 11.3. *Let p be a prime. Let $n = p^e m$ where $e > 0$ and m is not divisible by p. The number of subsets of order p^e in a set of order n is not divisible by p.*

Proof. Omitted. □

Problem 1.

(a) Let U be a subset of G, and consider the action of G on the subsets of G. Prove that the order of $\mathrm{Stab}([U])$ divides both $|G|$ and $|U|$.

Hint 1. Recall the proof of Lagrange's theorem.

Hint 2. Let $H = \mathrm{Stab}([U])$. Show that U can be partitioned into right cosets Hu.

(b) Prove the First Sylow Theorem.

Hint 1. Recall the formula $|S| = |O_1| + \cdots + |O_k|$ where $O_1, \ldots, O_k$ are the orbits of an action of G on S.

Hint 2. Let S be the set of subsets of G of order p^e, on which G acts by left multiplication. Use the orbit-stabilizer theorem.

11.3 Proof of Second Sylow Theorem [7.7]

Theorem 11.4. *Let G be a group whose order is divisible by p.*

(a) Any two Sylow p-subgroups of G are conjugate subgroups.

(b) Every subgroup of G which is a p-group is contained in a Sylow p-subgroup.

Let H be a subgroup of G. Recall that there is an action of G on G/H, the left cosets of H, given by $g[aH] = [gaH]$. This action is transitive, and the stabilizer of $[H]$ is H.

Problem 2.

(a) With notation as above, what is the stabilizer of $[aH]$ under the action of G on left cosets?

(b) Suppose $|G| = p^e m$, and let H be a Sylow p-subgroup of G. What is the order of G/H?

(c) Prove the Fixed Point Theorem: let K be a p-group acting on a set S such that $|S|$ is not divisible by p. Then there exists an element $s \in S$ whose stabilizer is the whole group K.

Hint. Use the formula $|S| = |O_1| + \cdots + |O_k|$ again.

(d) Let H be a Sylow p-subgroup of G. Let K be a subgroup of G which is a p-group. Show that K is contained in some conjugate gHg^{-1} of H.

Hint. Consider the action of K on G/H. (Since G acts on G/H we can just restrict this action to K.)

(e) Conclude that both parts of the Second Sylow Theorem are true.

11.4 Normalizers [7.6]

Definition 11.5. Let H be a subgroup of G. The *normalizer* of H in G, denoted $N(H)$ is the set of all $g \in G$ such that

$$H = gHg^{-1} := \{ghg^{-1} : h \in H\}.$$

Proposition 11.6. *The normalizer $N(H)$ is a subgroup of G, and H is a normal subgroup of $N(H)$.*

11.5 Proof of Third Sylow Theorem [7.7]

Theorem 11.7. *Let G be a group of order $p^e m$, with m not divisible by p. Let n_p be the number of Sylow p-subgroups of G. Then $n_p | m$ and $n_p \equiv 1 \pmod{p}$.*

Problem 3.

(a) Let S be the set of all Sylow p-subgroups of G. By considering the conjugation action of G on S given by

$$(g, H) \mapsto gHg^{-1},$$

show that $n_p | m$.

Hint. Let H be a Sylow p-subgroup. What are the orbit and stabilizer of $[H]$ under this action?

(b) Now let H be a Sylow p-subgroup, and let S be as above. Consider the conjugation action of H on S. Let H' be a Sylow p-subgroup different from G. Show that the order of the orbit $O_{[H']}$ is a multiple of p.

Hint 1. Show that this is equivalent to showing that H is not contained in $N(H')$.

Hint 2. Use the Second Sylow Theorem on the group $N(H)$ instead of G.

(c) Conclude that $n_p \equiv 1 \pmod{p}$.

11.6 Answers

Problem 1.

(a) Let $H = \operatorname{Stab}([U])$. By definition of stabilizer, we have $hu \in U$ for any $u \in U$ and $h \in H$, so the right coset $Hu \subseteq U$ for any $u \in H$. Thus, U is a disjoint union of right cosets of H. Since each right coset has $|H|$ elements, this means $|U|$ is a multiple of $|H|$.

(b) Let S be as in the hint. Since $|S|$ is not a multiple of p, some orbit of the action of G on S has order not divisible by p. Suppose $O_{[U]}$ is such an orbit for $[U] \in S$. By the orbit-stabilizer theorem,

$$|O_{[U]}| = \frac{p^e m}{|\operatorname{Stab}([U])|},$$

which means p^e divides $|\operatorname{Stab}([U])|$ since $|O_{[U]}|$ is not divisible by p. But by part (a), the orbit of the stabilizer also divides $|U| = p^e$, so $|\operatorname{Stab}([U])| = p^e$, so $\operatorname{Stab}([U])$ is a Sylow p-subgroup.

Problem 2.

(a) We showed previously that the stabilizers of elements in the same orbit are conjugates of each other. The stabilizer of $[aH]$ is aHa^{-1}.

(b) The number of cosets is $|G|/|H| = p^e m/p^e = m$.

(c) Using the formula in the hint, we conclude that there exists some $s \in S$ such that $|O_s|$ is not a multiple of p. By the orbit stabilizer theorem,

$$|O_s| = |K|/|\operatorname{Stab}(s)| = p^f/|\operatorname{Stab}(s)|$$

where $|K| = p^f$. Thus $|O_s|$ must be a power of p, but since it's not a multiple of p, we must have $|O_s| = 1$, and $\mathrm{Stab}(s) = K$, i.e., s is a fixed point of K.

(d) We consider the action in the hint. By parts (b) and (c), we know there is a coset, say $[gH]$, which is fixed by every element of K. So K is contained in the stabilizer of $[gH]$, which by part (a) is gHg^{-1}.

(e) With notation as in part (d), since gHg^{-1} is also a Sylow p-subgroup, every subgroup K which is a p-group is contained in a Sylow p-subgroup.

If K is also a Sylow p-subgroup, then K must be equal to gHg^{-1} since the two subgroups have the same order, so K is conjugate to H.

Problem 3.

(a) By the Second Sylow Theorem, this action is transitive, so the only orbit is S, and $|S| = n_p$. The stabilizer of H is $N(H)$. By the orbit-stabilizer theorem,

$$n_p = |G|/|N(H)| = p^e m/|N(H)|.$$

Since $N(H)$ contains H, $|N(H)|$ is a multiple of $|H| = p^e$. Thus, by the above displayed equation n_p divides m.

(b) The stabilizer of H' in H is $H \cap N(H')$. By the orbit-stabilizer theorem,

$$|O_{[H']}| = p^e m/|H \cap N(H')|.$$

Now $|H \cap N(H')|$ divides $|H| = p^e$, so in order to show that $|O_{[H']}|$ is divisible by p, it suffices to show that $|H \cap N(H')|$ is not equal to p^e, which is equivalent to showing that H is not contained in $N(H')$.

Assume for the sake of contradiction that H is contained in $N(H')$. Then H is a Sylow p-subgroup of $N(H')$. But H' is also a Sylow p-subgroup of $N(H')$. Furthermore, H' is normal in $N(H')$ by Proposition 11.6, so by the Second Sylow Theorem, H' should be the only Sylow p-subgroup in $N(H')$. This contradicts the assumption that $H \neq H'$.

(c) By part (b), every orbit of the action of G on S given in part (a) has order divisible by p, except for the orbit $O_{[H]} = \{H\}$, which has order 1. Thus, the total number of elements of S is congruent to 1 $\pmod{p}$.

12 Feb 23: Rings and homomorphisms

12.1 Gaussian integers [11.1]

The *Gaussian integers* are the complex numbers of the form $a + bi$ for $a, b \in \mathbb{Z}$. We denote the set of Gaussian integers by

$$\mathbb{Z}[i] \coloneqq \{a + bi : a, b \in \mathbb{Z}\}.$$

Note that this set is an additive subgroup of $\mathbb{C}$. Furthermore, it contains the multiplicative identity 1, and is closed under multiplication.

12.2 Definition of ring [11.1]

Definition 12.1. A *ring* is a set R equipped with two laws of composition, $+$ and $\times$, such that

1. $(R, +)$ is an abelian group; we denote its identity by 0.
2. $\times$ is commutative and associative, and has a multiplicative identity which we denote by 1.
3. The distributive law holds: for all $a, b, c \in R$, we have $a(b + c) = ab + ac$.

Note: There is also the notion of a *noncommutative ring*, in which $\times$ is not required to be commutative. In this class, we use the term "ring" to mean a commutative ring.

Definition 12.2. A *subring* of R is a subset which is an additive subgroup of R, is closed under multiplication, and contains 1.

Example 12.3.

1. $\mathbb{Z}$ and $\mathbb{Z}[i]$ are rings. $\mathbb{Z}$ is a subring of $\mathbb{Z}[i]$.
2. Under the usual addition and multiplication operations, $\mathbb{Z}/n\mathbb{Z}$ is a ring.
3. Like $\mathbb{Z}[i]$, the set

   $$\mathbb{Z}[\sqrt[3]{2}] = \{a + b\sqrt[3]{2} + c\sqrt[3]{4} : a, b, c \in \mathbb{Z}\}$$

   is a subring of $\mathbb{C}$. This is is the set of all integer linear combinations of powers of $\sqrt[3]{2}$. Note that we do not need to bother with adding $d\sqrt[3]{8}$ since $\sqrt[3]{8} = 2$ which is already in $\mathbb{Z}$.
4. The ring which contains only one element, 0, is called the *zero ring*.

12.3 Fields [3.2]

Definition 12.4. An element $u \in R$ is called a *unit* if it has a multiplicative inverse, i.e., if there exists $x \in R$ such that $ux = 1$.

Example 12.5.

1. The units of $\mathbb{Z}$ are 1 and -1.
2. The units of $\mathbb{Z}/n\mathbb{Z}$ are the elements of the group $(\mathbb{Z}/n\mathbb{Z})^\times$, i.e., the residues $\overline{a}$ where $\gcd(a, n) = 1$.
3. The units of $\mathbb{Z}[i]$ are 1, -1, i, $-i$.

Definition 12.6. A *field* is a ring F in which every *nonzero* element of F is a unit.

Example 12.7.

1. The real numbers $\mathbb{R}$, the complex numbers $\mathbb{C}$, and the rational numbers $\mathbb{Q}$ are fields.
2. Let p be a prime. In Lecture 1, we showed that every nonzero element of $\mathbb{Z}/p\mathbb{Z}$ has a multiplicative inverse. Thus, $\mathbb{Z}/p\mathbb{Z}$ is a field, which we denote by $\mathbb{F}_p$.

12.4 Polynomials [11.2]

Definition 12.8. Let R be a ring. The *polynomial ring* $R[x]$ consists of all polynomials with coefficients in R:

$$R[x] = \{a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0 \mid a_0, a_1, \ldots, a_n \in R\}.$$

It is a ring under the usual addition and multiplication operations on polynomials.

Example 12.9. If $R = \mathbb{Z}[x]$, then $R[y]$ is the ring of all two-variable polynomials in x and y. It consists of expressions which are finite sums of the form

$$f(x, y) = \sum a_{ij} x^i y^j, \quad a_{ij} \in \mathbb{Z}.$$

We denote this ring by $\mathbb{Z}[x, y]$.

We can similarly define $R[x_1, \ldots, x_n]$ for any ring R and any number of variables n.

12.5 Division with remainder [11.2]

Definition 12.10. Let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ be a polynomial. We denote by $\deg f$ the *degree* of f; it is the largest integer n such that the coefficient $a_n \neq 0$. A polynomial is called *monic* if the leading coefficient $a_n = 1$.

Proposition 12.11 (Division with remainder)**.** *Let R be a ring, and let $f(x)$ be a* monic *polynomial with coefficients in R. Let $g(x)$ be any polynomial with coefficients in R. Then there exist uniquely determined $q, r \in R[x]$ such that*

$$g(x) = f(x)q(x) + r(x),$$

and if $r \neq 0$, then $\deg r < \deg f$. Moreover, $r = 0$ if and only if f divides g.

In fact, division with remainder works as long as the leading coefficient of $f(x)$ is a unit in R, which is the case for all nonzero polynomials when the coefficient ring is a field.

Corollary 12.12. *Let $a \in R$. A polynomial $f \in R[x]$ is divisible by $(x - a)$ if and only if $f(a) = 0$.*

Proof. By division with remainder, we can write

$$f(x) = (x - a)q(x) + r$$

for some constant polynomial $r \in R$. Plugging in $x = a$ gives $f(a) = r$, so $r = 0$ if and only if $f(a) = 0$. □

12.6 Homomorphisms [11.3]

Definition 12.13. Let R and R' be rings. A *homomorphism* is a map $\varphi\colon R \to R'$ which satisfies

1. $\varphi(a + b) = \varphi(a) + \varphi(b)$ for all $a, b \in R$.
2. $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in R$.
3. $\varphi(1_R) = 1_{R'}$.

Definition 12.14. An *isomorphism* of rings is a bijective homomorphism.

Example 12.15. The reduction modulo n map $\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ which sends a to $\overline{a}$ is a ring homomorphism.

Example 12.16. The *evaluation map*

$$\mathbb{R}[x] \to \mathbb{R}, \quad f(x) \mapsto f(3)$$

is a ring homomorphism. Similarly, the map

$$\mathbb{R}[x] \to \mathbb{C}, \quad f(x) \mapsto f(i)$$

is a ring homomorphism.

The following proposition is a generalization Example 12.16

Proposition 12.17 (Substitution principle)**.** *Let $\varphi\colon R \to R'$ be a ring homomorphism. Let $\alpha \in R'$ be any element. There exists a unique homomorphism $\Phi\colon R[x] \to R'$ such that Φ agrees with φ on constant polynomials, and $\Phi(x) = \alpha$.*

Proof. Suppose Φ is a homomorphism which satisfies $\Phi(x) = \alpha$ and agrees with φ on R. Then for any polynomial $f(x) = a_n x^n + \cdots + a_0$ in $R(x)$, then conditions 1. and 2. of the definition of ring homomorphism force

$$\Phi(f) = \varphi(a_n)\alpha^n + \cdots + \varphi(a_0). \tag{$\heartsuit$}$$

In other words, the restriction of Φ to R and the value of $\Phi(x)$ completely determine the rest of the homomorphism for the polynomial ring $R[x]$

It is straightforward that for any $\alpha \in R'$, the formula given by ($\heartsuit$) is indeed a homomorphism. □

The substitution principle also works for multivariate polynomial rings.

Proposition 12.18. *Let $\varphi\colon R \to R'$ be a ring homomorphism. Let $\alpha_1, \ldots, \alpha_n \in R'$. There exists a unique homomorphism $\Phi\colon R[x_1, \ldots, x_n] \to R'$ such that Φ agrees with φ on R, and $\Phi(x_i) = \alpha_i$ for $i = 1, \ldots, n$.*

Proposition 12.19. *Let R be any ring. There is exactly one homomorphism $\mathbb{Z} \to R$.*

Proof. Let $\varphi\colon \mathbb{Z} \to R$ be a homomorphism. By definition of ring homomorphism, we must have $\varphi(1) = 1_R$. Also by definition of ring homomorphism, φ is a group homomorphism from the additive group $\mathbb{Z}$ to R under addition.

This means that for any integer $n \geq 1$, we have

$$\varphi(n) = \varphi(1 + \cdots + 1) = 1_R + \cdots + 1_R,$$

where there are n copies of the identity in both sums.

Since group homomorphisms send inverses to inverses, we also have

$$\varphi(-n) = -(1_R + \cdots + 1_R),$$

where again there are n copies of 1_R in the sum on the right.

In addition, $\varphi(0) = 0_R$ since group homomorphisms send the identity to the identity.

Thus, there is only one possible value of $\varphi(n)$ for every element $n \in \mathbb{Z}$. It is clear that the map defined by $\varphi(n) = 1_R + \cdots + 1_R$ (n terms) and and $\varphi(-n) = -\varphi(n)$ for $n \geq 0$ is indeed a homomorphism. □

Definition 12.20. Let R be a ring. The *characteristic* of R is the smallest positive integer n such that the sum $1 + \cdots + 1$ of n copies of 1 is equal to 0, if such an n exists.

If no such n exists, then we say that R has *characteristic zero*.

We usually denote by n the sum of n copies of 1 in R. For example, 3 denotes the element $1 + 1 + 1$ in any ring R.

13 Feb 25: Ideals and quotients

13.1 Ideals [11.3]

Definition 13.1. The *kernel* of a ring homomorphism $\varphi\colon R \to R'$ is the set

$$\ker(\varphi) \coloneqq \{a \in R : \varphi(a) = 0_{R'}\}.$$

Recall that for groups, kernel of homomorphism are normal subgroups, and we can quotient by normal subgroups. The analogue of a normal subgroup for rings is an *ideal*. Rather than quotienting by subrings, we quotient by ideals.

Definition 13.2. An ideal of a ring R is a nonempty subset I of R such that

1. I is closed under addition, i.e., if $x, y \in I$, then $x + y$ is in I as well.
2. for any $x \in I$ and any $r \in R$, the element rx is also in I.

Note that the second condition is stronger than saying that I is closed under multiplication. Also note that ideals are additive subgroups of R.

Proposition 13.3. *The kernel of a ring homomorphism $\varphi\colon R \to R'$ is an ideal.*

Proof. If $x, y \in \ker(\varphi)$, then

$$\varphi(x + y) = \varphi(x) + \varphi(y) = 0_{R'},$$

so φ is closed under addition. If $x \in \ker(\varphi)$ and $r \in R$, then

$$\varphi(rx) = \varphi(r)\varphi(x) = \varphi(r) \cdot 0_{R'} = 0_{R'}.$$

Thus, $\ker(\varphi)$ satisfies the second condition for being an ideal as well. □

13.2 Principal ideals [11.3]

Definition 13.4. For any $a \in R$, the *principal ideal* generated by a is the set

$$(a) \coloneqq \{ar : r \in R\}.$$

In other words, it is the set of all multiples of a in R. It is easy to check that (a) is an ideal.

The *zero ideal* is the ideal (0) containing only the element 0, and the *unit ideal* is the ideal (1), which is the whole ring R. A ideal is *proper* if it is not equal to (0) or (1).

Proposition 13.5. *Every ideal of $\mathbb{Z}$ is equal to (n) for some integer n.*

Proof. Let I be an ideal of $\mathbb{Z}$. If $I = \{0\}$, then $I = (0)$. Otherwise, I contains at least one positive integer.

Let n be the smallest positive integer n such that $n \in I$. Then $(n) \subseteq I$ by condition 2. for being in ideal. On the other hand, suppose $m \in I$. By division with remainder, we can write

$$m = nq + r$$

for some $0 \leq r < n$. Since $m, n \in I$ and I is an ideal, we must have $r \in I$ as well. Since n is the smallest positive integer in I, we must have $r = 0$, so $m \in (n)$. Thus, $(n) \subseteq I$ and $I \subseteq (n)$, so $I = (n)$. □

Proposition 13.6. *Let F be a field. Every ideal of $F[x]$ is equal to (f) for some monic polynomial f.*

Proof. Let I be an ideal of $F[x]$. If $I \neq (0)$, then it contains at least one monic polynomial. Let f be a monic polynomial in I of minimal degree. Then $(f) \subseteq I$.

On the other hand, suppose $g \in I$. By division with remainder, we can write

$$g = fq + r$$

for $r \in F[x]$ with $r = 0$ or $\deg r < \deg f$. However, since $f, g \in I$, we also have $r \in I$. By minimality of the degree of f, we must have $r = 0$, so $g \in (f)$. As in the proof of the previous proposition, we conclude that $I = (f)$. □

13.3 Generating ideals [11.3]

In general, not all ideals are principal.

Definition 13.7. Let R be a ring. The *ideal generated by* elements $a_1, \ldots, a_n \in R$ is the set

$$(a_1, \ldots, a_n) \coloneqq \{r_1 a_1 + \cdots + r_n a_n : r_i \in R\}.$$

It is the smallest ideal which contains $a_1, \ldots, a_n$.

Example 13.8.

1. The ideal $(5, x)$ of $\mathbb{Z}[x]$ consists of all integer polynomials f such that the constant term $f(0)$ is a multiple of 5. It is not a principal ideal.

2. The ideal (x, y) of $\mathbb{C}[x, y]$ is not principal. It consists of all polynomials $f \in \mathbb{C}[x, y]$ such that the constant term $f(0, 0)$ is 0.

13.4 Definition of quotient ring [11.4]

Let R be a ring, and let I be an ideal of R. In this subsection, we define the *quotient ring* R/I.

Note that since I is an ideal, it is an additive subgroup of R under addition. Furthermore, since addition is commutative, I is a normal subgroup of the additive group R.

Definition 13.9. The elements of R/I are the cosets $a+I$ in the additive group R. Addition in the ring R/I is the same as addition in the quotient group R/I. Multiplication is defined by

$$(a+I)(b+I) = ab + I.$$

As was the case when we defined modular arithmetic, we need to check that multiplication is well-defined. The proof is similar.

Lemma 13.10. *If $a+I = a'+I$ and $b+I = b'+I$, then $ab+I = a'b'+I$.*

Proof. Suppose $a' = a+u$ and $b' = b+v$ for $u, v \in I$, then

$$a'b' = (a+u)(b+v) = ab + (av + bu + uv).$$

Since I is an ideal, we also have $av + bu + uv \in I$. Thus, $a'b' + I = ab + I$. □

We also need to check that multiplication is associative, has a multiplicative identity, and satisfies the distributive property. These follow easily from the ring axioms for R.

The multiplicative identity of R/I is the coset $1+I$, and the additive identity is $0+I$. We denote by $\overline{a}$ the element $a+I$ of R/I.

13.5 Mapping property of quotients [11.4]

There is a canonical *quotient map*

$$\pi\colon R \to R/I, \quad \pi(a) = a + I,$$

It is easy to check that this is a surjective ring homomorphism, and that the kernel of π is I.

Proposition 13.11 (Universal property of quotients)**.** *For any ring R', and any ring homomorphism $f\colon R \to R'$ such that $I \subseteq \ker(f)$, there exists a unique homomorphism $\overline{f}\colon R/I \to R'$ such that $f = \overline{f} \circ \pi$.*

Proof. The homomorphism $\overline{f}$ is given by $\overline{f}(a+I) = f(a)$. The fact that it is well-defined follows from the fact that $I \subseteq \ker(f)$. It is easy to check that it is a homomorphism and satisfies $f = \overline{f} \circ \pi$. □

Example 13.12. Since there is a unique homomorphism $\mathbb{Z} \to \mathbb{C}$, giving a homomorphism

$$\overline{\varphi}\colon \mathbb{Z}[x]/(x^3-2) \to \mathbb{C}$$

is the same as giving a homomorphism $\varphi\colon \mathbb{Z}[x] \to \mathbb{C}$ whose kernel contains the ideal (x^3-2). This is the same as specifying an element $\varphi(x) = \alpha \in \mathbb{C}$ such that $\alpha^3 = 2$. There are three cube roots of 2 in $\mathbb{C}$, so there are three homomorphisms from $\mathbb{Z}[x]/(x^3-2)$ to $\mathbb{C}$.

Proposition 13.13 (First isomorphism theorem)**.** *If $f\colon R \to R'$ is a surjective ring homomorphism such that* $\ker(f) = I$*. Then the induced map $\overline{f}\colon R/I \to R'$ is an isomorphism.*

Proof. By the definition of $\overline{f}$ in the proof above, since f is surjective, $\overline{f}$ is also surjective. It is injective because $\overline{f}(a + I) = 0$ is equivalent to $f(a) = 0$, which is true if and only if $a \in I$. $\square$

14 Mar 2: Correspondence theorem and adjoining elements

14.1 Correspondence theorem [11.4]

Let R be a ring, and let I be an ideal of R. The following theorem describes ideals of the quotient ring R/I.

Theorem 14.1 (Correspondence theorem)**.** *Let I be an ideal of R, and let $\pi\colon R \to R/I$ be the quotient homomorphism. There is a bijection between the sets*

$$\{\text{ideals of } R \text{ which contain } I\} \quad \textit{and} \quad \{\text{ideals of } R/I\},$$

given by $J \mapsto \pi(J)$, with inverse $\overline{J} \mapsto \pi^{-1}(\overline{J})$.

Proof.

1. First, we check that for any ideal J of R, the image

$$\pi(J) = \{\overline{a} : a \in J\}$$

is an ideal of R/I. It is an additive subgroup of R/I since I is an additive subgroup of R, so we just need to check that $\overline{a} \cdot \overline{r} \in \varphi(I)$ for any $a \in I$ and $r \in R$. This is true because $\overline{a} \cdot \overline{r} = \overline{ar}$, and $ar \in I$ because I is an ideal.

2. Now, we check that for any ideal $\overline{J}$ of R/I, the preimage

$$\pi^{-1}(\overline{J}) = \{a \in R : \overline{a} \in \overline{J}\}$$

is an ideal of R which contains I. Indeed, it contains I because $\overline{a} = \overline{0}$ for any $a \in I$, and every ideal of R/I contains $\overline{0}$.

To check that $\pi^{-1}(\overline{J})$ is an ideal of R, we note that it is an additive subgroup of R because $\overline{J}$ is an additive subgroup of R/I. In addition, for any $a \in \pi^{-1}(\overline{J})$ and $r \in R$, we have $\overline{ar} = \overline{a} \cdot \overline{r} \in \overline{J}$, so $ar \in \pi^{-1}(\overline{J})$ as well.

3. Finally, we check that the two maps defined are inverses.

   - Let J be an ideal of R which contains I. It is clear that $J \subseteq \pi^{-1}(\pi(J))$. On the other hand, if $r \in \pi^{-1}(\pi(J))$, this means that $\overline{r} = \overline{a}$ for some $a \in J$, i.e., $r - a \in I$ for some $a \in J$. Since $I \subseteq J$, we have $r \in J$ as well.
   - Let $\overline{J}$ be an ideal of R/I. It is clear that $\pi(\pi^{-1}(\overline{J})) \subseteq \overline{J}$. On the other hand, since the quotient map π is surjective, any element of $\overline{J}$ is equal to $\overline{a}$ for some $a \in R$, and in fact such a is by definition an element of $\pi^{-1}(\overline{J})$, so $\overline{a} \in \pi(\pi^{-1}(\overline{J}))$. $\square$

Example 14.2. The ideals of the ring $\mathbb{C}[x]/(x^2 - 1)$ correspond to ideals I of $\mathbb{C}[x]$ such that $(x^2 - 1) \subseteq I$. Since ideal of $\mathbb{C}[x]$ is principal, $I = (f)$ for some monic polynomial $f \in \mathbb{C}[x]$. Now, $(x^2 - 1) \subseteq (f)$ if and only if f divides $x^2 - 1 = (x - 1)(x + 1)$, so $f = 1$, $x - 1$, $x + 1$, or $x^2 - 1$. Thus, the quotient ring $\mathbb{C}[x]/(x^2 - 1)$ has 4 ideals.

14.2 Adjoining elements [11.4, 11.5]

Previously, we have defined $\mathbb{Z}[i]$ as an explicit subset of $\mathbb{C}$ consisting of the complex numbers $a + bi$ with $a, b \in \mathbb{Z}$. Note that in this ring, the element i satisfies $i^2 = -1$.

However, we can also consider an abstract ring $\mathbb{Z}[\alpha]$, whose elements are polynomials in α, similar to the ring $\mathbb{Z}[x]$, except the element α satisfies $\alpha^2 = -1$. Thus, every element of $\mathbb{Z}[\alpha]$ can be expressed in the form

$$a + b\alpha$$

for some $a, b \in \mathbb{Z}$. For example, using division with remainder, we have

$$\alpha^3 + 2\alpha^2 + 4\alpha + 1 = (\alpha^2 + 1)(\alpha + 2) + 3\alpha - 1 = 3\alpha - 1,$$

since $\alpha^2 + 1 = 0$. It is easy to check that this ring $\mathbb{Z}[\alpha]$ is isomorphic to $\mathbb{Z}[i]$ via the map $\alpha \mapsto i$.

The quotient ring construction is a way of formally defining "the ring $\mathbb{Z}[\alpha]$ with $\alpha^2 = -1$." Specifically, we define it as the ring $\mathbb{Z}[x]/(f)$, where $f(x) = x^2+1$. In this example, we have $\mathbb{Z}[x]/(x^2+1) \simeq \mathbb{Z}[i]$ via the isomorphism $x \mapsto i$.

In general, we can consider the ring $R[x]/(f)$ for any ring R and any polynomial $f \in R[x]$. We are most interested in the case when f is monic.

Proposition 14.3. *Let R be a ring and let f be a degree n monic polynomial in $R[x]$. Then every element of $R[x]/(f)$ can be written uniquely as a linear combination*

$$a_{n-1}\alpha^{n-1} + \cdots + a_0, \quad a_0, \ldots, a_{n-1} \in R,$$

where α denotes the image of x in $R[x]/(f)$.

Example 14.4. We show that the ring

$$\mathbb{Q}(\sqrt[3]{2}) := \{a + b\sqrt[3]{2} + c\sqrt[3]{4} : a, b, c \in \mathbb{Q}\}$$

is isomorphic to $\mathbb{Q}[x]/(x^3 - 2)$. Indeed, the map

$$\varphi\colon \mathbb{Q}[x] \mapsto \mathbb{Q}(\sqrt[3]{2}), \quad x \mapsto \sqrt[3]{2}$$

is surjective, and its kernel contains $(x^3 - 2)$. Since $\sqrt[3]{2}$ is irrational, the polynomial $x^3 - 2$ has no monic factors other than 1 and itself. Since every ideal of $\mathbb{Q}[x]$ is of the form (f) for some monic $f \in \mathbb{Q}[x]$, this means that $\ker(\varphi) = (x^3 - 2)$. By the first isomorphism theorem, φ induces an isomorphism $\mathbb{Q}[x]/(x^3 - 2) \to \mathbb{Q}(\sqrt[3]{2})$.

Lemma 14.5. *Let f be a monic polynomial in $\mathbb{Z}[x]$, and Let g be any element of $\mathbb{Z}[x]$. Then f divides g in $\mathbb{Z}[x]$ if and only if f divides g in $\mathbb{Q}[x]$.*

Proof. Since f is monic, by division with remainder, we can write $g = fq + r$ for $q, r \in \mathbb{Z}[x]$ with $\deg r < \deg f$ or $r = 0$. But this equation also holds in $\mathbb{Q}[x]$, so if f divides g in $\mathbb{Q}[x]$, then $r = 0$. This means that f divides g in $\mathbb{Z}[x]$ as well. $\square$

Example 14.6. Let

$$\mathbb{Z}[\sqrt[3]{2}] \coloneqq \{a + b\sqrt[3]{2} + c\sqrt[3]{4} : a, b, c \in \mathbb{Z}\}.$$

We show that the kernel of the homomorphism

$$\varphi\colon \mathbb{Z}[x] \to \mathbb{Z}[\sqrt[3]{2}], \quad x \mapsto \sqrt[3]{2}$$

is also (x^3-2). This would imply that φ induces an isomorphism $\mathbb{Z}[x]/(x^3-2) \to \mathbb{Z}[\sqrt[3]{2}]$.

Suppose $f \in \ker(\varphi)$. By the previous example, $x^3 - 2$ must divide f in $\mathbb{Q}[x]$. But by the above lemma, this means that $x^3 - 2$ divides f in $\mathbb{Z}[x]$ as well, so $\ker(\varphi) = (x^3 - 2)$.

Example 14.7. Since $\mathbb{Z}[x]/(x^2+1)$ is isomorphic to $\mathbb{Z}[i]$ under an isomorphism which sends $x \mapsto i$, the quotient ring $\mathbb{Z}[i]/(i-2)$ is isomorphic to $\mathbb{Z}[x]/(x^2 + 1, x - 2)$.

However, we could also quotient out by $(x-2)$ first, to get that $\mathbb{Z}[x]/(x-2)$ is isomorphic to $\mathbb{Z}[x]$, under an isomorphism which sends $x \mapsto 2$, so the quotient $\mathbb{Z}[x]/(x-2, x^2+1)$ is isomorphic to $\mathbb{Z}/(2^2+1) = \mathbb{Z}/(5)$. Thus, $\mathbb{Z}[i]/(i-2) \simeq \mathbb{F}_5$.

Example 14.8. The ring $\mathbb{F}_5[x]/(x^2-3)$ is isomorphic to a ring $\mathbb{F}_5[\delta]$ with δ^2-3. The elements of this ring can be written in the form $a + b\delta$ for $a, b \in \delta$.

We claim that $\mathbb{F}_5[\delta]$ is actually a field. We need to show that every element $a + b\delta$ with a, b not both zero has an inverse. We have

$$(a + b\delta)(a - b\delta) = a^2 - b^2\delta^2 = a^2 - 3b^2.$$

Now, $a^2-3b^2 \neq 0 \in \mathbb{F}_5$ since 3 is not a square modulo 5, so a^2-3b^2 is invertible. Thus, $(a + b\delta)$ has inverse $(a^2 - 3b^2)^{-1}(a - b\delta)$.

The following is an example of $R[x]/(f)$ when f is not monic.

Example 14.9. The ring $\mathbb{Z}[x]/(2x-1)$ is isomorphic to

$$\mathbb{Z}\left[\tfrac{1}{2}\right] = \{a_n \left(\tfrac{1}{2}\right)^n + \cdots + a_0 : a_0, \ldots, a_n \in \mathbb{Z}\}.$$

Essentially, we are adjoining an inverse of 2, which is not a unit in $\mathbb{Z}$, but is a unit in $\mathbb{Z}\left[\frac{1}{2}\right]$.

15 Mar 4: Product rings, integral domains, maximal ideals

15.1 Product rings [11.6]

Definition 15.1. Let R and R' be rings. The *product ring* $R \times R'$ is the set of pairs

$$\{(x, x') : x \in R, x' \in R'\}$$

with addition and multiplication given by

$$(x, x') + (y, y') = (x + y, x' + y'), \quad (x, x') \cdot (y, y') = (xy, x'y').$$

The additive identity is $(0, 0)$, and the multiplicative identity is $(1, 1)$.

The product ring comes with two *projection homomorphisms* $\pi\colon R \times R' \to R$ and $\pi'\colon R \times R' \to R'$, given by $\pi(x, x') = x$ and $\pi'(x, x') = x'$.

An *idempotent* of a ring is an element e such that $e^2 = e$. It is easy to see that the elements $(1, 0)$ and $(0, 1)$ of a product ring $R \times R'$ are idempotents.

Example 15.2. The ring $\mathbb{C}[x]/(x^2 - 1)$ is isomorphic to the product ring $\mathbb{C} \times \mathbb{C}$ via the isomorphism

$$\varphi\colon \mathbb{C}[x]/(x^2 - 1) \to \mathbb{C} \times \mathbb{C}, \quad x \mapsto (1, -1).$$

Indeed, this map sends any $f \in \mathbb{C}[x]$ to $(f(1), f(-1))$, so $\ker(\varphi)$ consists of polynomials f such that $f(1) = f(-1) = 0$.

The condition $f(1) = 0$ implies that $f = (x - 1)g$ for some g, and then $f(-1) = 0$ implies $-2g(-1) = 0$, so $g = (x + 1)h$ for some h. Thus, $f = (x^2 - 1)h = 0$, so φ is injective.

Furthermore, φ is surjective because for any $a, b \in \mathbb{C}$, we have $\varphi(f) = (a, b)$ for $f(x) = b(1 - x)/2 + a(x + 1)/2$. Thus, φ is an isomorphism.

Example 15.3. The ring $\mathbb{F}_{11}(x)/(x^2 - 3)$ is isomorphic to the product ring $\mathbb{F}_{11} \times \mathbb{F}_{11}$. This is because $3 = 5^2$ in $\mathbb{F}_{11}$, so we have a map

$$\varphi\colon \mathbb{F}[x]/(x^2 - 3) \to \mathbb{F}_{11} \times \mathbb{F}_{11}, \quad x \mapsto (5, -5).$$

By the same argument as in the previous example, this map is an isomorphism.

15.2 Chinese Remainder Theorem

Definition 15.4. Let R be a ring, and let I and J be ideals of R.

1. The sum of I and J is defined to be the ideal $I + J = \{a + b : a \in I, b \in J\}$.
2. The product IJ is defined to be the ideal *generated* by the set $\{ab : a \in I, b \in J\}$. Note that the set of products is not itself an ideal.
3. The intersection $I \cap J$ is also an ideal. Note that $I \cap J$ contains IJ.

We will check that $I + J$ and $I \cap J$ are ideals on the problem set.

Definition 15.5. Ideals I and J of a ring R are said to be *coprime* if $I + J = R$.

Theorem 15.6. *Let R be a ring and let I and J be coprime ideals. Then $R/I \cap J$ is isomorphic to $R/I \times R/J$.*

Proof. Consider the ring homomorphism

$$\varphi\colon R \to R/I \times R/J, \quad x \mapsto (x + I, x + J)$$

It is clear that the kernel of φ is $I \cap J$.

We claim that φ is surjective. Since $I + J = R$, there exists $u \in I$ and $v \in J$ such that $u + v = 1$. Note that

$$\varphi(u) = (0, 1), \quad \varphi(v) = (1, 0).$$

Let $a, b \in R$. We want to show that there exists $x \in R$ such that $\varphi(x) = (\overline{a}, \overline{b})$. Take

$$x = av + bu.$$

Thus, by the first isomorphism theorem, $R/I \cap J$ is isomorphic to $R/I \times R/J$. $\square$

15.3 Integral domains [11.7]

Definition 15.7. A ring R is an *integral domain* if it is not the zero ring, and it has the property that if $ab = 0$ for any $a, b \in R$, then either $a = 0$ or $b = 0$.

Definition 15.8. A *zero divisor* of a ring R is a nonzero element $a \in R$ such that there exists nonzero $b \in R$ satisfying $ab = 0$.

An integral domain is a ring which contains no zero divisors. Integral domains satisfy the *cancellation law.* If R is an integral domain and $a, b, c \in R$, then

$$ab = ac \text{ and } a \neq 0 \implies b = c.$$

This is because $ab = ac$ implies $a(b - c) = 0$, so either $a = 0$ or $b - c = 0$.

Example 15.9.

1. $\mathbb{Z}$ is an integral domain.
2. $\mathbb{Z}/15\mathbb{Z}$ is not an integral domain, since $3 \cdot 5 = 0$ in $\mathbb{Z}/15\mathbb{Z}$, but $3 \neq 0$ and $5 \neq 0$ in $\mathbb{Z}/15\mathbb{Z}$.
3. Fields are integral domains, so $\mathbb{Z}/7\mathbb{Z}$ is an integral domain.

Definition 15.10. Let R be an integral domain. The *fraction field* of R, denoted $\mathrm{Frac}(R)$ is a field whose elements are equivalence classes of "fractions"

$$\mathrm{Frac}(R) := \left\{ \frac{a}{b} : a, b \in R,\ b \neq 0 \right\},$$

under the equivalence relation

$$\frac{a}{b} \sim \frac{c}{d} \quad \text{if} \quad ad = bc.$$

Addition and multiplication are given by

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd},$$

as you would expect.

Remark 15.11. It is straightforward to check that $\sim$ is in fact an equivalence relation, and that addition and multiplication are well-defined.

For example, to check that $\sim$ is transitive, we note that if

$$\frac{a}{b} \sim \frac{c}{d}, \quad \frac{c}{d} \sim \frac{e}{f},$$

then $ad = bc$ and $cf = de$, so $adf = bcf = bde$. Since R is an integral domain, by cancellation, we have $af = be$. This shows that

$$\frac{a}{b} \sim \frac{e}{f}.$$

If R is not an integral domain the definition of fraction field does not work.

Example 15.12.

1. The fraction field of $\mathbb{Z}$ in $\mathbb{Q}$.
2. The fraction field of $\mathbb{R}[x]$ consists of *rational functions* f/g, where f and g are polynomials, and $g \neq 0$. For example $\frac{2x}{x^2-3}$ is a rational function.
3. The fraction field of $\mathbb{Z}[i]$ is

   $$\mathbb{Q}(i) := \{a + bi : a, b \in \mathbb{Q}\}.$$

   This is because every element of $\mathrm{Frac}(\mathbb{Z}[i])$ is equal to

   $$\frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{c^2 + d^2} = \frac{ac + bd}{c^2 + d^2} + \frac{(bc - ad)i}{c^2 + d^2} \in \mathbb{Q}(i),$$

   for some $a, b, c, d \in \mathbb{Q}$ with $(c, d) \neq (0, 0)$.

15.4 Maximal ideals [11.8]

Definition 15.13. A *maximal ideal* of a ring R is an ideal I which is not equal to R, and is not contained in any ideal other than I and R.

Proposition 15.14. *Let I be an ideal of a ring R. Then I is maximal if and only if R/I is a field.*

To prove this Proposition 15.14, we first observe the following.

Proposition 15.15. *A ring R has only two ideals (0) and R if and only if R is a field.*

Proof. Suppose R has only the two ideals (0) and R. Let a be a nonzero element of R. Then the principal ideal (a) must be equal to R, which means that $1 \in (a)$. This means that $ax = 1$ for some $x \in R$, so a has an inverse.

On the other hand, if R is a field, the any ideal I which contains a nonzero element $a \in R$ also contains $1 = aa^{-1}$, so $I = R$. □

Corollary 15.16. *The zero ideal of a ring R is maximal if and only if R is a field.*

Proof. This is just because (0) is contained in every other ideal, so it is maximal if and only if it is the only ideal of R other than R. □

Proof of Proposition 15.14. Recall that the correspondence theorem gives a bijection

$$\{\text{ideals of } R \text{ which contain } I\} \leftrightarrow \{\text{ideals of } R/I\}.$$

This correspondence is inclusion-preserving, and the ideal I corresponds to the zero ideal of R/I. Thus, I is a maximal ideal of R if and only if (0) is a maximal ideal of R/I. By the above corollary, this is true if and only if R/I is a field. □

Example 15.17. Consider the ideal $I = (x, 5)$ of $\mathbb{Z}[x]$. We have

$$\mathbb{Z}[x]/(x, 5) \simeq \mathbb{Z}/(5) = \mathbb{F}_5,$$

which is a field, so I is a maximal ideal of $\mathbb{Z}[x]$.

Proposition 15.18. *The maximal ideals of $\mathbb{Z}$ are the ideals (p) where p is a prime.*

Proof. First, we note that the ideal (p) is prime because $\mathbb{Z}/(p) = \mathbb{F}_p$, which is a field.

On the other hand, every ideal of $\mathbb{Z}$ is equal to (n) for some integer $n \geq 0$. The ideal $(1) = \mathbb{Z}$ is not maximal, because maximal ideals cannot be equal to the whole ring. The ideal (0) is not maximal because $\mathbb{Z}$ is not a field.

Finally, if n is not prime, so $n = ab$ for some integers a, b with $1 < a < n$, then (n) is strictly contained in (a), which is a proper ideal of $\mathbb{Z}$, so (n) is not maximal. □

16 Mar 16: Maximal ideals and prime ideals

16.1 Maximal ideals in $F[x]$ [11.8]

- Recall that a *maximal ideal* of a ring R is an ideal $I \neq R$ which is not contained in any other ideal other than R.
- We showed that an ideal I is maximal if and only if R/I is a field.

Definition 16.1. Let F be a field. A polynomial $f \in F[x]$ is *irreducible* if it is not constant, and is not the product of two nonconstant polynomials with coefficients in F.

Example 16.2.

1. In $\mathbb{R}[x]$, the polynomial $x^2 + 1$ is irreducible, but $x^2 - 1 = (x-1)(x+1)$ is not irreducible.
2. In $\mathbb{C}[x]$, the polynomial $x^2 + 1 = (x+i)(x-i)$ is not irreducible.
3. The constant polynomial $f(x) = 5$ is a unit in $\mathbb{C}[x]$.

Proposition 16.3. *Let F be a field. The maximal ideals of $F[x]$ are the ideals (f) for monic irreducible polynomials f.*

Proof. We know that every nonzero ideal of $F[x]$ is equal to (f) for some monic polynomial f.

- If $f = gh$ for nonconstant polynomials g, h, then $(f) \subseteq (g)$, and $(g) \neq F[x]$, since g is nonconstant, so f is not maximal.
- On the other hand, $(f) \subseteq (g)$ if and only if g divides f. If f is irreducible, then g divides f if and only if g is constant, or a constant multiple of f.

  In the first case, $(g) = F[x]$, and in the second case $(g) = (f)$. Thus, if f is irreducible, (f) is a maximal ideal. □

Example 16.4. Previously, we showed that the quotient ring $\mathbb{F}_5[x]/(x^2-3)$ is a field, while $\mathbb{F}_{11}[x]/(x^2-3)$ is not a field.

This is because $x^2 - 3$ is irreducible in $\mathbb{F}_5[x]$, but factors as $(x-5)(x+5)$ in $\mathbb{F}_{11}[x]$.

16.2 Maximal ideals in $\mathbb{C}[x_1, \ldots, x_n]$ [11.8, 11.9]

Proposition 16.5. *The maximal ideals of $\mathbb{C}[x]$ are the ideals $(x-a)$ for complex numbers $a \in \mathbb{C}$.*

Proof. Since every monic polynomial $f \in \mathbb{C}[x]$ factors as $f = (x-a_1)\cdots(x-a_n)$ for the roots $a_1, \ldots, a_n$ of f, the monic irreducible polynomials in $\mathbb{C}[x]$ are the polynomials $x - a$ for $a \in \mathbb{C}$. □

Theorem 16.6 (Hilbert's Nullstellensatz)**.** *The maximal ideals of* $\mathbb{C}[x_1, \ldots, x_n]$ *are the ideals* $(x_1 - a_1, \ldots, x_n - a_n)$ *for* $(a_1, \ldots, a_n) \in \mathbb{C}^n$*.*

Thus, maximal ideals of $\mathbb{C}[x_1, \ldots, x_n]$ correspond to points in complex n-dimensional space $\mathbb{C}^n$.

Lemma 16.7. *Let* I *be an ideal of a ring* R*. Maximal ideals of* R/I *correspond to maximal ideals of* R *which contain* I*.*

Proof. This follows from the correspondence theorem, because the correspondence is inclusion-preserving. □

Example 16.8. Maximal ideals of $\mathbb{C}[x, y]/(x - y, y - x^2)$ correspond to ideals $(x - a, y - b)$ which contain $x - y$ and $y - x^2$.

Note that $(x - a, y - b)$ is the kernel of the evaluation map

$$\mathbb{C}[x, y] \to \mathbb{C}, \quad f \mapsto f(a, b).$$

(This is the map which sends $x \mapsto a$ and $y \mapsto b$.) Thus, $(x - a, y - b)$ contains bot h$x - y$ and $y - x^2$ if and only if

$$a - b = 0, \quad b - a^2 = 0.$$

This is true if and only if $(a, b) = (0, 0)$ or $(a, b) = (1, 1)$, so the quoteint ring $\mathbb{C}[x]/(x - y, y - x^2)$ has two maximal ideals.

16.3 Factorization of integers [12.1]

We are familiar with the fact that every positive integer can be uniquely factored as a product of primes. This is called the *fundamental theorem of arithmetic*.

The uniqueness is proved as follows. Suppose

$$p_1 p_2 \cdots p_k = q_1 q_2 \cdots q_\ell \tag{1}$$

for primes $p_1, \ldots, p_k$ and $q_1, \ldots q_\ell$. We wish to show that $k = \ell$, and that $p_1, \ldots, p_k$ is just a rearrangement of $q_1, \ldots q_\ell$.

- Recall the key fact that that if a prime p divides ab for some integers a and b, then p divides at least one of a and b.

We apply this to Equation 1 by noting that p_1 divides $q_1 q_2 \cdots q_\ell$, so it divides (and hence is equal to) one of $q_1, q_2, \ldots, q_\ell$. Without loss of generality, assume that $p_1 = q_1$.

Thus, we can cancel $p_1 = q_1$ from both sides, and get

$$p_2 \cdots p_k = q_2 \cdots q_\ell.$$

The result follows from induction (say, on k).

16.4 Prime ideals

Definition 16.9. Let R be a ring. An ideal $I \neq R$ is said to be a *prime ideal* if whenever $ab \in I$ for elements $a, b \in R$, then either $a \in I$ or $b \in I$.

Example 16.10.

1. The ideal (0) of a ring R is prime if and only if R is an integral domain. This follows from definition.
2. The prime ideals of $\mathbb{Z}$ are (0) and (p) for all primes p.
3. For a field F, the prime ideals of $F[x]$ are (0) and (f) for monic irreducible polynomials f.

Example 16.11. The ideal (5) is a prime ideal of $\mathbb{Z}$, but it is not a prime ideal of $\mathbb{Z}[i]$. This is because for $a = 1 + 2i$ and $b = 1 - 2i$, we have

$$ab = 1 - (2i)^2 = 5 \in (5),$$

but it is clear that $a, b \notin (5)$, since $\frac{1}{5} + \frac{2}{5}i$ and $\frac{1}{5} - \frac{2}{5}i$ are not elements of $\mathbb{Z}[i]$.

16.5 R/I is integral domain for prime I

Recall that an ideal I of R is maximal if and only if R/I is a field. We have a similar statement for prime ideals.

Proposition 16.12. *Let I be an ideal of a ring R. Then I is a prime ideal of R if and only if R/I is an integral domain.*

Proof. Suppose I is a prime ideal of R. Suppose $\overline{a}, \overline{b} \in R/I$ satisfy $\overline{a} \cdot \overline{b} = 0$ in R/I. Note that

$$\overline{a} \cdot \overline{b} = 0 \iff \overline{ab} = 0 \iff ab \in I.$$

Since I is a prime ideal, we have either $a \in I$ or $b \in I$. This is equivalent to either $\overline{a} = 0$ or $\overline{b} = 0$ in R/I, so R/I is an integral domain.

Conversely, suppose that R/I is an integral domain. Then $ab \in I$ implies $\overline{a} \cdot \overline{b} = 0$, which implies $\overline{a} = 0$ or $\overline{b} = 0$, which is equivalent to $a \in I$ or $b \in I$. Thus, I is a prime ideal. $\square$

Corollary 16.13. *Let I be a maximal ideal of a ring R. Then I is prime.*

Proof. If I is a maximal ideal, then R/I is a field, so it is an integral domain. $\square$

Example 16.14.

1. Let $R = \mathbb{Z}[x]$, and let $I = (x)$. Then

$$R/I = \mathbb{Z}[x]/(x) \simeq \mathbb{Z},$$

which is an integral domain, so (x) is a prime ideal of $\mathbb{Z}[x]$. However, $\mathbb{Z}$ is not a field, so (x) is not a maximal ideal of $\mathbb{Z}[x]$.

2. Indeed, (x) is contained in the ideal $I' = (5, x)$ of $\mathbb{Z}[x]$. Since

$$R/I' = \mathbb{Z}[x]/(5, x) \simeq \mathbb{F}_5[x]/(x) \simeq \mathbb{F}_5,$$

which is a field, $(5, x)$ is a maximal ideal, and hence also a prime ideal of $\mathbb{Z}[x]$.

3. The ideal (x) of $\mathbb{C}[x, y]$ is a prime ideal, since $\mathbb{C}[x, y]/(x) \simeq \mathbb{C}[y]$ is an integral domain, but (x) is not a maximal ideal. For example, it is contained it (x, y).

16.6 Correspondence theorem for prime ideals

Proposition 16.15. *Let R be a ring, and let I be an ideal of R. The correspondence theorem gives a bijection between*

$$\{\text{prime ideals of } R \text{ which contain } I\} \quad \textit{and} \quad \{\text{prime ideals of } R/I\}.$$

Proof. Recall that the bijection given by the correspondence theorem sends an ideal $J \supseteq I$ of R to $\pi(J)$, where

$$\pi\colon R \to R/I, \quad x \mapsto \overline{x}$$

is the quotient map, and the inverse sends an ideal J' of R/I to $\pi^{-1}(J')$.

1. We check that if J is a prime ideal of R, then $\pi(J)$ is a prime ideal of R/I. Suppose $\overline{a}, \overline{b} \in R/I$ satisfy $\overline{a} \cdot \overline{b} \in \pi(J)$. This means $\overline{ab} \in \pi(J)$. Since J contains $\ker(\pi) = I$, this implies that $ab \in J$.

   Since J is a prime ideal, $ab \in J$ implies either $a \in J$ or $b \in J$, which implies either $\overline{a} \in \pi(J)$ or $\overline{b} \in \pi(J)$. Thus, $\pi(J)$ is a prime ideal.

2. We check that if J' is a prime ideal of R/I, then $\pi^{-1}(J')$ is a prime ideal of R. Indeed, this is true for any ring homomorphism $\pi\colon R \to S$.

   Suppose $a, b \in R$ satisfy $ab \in \pi^{-1}(J')$. This means that $\overline{a} \cdot \overline{b} \in J'$, and since J' is prime, either $\overline{a} \in J'$ or $\overline{b} \in J'$. This is equivalent to either $a \in \pi^{-1}(J')$ or $b \in \pi^{-1}(J')$. Thus, $\pi^{-1}(J')$ is prime.

□

Example 16.16. Prime ideals of $\mathbb{C}[x, y]/(xy)$ correspond to prime ideals of $\mathbb{C}[x, y]$ which contain (xy). But if a prime ideal I contains xy, then it must contain either x or y.

The prime ideals of $\mathbb{C}[x, y]$ containing (xy) are the ideals

$$(x),\ (y),\ (x, y - b),\ (x - a, y),$$

for $a, b \in \mathbb{C}$. The ideals (x) and (y) are not maximal, since they are both contained in (x, y).

17 Mar 18: Unique factorization domains

17.1 Unique factorization domains [12.2]

We first discuss how to generalize the statement of the fundamental theorem arithmetic of arithmetic to other rings.

Definition 17.1. Let R be an integral domain, and let a and b be elements of R. We say that

- a *divides* b if $b = aq$ for some $q \in R$.
- a is a *proper divisor* of b if $b = aq$ and neither a nor q are units.
- a and b are *associates* if $b = ua$ for a unit u; equivalently, if a divides b and b divides a.
- a is *irreducible* if it has no proper divisors.

Given an element $a \in R$, we can try to repeatedly decompose a into a product of other elements until all factors are irreducible. We say that *factoring terminates* in R if this process always ends, so that every element can be written as a product of irreducible elements.

Definition 17.2. A *unique factorization domain* (abbreviated UFD) is an integral domain in which factoring terminates, and the factorization is unique in the sense that if

$$p_1 \cdots p_k = q_1 \cdots q_\ell$$

for irreducible elemens $p_1, \ldots, p_k$ and $q_1, \ldots, q_\ell$, then $k = \ell$, and the elements in the products can be rearranged so that p_i is an associate of q_i for $i = 1, \ldots, k$.

Example 17.3. The ring $\mathbb{Z}[\sqrt{-5}]$ is not a unique factorization domain. Indeed,

$$2 \cdot 3 = 6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$$

are two different factorizations of 6. It is easy to check that 2, 3, $1 + \sqrt{-5}$ and $1 - \sqrt{-5}$ are all irreducible.

We already know that $\mathbb{Z}$ is a UFD. Today, we will show that all principal ideal domains are UFD's. For example, we'll see that this implies that $\mathbb{Q}[x]$ and $\mathbb{Z}[i]$ are unique factorization domains.

17.2 Prime elements [12.2]

Definition 17.4. Let R be an integral domain. A nonzero element $p \in R$ is called *prime* if whenever p divides ab, then p divides a or p divides b.

(This is equivalent to saying that the prinicpal ideal (p) is prime.)

Lemma 17.5. *Every prime element of R is irreducible.*

Proof. Let p be a prime element of R. Suppose

$$p = ab.$$

Since p is prime, either p divides a or p divides b. Suppose p divides a. Then we also have that a divides p, so p and a are associates, and b is a unit. Thus p has no proper divisors, so it is irreducible. □

Example 17.6. The element 2 of $\mathbb{Z}[\sqrt{-5}]$ is irreducible, but it is not prime, because 2 divides $(1-\sqrt{-5})(1+\sqrt{-5}) = 6$, but it does not divide either $1-\sqrt{-5}$ or $1+\sqrt{-5}$.

Proposition 17.7. *Let R be an integral domain in which factoring terminates. Then R is a UFD if and only if every irreducible element is prime.*

Proof.

1. First, suppose that R is a UFD, and let p be an irreducible element of R. Suppose that p divides ab for some elements $a, b \in R$, so $ab = pc$ for some $c \in R$.

   Suppose a, b, and c have irreducible factorizations $a = p_1 \cdots p_k$, $b = q_1 \cdots q_\ell$, and $c = r_1 \cdots r_m$. Then

   $$p_1 \cdots p_k \cdot q_1 \cdots q_\ell = ab = p \cdot r_1 \cdots r_m,$$

   are two irreducible factorizations of ab, so p must be equal to one of the terms $p_1, \ldots, p_k$ or $q_1, \ldots, q_\ell$. This implies that p must divide at least one of a and b, so p is a prime element.

2. Suppose that every irreducible element is prime. The proof that irreducible factorizations are unique is the same as the proof of the fundamental theorem of arithmetic. □

17.3 Principal ideal domains [12.2]

Definition 17.8. A *principal ideal domain* (abbreviated PID) is an integral domain in which every ideal is principal.

Example 17.9.

1. $\mathbb{Z}$ is a PID.

2. $\mathbb{Z}/6\mathbb{Z}$ is not a PID. Every ideal is principal, but it is not an integral domain.

3. Let F be a field. The polynomial ring $F[x]$ is a PID.

4. The two-variable polynomial ring $\mathbb{C}[x, y]$ is not a PID. The ideal (x, y) is not principal.

5. The polynomial ring $\mathbb{Z}[x]$ is not a PID. The ideal $(x, 5)$ is not principal.

17.4 PID's are UFD's [12.2]

The goal of this subsection is to prove that all principal ideal domains are unique factorization domains.

Definition 17.10. If R is a PID, a *greatest common divisor* of two elements $a, b \in R$ is an element $d \in R$ such that $(a, b) = (d)$.

It is clear that $\gcd(a, b)$ divides both a and b, and it is unique up to associates.

Proposition 17.11. *In a principal ideal domain, every irreducible element is prime.*

Proof. Let q be an irreducible element of R. Suppose that q divides ab for $a, b \in R$. We want to show that q divides either a or b.

Let d be a greatest common divisor of q and a. Since q is irreducible, d is either a unit or an associate of q. If d is an associate of q, then q divides a, so we are done.

If d is a unit, we can assume that $d = 1$. Since $1 \in (q, a)$, we have

$$1 = rq + sa$$

for some $r, s \in R$. Multiplying both sides by b gives

$$b = rqb + sab.$$

But q divides the right hand side of this equation, so q divides b. □

Corollary 17.12. *A principal ideal domain is a unique factorization domain.*

Proof. Let R be a PID. By Propositions 17.7 and 17.11, it suffices to prove that factoring terminates in R. □

Corollary 17.13. *Let F be a field. The polynomial ring $F[x]$ is a unique factorization domain.*

17.5 Euclidean domains [12.2]

Previously, we have shown that $\mathbb{Z}$ and $F[x]$ are PID's, for F a field. The proofs of these facts essentially used division with remainder. We now abstract this idea, and use the same method to show that $\mathbb{Z}[i]$ is a PID.

Definition 17.14. A *Euclidean domain* is an integral domain R with a *size function*

$$\sigma \colon R - \{0\} \to \mathbb{Z}_{\geq 0}$$

such that *division with remainder* is possible. That is, for any $a, b \in R$ with a nonzero, there exist $q, r \in R$ such that

$$b = aq + r, \quad r = 0 \quad \text{or} \quad \sigma(r) < \sigma(a).$$

Proposition 17.15.

1. *$\mathbb{Z}$ is a Euclidean domain, with size function $\sigma(a) = |a|$.*
2. *For F a field, $F[x]$ is a Euclidean domain, with size function $\sigma(f) = \deg f$.*
3. *$\mathbb{Z}[i]$ is a Euclidean domain, with size function*

$$\sigma(a + bi) = |a + bi|^2 = a^2 + b^2.$$

Proof. Let $\alpha \in \mathbb{Z}[i]$ be nonzero. The multiples of α form a square lattice in $\mathbb{C}$ with side length $|\alpha|$. Thus, for any $\beta \in \mathbb{Z}[i]$, there exists $\gamma = \alpha q \in (\alpha)$ such that

$$|\beta - \gamma| < \tfrac{\sqrt{2}}{2} \cdot |\alpha| < |\alpha|.$$

Letting $r = \beta - \gamma$, we have

$$\beta = \alpha q + r$$

and $\sigma(r) = |r|^2 < |\alpha|^2 = \sigma(\alpha)$. □

Proposition 17.16. *A Euclidean domain is a principal ideal domain.*

Proof. Let R be a Euclidean domain, with size function σ. Let I be an ideal of R. Since the codomain of σ is the nonnegative integers, either $I = (0)$, or there exists nonzero $a \in I$ such that $\sigma(a)$ is minimal (among all nonzero $a \in I$).

Then we claim that $I = (a)$. Indeed, for any $b \in a$, we can write $b = aq + r$ for some r such that $r = 0$ or $\sigma(r) < \sigma(a)$. But since $a, b \in I$, we also have $r \in I$, so by the minimality of $\sigma(r)$, we must have $r = 0$. Thus $b \in (a)$, so $I \subseteq (a)$. Clearly, $(a) \subseteq I$, so $I = (a)$, as desired. □

Corollary 17.17. *The ring $\mathbb{Z}[i]$ is a unique factorization domain.*

17.6 Aside on factoring ideals

In the ring $\mathbb{Z}[\sqrt{-5}]$, which is not a PID, the element 6 has two factorizations

$$2 \cdot 3 = 6 = (1 + \sqrt{-5}) \cdot (1 - \sqrt{-5}).$$

However, the ideal $(3, 1 + \sqrt{-5})$ is not principal, so we cannot talk about the GCD of 3 and $1 + \sqrt{-5}$. This is related to the fact that unique factorization fails in $\mathbb{Z}[\sqrt{-5}]$.

However, it turns out that $(3, 1 + \sqrt{-5})$, $(3, 1 - \sqrt{-5})$, and $(2, 1 + \sqrt{-5})$ are all prime ideals, and we have the factorization of the *ideal* (6) into the product of prime ideals

$$(6) = (2, 1 + \sqrt{-5})^2 (3, 1 + \sqrt{-5})(3, 1 - \sqrt{-5}) = \mathfrak{p}^2 \mathfrak{q}_1 \mathfrak{q}_2,$$

and this factorization is unique. Indeed, one can check that

$$(2) = \mathfrak{p}^2, \quad (3) = \mathfrak{q}_1 \mathfrak{q}_2, \quad (1 + \sqrt{-5}) = \mathfrak{p}\mathfrak{q}_1, \quad (1 - \sqrt{-5}) = \mathfrak{p}\mathfrak{q}_2.$$

18 Mar 23: Gauss's lemma

18.1 Factorization in $\mathbb{Q}[x]$ vs $\mathbb{Z}[x]$ [12.3]

Last time, we showed that for any field F, the ring $F[x]$ is a unique factorization domain, because it is a PID. For example, we know that $\mathbb{Q}[x]$ is a UFD.

The ring $\mathbb{Z}[x]$ is not a PID (one example of a non-principal ideal of this ring would be $(5, x)$). However, we have an inclusion

$$\mathbb{Z}[x] \subseteq \mathbb{Q}[x],$$

and factorization of polynomials in $\mathbb{Z}[x]$ is closely related to factorization in $\mathbb{Q}[x]$. Today, we will show that $\mathbb{Z}[x]$ is also a UFD.

Example 18.1. If we want to factor the polynomial $f(x) = 4x^2 + 10x + 6$ in $\mathbb{Q}[x]$, we could write

$$4x^2 + 10x + 6 = 4(x^2 + \tfrac{5}{2}x + \tfrac{3}{2}) = 4(x + 1)(x + \tfrac{3}{2}).$$

Since 4 is a unit in $\mathbb{Q}[x]$, some irreducible factorizations of f would be

$$f(x) = (4x + 4)(x + \tfrac{3}{2}) \quad \text{or} \quad (x + 1)(4x + 6),$$

which are the same factorization up to associates. Note that the factorization on the right is not a factorization in $\mathbb{Z}[x]$, while the factorization on the left is.

The irreducible factorization of f in $\mathbb{Z}[x]$ would be

$$f(x) = 2(x + 1)(2x + 3).$$

Here, note that 2 is an irreducible element of $\mathbb{Z}[x]$, not a unit. It is also no longer enough to consider only monic polynomials.

Nevertheless, we can see that the factorization of f in $\mathbb{Z}[x]$ is closely related to its factorization in $\mathbb{Q}[x]$.

18.2 Primitive polynomials [12.3]

Question 18.2. Let $f \in \mathbb{Z}[x]$. Does f being irreducible in $\mathbb{Q}[x]$ imply that f is irreducible in $\mathbb{Z}[x]$?

Suppose f is irreducible in $\mathbb{Q}[x]$, and $f = gh$ for $g, h \in \mathbb{Z}[x]$. Then one of g and h must be a unit in $\mathbb{Q}[x]$, i.e., a constant polynomial.

So, f might not be irreducible in $\mathbb{Z}[x]$, but the only way this can happen is if

$$f = c \cdot g$$

for some nonzero integer $c \in \mathbb{Z}$ which is not a unit, and some integer polynomial $g \in \mathbb{Z}[x]$.

Example 18.3. The polynomial $f(x) = 8x^3 + 16x^2 + 12$ is irreducible in $\mathbb{Q}[x]$, but it factors as

$$f(x) = 2 \cdot (4x^3 + 8x^2 + 6)$$

in $\mathbb{Z}[x]$. In fact, $4x^3 + 8x^2 + 6$ is still reducible, but if we divide out by the greatest common divisor of all of the coefficients, then

$$f_0(x) = 2x^3 + 4x^2 + 3$$

is irreducible in $\mathbb{Z}[x]$.

Definition 18.4. A *primitive polynomial* is a polynomial

$$f(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$$

such that $\gcd(a_0, \ldots, a_n) = 1$ and the leading coefficient a_n is positive.

Lemma 18.5. *Let $f \in \mathbb{Q}[x]$. There exist* unique *$c \in \mathbb{Q}$ and $f_0 \in \mathbb{Z}[x]$ such that $f = c \cdot f_0$, and f_0 is primitive. Furthermore, $f \in \mathbb{Z}[x]$ if and only if $c \in \mathbb{Z}$.*

Example 18.6. Suppose $f(x) = \frac{2}{3}x^2 + 6x + \frac{4}{5}$. Then we have

$$f(x) = \tfrac{1}{15}(10x^2 + 90x + 12) = \tfrac{2}{15}(5x^2 + 45x + 6) = c \cdot f_0(x).$$

In the first step above, we cleared denominators. In the second step, we factored out the GCD of the coefficients of $10x^2 + 90x + 12$.

The uniqueness essentially follows from the fact that after clearing denominators, the constant c we choose must be this GCD (up to sign).

Question 18.7. Let $f \in \mathbb{Z}[x]$. Does f being irreducible in $\mathbb{Z}[x]$ imply that f is irreducible in $\mathbb{Q}[x]$?

18.3 Gauss's lemma [12.3]

Proposition 18.8 (Gauss's lemma)**.** *The product of two primitive polynomials is primitive.*

Let p be a prime. We have a *reduction modulo p* map

$$\psi_p \colon \mathbb{Z}[x] \mapsto \mathbb{F}_p[x]$$

which sends each coefficient to its residue in $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$.

For example, the reduction modulo 7 map sends $f(x) = 10x^2 + 90x + 12$ to

$$\overline{f}(x) = \overline{3}x^2 + \overline{6}x + \overline{5} \in \mathbb{F}_7[x].$$

The map ψ_p is a ring homomorphism, and $\ker(\psi_p)$ consists of integer polynomials whose coefficients are all divisible by p.

Proof of Gauss's lemma. Let f_0 and g_0 be primitive polynomials. We want to show that the coefficients of $f_0 g_0$ are coprime.

Assume for the sake of contradiction that there exists some prime p such that p divides every coefficient of $f_0 g_0$. This is equivalent to saying that

$$\psi_p(f_0 g_0) = \psi_p(f_0)\psi_p(g_0) = 0.$$

But $\mathbb{F}_p[x]$ is an integral domain, so either $f_0 \in \ker(\psi_p)$ or $g_0 \in \ker(\psi_p)$.

This means that at least one of f_0 or g_0 has coefficients all divisible by p, but this contradicts the assumption than f_0 and g_0 are both primitive. □

Theorem 18.9. *Let f_0 be a primitive polynomial, and let g be any element of $\mathbb{Z}[x]$. If f_0 divides g in $\mathbb{Q}[x]$, then f_0 divides g in $\mathbb{Z}[x]$*

Proof. Suppose $g = f_0 h$ for $h \in \mathbb{Q}[x]$. We want to show that $h \in \mathbb{Z}[x]$.

Write $h = ch_0$ for $c \in \mathbb{Q}$ and h_0 primitive. Then $g = cf_0 h_0$, and by Gauss's lemma, $f_0 h_0$ is primitive. Then by Lemma 18.5, since $g \in \mathbb{Z}[x]$, we have $c \in \mathbb{Z}$. Thus, $h = ch_0 \in \mathbb{Z}[x]$. □

Corollary 18.10. *Let $f, g \in \mathbb{Z}[x]$. If f and g have a common factor in $\mathbb{Q}[x]$, then they have a common factor in $\mathbb{Z}[x]$.*

Proof. By Lemma 18.5, we can take a common factor h_0 which is a primitive polynomial, and then it is a common factor in $\mathbb{Z}[x]$ as well by Theorem 18.9. □

18.4 $\mathbb{Z}[x]$ is UFD [12.3]

Proposition 18.11. *Let f be an irreducible element of $\mathbb{Z}[x]$ with positive leading coefficient. Then either f is a constant polynomial p for p a prime, or f is a primitive polynomial which is irreducible in $\mathbb{Q}[x]$.*

Proof.

1. It is clear that $f(x) = n$ is irreducible if and only if it n is a prime integer.

2. By our previous discussion, if f_0 is nonconstant, primitive, and irreducible in $\mathbb{Q}[x]$, then it is also irreducible in $\mathbb{Z}[x]$.

3. The last thing we need to check is that if f is irreducible in $\mathbb{Z}[x]$, then it is primitive and irreducible in $\mathbb{Q}[x]$. By definition, it must be primitive.

   Assume for the sake of contradiction that $f = gh$ for nonconstant $g, h \in \mathbb{Q}[x]$. Write $g = c \cdot g_0$ and $h = d \cdot h_0$ for $c, d \in \mathbb{Q}$ and primitive g_0, h_0. Then
   $$f = cd \cdot g_0 h_0.$$
   By Gauss's lemma, $g_0 h_0$ is primitive. Since f is also primitive, we must have $f = g_0 h_0$. This contradicts the assumption that f is irreducible in $\mathbb{Z}[x]$. □

Proposition 18.12. *Every irreducible element of $\mathbb{Z}[x]$ is prime.*

Proof. The fact that a prime integer p is prime in $\mathbb{Z}[x]$ was shown in the proof of Gauss's lemma. (Recall that being prime just means that if p divides fg in $\mathbb{Z}[x]$, then p divides either f or g in $\mathbb{Z}[x]$.)

Assume that f_0 is a primitive polynomial which is irreducible in $\mathbb{Q}[x]$. Suppose that f_0 divides gh for $g, h \in \mathbb{Z}[x]$.

Since irreducible elements are prime in $\mathbb{Q}[x]$, we have that f_0 divides either g or h in $\mathbb{Q}[x]$. By Theorem 18.9, this implies that f_0 divides g or h in $\mathbb{Z}[x]$. □

Theorem 18.13. *The polynomial ring $\mathbb{Z}[x]$ is a unique factorization domain.*

18.5 Generalization [12.3]

In any UFD R, one can define the greatest common divisor of elements using irreducible factorizations (even if R is not a PID). Thus, one can consider primitive polynomials in $R[x]$.

Theorem 18.14. *Let R be a unique factorization domain. Then the polynomial ring $R[x]$ is also a unique factorization domain.*

Corollary 18.15. *Let F be a field. The polynomial ring $F[x, y]$ is a unique factorization domain.*

Proof. Apply the above theorem to the ring $R = F[x]$. Note that $F[x, y]$ is not a PID. □

18.6 Factorization in $\mathbb{R}[x]$ [12.2]

Recall that every monic polynomial $f \in \mathbb{C}[x]$ has an irreducible factorization into linear factors

$$f(x) = (x - \alpha_1) \cdots (x - \alpha_n),$$

where $n = \deg f$, and $\alpha_1, \ldots, \alpha_n \in \mathbb{C}$ are the roots of f, with repetition.

Proposition 18.16. *Let $f \in \mathbb{R}[x]$ be a polynomial with real coefficients. If $\alpha \in \mathbb{C}$ is a root of f, then the complex conjugate $\overline{\alpha}$ is also a root of f, and the polynomial $(x - \alpha)(x - \overline{\alpha})$ has real coefficients and divides f.*

Thus, while a real polynomial may not factor into linear factors in $\mathbb{R}[x]$, its non-real roots come in complex conjugate pairs. So in $\mathbb{R}[x]$, every polynomial factors into irreducible linear and quadratic factors. For example,

$$x^3 - 1 = (x - 1)(x^2 + x + 1).$$

The quadratic polynomial $x^2 + x + 1$ is irreducible because its discriminant is $-3 < 0$.

Irreducible polynomials in $\mathbb{Q}[x]$ can have arbitrarily high degree.

18.7 Mar 25: Modules

18.8 Modules [14.1]

Recall that a vector space over $\mathbb{R}$ is a structure which looks like $\mathbb{R}^n$. A *module* is a generalization of this concept.

Definition 18.17. Let R be a ring. A *module over* R is an abelian group $(V, +)$ together with a scalar multiplication map

$$R \times V \to V, \quad (r, v) \mapsto rv$$

satisfying the axioms

$$1v = v, \quad (rs)v = r(sv), \quad (r + s)v = rv + sv, \quad r(u + v) = ru + rv$$

for any $r, s \in R$ and $u, v \in V$.

If R happens to be a field, we call modules over R vector spaces instead.

Definition 18.18. A *submodule* of a module V over R is a nonempty subset W of V which is closed under addition and scalar multiplication.

Definition 18.19. If F is a field, a *vector space over* F is a module over F.

Example 18.20.

1. $\mathbb{R}^n$ is a vector space over $\mathbb{R}$ with the usual addition and scalar multiplication of vectors.

2. The set $\mathbb{Z}^n$ of n-dimensional column vectors with entries in $\mathbb{Z}$ is a module over $\mathbb{Z}$.

3. The set
$$\{ax^2 + bx + c : a, b, c \in \mathbb{R}\}$$
of polynomials of degree at most 2 is a vector space over $\mathbb{R}$.

4. The set
$$\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$$
is a vector space over $\mathbb{Q}$.

5. The set
$$(2) = \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\} \subseteq \mathbb{Z}$$
is a module over $\mathbb{Z}$.

6. The set
$$\mathbb{Z}/5\mathbb{Z} = \{\overline{0}, \overline{1}, \overline{2}, \overline{3}, \overline{4}, \overline{5}\}$$
is a module over $\mathbb{Z}$.

The following are some more general examples.

Example 18.21.

1. For any ring R, the set R^n of n-dimensional column vector with entries in R is a module over R. In particular, $R = R^1$ is a module over R.

2. An ideal I of a ring R is a module over R. It is closed under scalar multiplication, because for any $v \in I$ and any $r \in R$, the element $rv \in I$ as well. It is a submodule of the module R itself.

   In fact, "submodule of R" and "ideal of R" mean the same thing.

3. For any ideal I of a ring R, the quotient R/I is a module over R.

4. Any abelian group $(G, +)$ is a module over $\mathbb{Z}$. Scalar multiplication is given by, e.g.,
$$3 \cdot v = v + v + v$$
   for $v \in G$.

   In fact, "abelian group" and "$\mathbb{Z}$-module" mean the same thing.

18.9 Direct sums and quotients [14.1]

Definition 18.22. The *direct sum* of modules V and W over R is the module $V \oplus W$ whose elements are pairs $(v, w) \in V \times W$, with addition and scalar multiplication given by

$$(v, w) + (v', w') = (v + v', w + w'), \quad r(v, w) = (rv, rw).$$

Example 18.23.

1. $\mathbb{R}^2 = \mathbb{R} \oplus \mathbb{R}$.

2. The vector space considered previously is equal to the direct sum

$$\{ax^2 + bx + c : a, b, c \in \mathbb{R}\} = \mathbb{R}x^2 \oplus \mathbb{R}x \oplus \mathbb{R},$$

   where

$$\mathbb{R}x^2 := \{ax^2 : a \in \mathbb{R}\}, \quad \mathbb{R}x := \{bx : b \in \mathbb{R}\}, \quad \mathbb{R} := \{c : c \in \mathbb{R}\}.$$

Definition 18.24. Let W be a submodule of V. The *quotient module* V/W is the group of additive cosets $\overline{v} = [v + W]$, with scalar multiplication given by

$$r[v + W] = [rv + W].$$

Quotients R/I of a R by an ideal are an example of quotient modules over R.

18.10 Homomorphisms [14.1]

Definition 18.25. Let R be a ring. A *homomorphism* from an R-module V to an R-module W is a map $\varphi\colon V \to W$ satisfying

$$\varphi(u+v) = \varphi(u) + \varphi(v), \quad \varphi(rv) = r\varphi(v)$$

for any $r \in R$ and $u, v \in R$. An *isomorphism* is a bijective homomorphism.

Definition 18.26. The *kernel* of a homomorphism $\varphi\colon V \to W$ is the submodule of V defined by

$$\ker(\varphi) = \{v \in V : \varphi(v) = 0\}.$$

The *image* of φ is a submodule of W.

Theorem 18.27. *Let $\varphi\colon V \to W$ be a surjective homomorphism of R-modules. Then φ induces an isomorphism*

$$V/\ker(\varphi) \to W, \quad \overline{v} = [v + \ker(\varphi)] \mapsto \varphi(v).$$

18.11 Bases [14.2]

Definition 18.28. Let V be a module over a ring R. Let $S = \{v_1, \dots, v_n\}$ be a subset of V.

- We say that S is *linearly independent* if

  $$r_1 v_1 + \cdots + r_2 v_2 = 0$$

  for $r_1, \dots, r_n \in R$ implies $r_1 = \cdots = r_n = 0$.

- We say that S is a *spanning set* or a *generating set* for V if every element $v \in V$ can be expressed as a linear combination

  $$v = r_1 v_1 + \cdots r_n v_n$$

  for some $r_1, \dots, r_n \in R$.

- We say that S is a *basis* of V if it is a linearly independent spanning set.

  Equivalently, S is a basis if every element of V can be expressed *uniquely* as a linear combination of $v_1, \dots, v_n$.

Definition 18.29. A *free module* over R is a module which has a basis.

Example 18.30.

1. The $\mathbb{Q}$-vector space $\mathbb{Q}[\sqrt{2}]$ has basis $\{1, \sqrt{2}\}$.
2. The module $\mathbb{Z}^3$ has basis $\{(1,0,0)^\top, (0,1,0)^\top, (0,0,1)^\top\}$.
3. The set of polynomials $\mathbb{R}[x]$ as an $\mathbb{R}$-vector space has basis $\{1, x, x^2, \dots\}$. This is an infinite basis.

Note that if V has a finite basis $\{v_1, \ldots, v_n\}$, then V is isomorphic to R^n.

Example 18.31. The $\mathbb{Z}$-module $\mathbb{Z}/5\mathbb{Z}$ does not have a basis! No set can be linearly independent because $5v = 0$ for any element $v \in \mathbb{Z}/5\mathbb{Z}$.

Every vector space has a basis, but for general rings, most modules are not free.

Theorem 18.32. *Let V be a free module over R. Any two bases of V have the same number of elements.*

The size of a basis of a free module V is the *dimension* of V.

18.12 Matrices [14.2]

Let V and W be free R-modules, and let $\varphi\colon V \to W$ be a homomorphism. Since V and W are free, we can find bases $\{v_1, \ldots, v_n\}$ and $\{w_1, \ldots, w_m\}$ of V and W, respectively.

Since φ is a homomorphism, in order to specify φ, it suffices to specify $\varphi(v_j)$ for each basis element v_j of V. Since $\{w_1, \ldots, w_m\}$ is a basis of W, for each j, we can write

$$\varphi(v_j) = a_{1j}w_1 + \cdots + a_{mj}w_m$$

for some $a_{1j}, \ldots, a_{mj} \in R$. The $m \times n$ matrix A with (i, j) entry a_{ij} is the matrix of the linear map φ.

If $V = W$ and we pick the same basis of V on both sides, the *determinant* of φ is the determinant of the matrix A. Similarly, the *trace* of φ is the trace of the matrix A. This turns out to be independent of the choice of basis!

18.13 Structure theorem for finitely generated modules over a PID [14.7]

Definition 18.33. A module V over a ring R is called *finitely generated* if it has a finite generating set.

Example 18.34.

1. The $\mathbb{Z}$-module $\mathbb{Z}/5\mathbb{Z}$ is generated by the element $\overline{1}$.

2. The $\mathbb{Z}$-module
$$V = \mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z} \oplus \mathbb{Z}/10\mathbb{Z}$$
is finitely generated. The set $\{(1, \overline{0}, \overline{0}), (0, \overline{1}, \overline{0}), (0, \overline{0}, \overline{1})\}$ is a generating set for V, although it is not a basis.

Theorem 18.35. *Let R be a PID, and let V be a finitely generated module over R. Then*

$$V \cong \bigoplus_{i=1}^{n} R/(s_i)$$

for some $s_1, \ldots, s_n \in R$ such that $s_1 | s_2 | \cdots | s_n$.

Corollary 18.36. *Let R be a PID, and let V be a finitely generated module over R. Then*

$$V \cong R^{\oplus r} \oplus R/(q_1) \oplus \cdots \oplus R/(q_m),$$

for some $q_i = p_i^{e_i}$, where p_i is a prime element of R, and $e_i \geq 1$.

In both the theorem and the corollary, the choice of s_i and q_i is unique (up to permutation).

Corollary 18.37. *Every finite abelian group is isomorphic to a direct sum of cyclic groups of prime power orders.*

Example 18.38. By the Chinese Remainder Theorem,

$$\begin{aligned} \mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z} \oplus \mathbb{Z}/10\mathbb{Z} &\cong \mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/10\mathbb{Z} \\ &\cong \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/40\mathbb{Z}. \end{aligned}$$

19 Apr 1: Algebraic field extensions

19.1 Field extensions [15.1, 15.2]

A *field extension* is a pair of fields $F \subseteq K$, with F contained in K. We denote field extensions by the notation K/F.

Example 19.1.

1. $\mathbb{C}/\mathbb{R}$, $\mathbb{R}/\mathbb{Q}$, and $\mathbb{C}/\mathbb{Q}$ are all field extensions.
2. $\mathbb{Q}[\sqrt{2}]/\mathbb{Q}$ and $\mathbb{Q}[i]/\mathbb{Q}$ are field extensions, where

$$\mathbb{Q}[\sqrt{2}] \coloneqq \{a + b\sqrt{2} : a, b \in \mathbb{Q}\},$$

as we have seen before, and similarly,

$$\mathbb{Q}[i] \coloneqq \{a + bi : a, b \in \mathbb{Q}\}.$$

- If K/F is a field extension and $\alpha \in K$, then the *ring generated by* α is

$$F[\alpha] \coloneqq \{a_n\alpha^n + \cdots + a_0 : a_0, \ldots, a_n \in F\}.$$

In other words, $F[\alpha]$ consists of elements $\beta \in K$ which can be expressed as as polynomial in α; the degree of this polynomial can be anything. $F[\alpha]$ is the smallest subring of K which contains α.

- Similarly, define $F(\alpha)$ as the smallest subfield of K which contains α. We have

$$F(\alpha) = \mathrm{Frac}(F[\alpha]).$$

Elements of $F(\alpha)$ are quotients β_1/β_2 for $\beta_1, \beta_2 \in F[\alpha]$, with $\beta_2 \neq 0$.

- For $\alpha_1, \ldots, \alpha_n \in K$, let $F[\alpha_1, \ldots, \alpha_n]$ (resp. $F(\alpha_1, \ldots, \alpha)$) be the smallest subring (resp. subfield) of K containing $\alpha_1, \ldots, \alpha_n$.

The rings $\mathbb{Q}[\sqrt{2}]$ and $\mathbb{Q}[i]$ are examples of the $F[\alpha]$ construction in the extension $\mathbb{C}/\mathbb{Q}$. Since they are both fields, they are also instances of the $\mathbb{Q}(\alpha)$ construction! Indeed,

$$\mathbb{Q}[\sqrt{2}] = \mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}[i] = \mathbb{Q}(i).$$

19.2 Algebraic and transcendental elements [15.2]

Definition 19.2. Let K/F be a field extensions, and let $\alpha \in K$. We say that α is *algebraic over* F if it is the root of some monic polynomial with coefficients in F. It is *transcendental* otherwise.

Definition 19.3. If $\alpha \in K$ is algebraic over F, the *irreducible polynomial* of α over F is the monic polynomial f of lowest degree in $F[x]$ such that $f(\alpha) = 0$.

Definition 19.4. If $\alpha \in K$ is algebraic over F, the *degree* of α over F is the of its irreducible polynomial.

Question 19.5. Why is the "irreducible polynomial" of α, as defined above, irreducible and unique?

Proposition 19.6. *The irreducible polynomial of α is the monic polynomial $f \in F[x]$ such that (f) is the kernel of the evaluation homomorphism*

$$\varphi\colon F[x] \to K, \quad x \mapsto \alpha.$$

In other words, if $g \in F[x]$ such that $g(\alpha) = 0$, then f divides g.

Example 19.7.

1. $\sqrt{2}$ and i are algebraic over $\mathbb{Q}$. Their irreducible polynomials over $\mathbb{Q}$ are $x^2 - 2$ and $x^2 + 1$, respectively.
2. π and e are transcendental over $\mathbb{Q}$.
3. The irreducible polynomial of $3 + 4i$ over $\mathbb{Q}$ is $x^2 - 8x + 25$.
4. The irreducible polynomial of $\sqrt{i}$ over $\mathbb{Q}$ is $x^4 + 1$, but the irreducible polynomial of $\sqrt{i}$ over $\mathbb{Q}(i)$ is $x^2 + i$.

As seen in 4. above, the irreducible polynomial of α over F depends on the field of coefficients F.

Proposition 19.8. *Let K/F be a field extension. If $\alpha \in K$ is algebraic over F with irreducible polynomial f, then $F[\alpha] \simeq F[x]/(f)$, and $F[\alpha]$ is a field. Therefore, $F[x] = F(\alpha)$.*

Proof. As noted above, the homomorphism

$$\varphi\colon F[x] \to K, \quad x \mapsto \alpha$$

has kernel (f). It is clear that the image of φ is $F[\alpha]$. Thus, by the first isomorphism theorem, φ induces a homomorphism $F[x]/(f) \to F[\alpha]$.

Since f is irreducible, (f) is a maximal ideal of $F[x]$, so $F[x]/(f)$ is a field. Thus, $F[\alpha]$ is a field. $\square$

More explicitly, if α has irreducible polynomial $f(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0$, then every element $\beta \in F[\alpha]$ can be expressed uniquely in the form

$$\beta = a_{n-1}\alpha^{n-1} + \cdots + a_0,$$

for some $a_0, \ldots, a_{n-1} \in F$.

Let $g(x) = a_{n-1}x^n + \cdots + a_0$. If $g \neq 0$, then since f is irreducible and $\deg g < \deg f$, the polynomials f and g are coprime, so there exist $r, s \in F[x]$ such that

$$rf + sg = 1.$$

Then if $s(x) = b_m x^m + \cdots + b_0$, then

$$\gamma = b_m \alpha^m + \cdots + b_0$$

is the multiplicative inverse of β.

Example 19.9 (Euclidean algorithm)**.** The polynomials r and s discussed above can be computed using repeated division with remainder. For example, if $f = x^3 + 2x^2 + 1$, and $g = x^2 + 1$, then

$$x^3 + 2x^2 + 1 = (x+2)(x^2+1) + (-x-1),$$

and

$$x^2 + 1 = (-x+1)(-x-1) + 2,$$

so

$$\begin{aligned} 2 &= (x^2+1) + (x-1)(-x-1) \\ &= (x^2+1) + (x-1)(x^3+2x^2+1 - (x+2)(x^2+1)) \\ &= (x-1)(x^3+2x^2+1) - (x^2+x-3)(x^2+1). \end{aligned}$$

So we can take $r = (x-1)/2$ and $s = -(x^2+x-3)/2$.

Proposition 19.10. *If $\alpha_1, \ldots, \alpha_n \in K$ are algebraic over F, then $F[\alpha_1, \ldots, \alpha_n] = F(\alpha_1, \ldots, \alpha_n)$.*

Proof. We use induction:

$$F[\alpha_1, \ldots, \alpha_n] = F[\alpha_1, \ldots, \alpha_{n-1}][\alpha_n] = F(\alpha_1, \ldots, \alpha_{n-1})[\alpha_n] = F(\alpha_1, \ldots, \alpha_n).$$

□

19.3 Degree of field extension [15.3]

- If K/F is a field extension, then K is an F-vector space.

Definition 19.11. The *degree* of a field extension K/F is the dimension of K as an F-vector space. It is denoted $[K : F]$.

A *finite extension* is a field extension of finite degree.

Proposition 19.12. *Let K/F be a field extension, and let $\alpha \in K$. Then α is algebraic over F if and only if $F(\alpha)$ has finite degree over F. If α is algebraic, then $[F(\alpha) : F]$ is equal to the degree of α over F.*

Proof.

1. If α is algebraic, with irreducible polynomial f of degree n, then

$$1, \alpha, \alpha^2, \ldots, \alpha^{n-1}$$

is a basis of $F(\alpha)$, so $[F(\alpha) : F] = n$.

2. If $[F(\alpha) : F] = n$, then then $n + 1$ elements

$$1, \alpha, \alpha^2, \ldots, \alpha^n$$

must be linearly dependent over F. This means that there exist $c_0, \ldots, c_n \in F$, not all zero, such that

$$c_n \alpha^n + \cdots + c_0 = 0.$$

Then α is a root of the polynomial $f(x) = c_n x^n + \cdots + c_0$, so it is algebraic over F. □

Theorem 19.13. *Let $F \subseteq K \subseteq L$ be field extensions. Then*

$$[L : F] = [L : K][K : F].$$

Proof. This theorem holds even when some of the extensions have infinite degree. We only discuss the proof when all degrees are finite.

Suppose $\{\alpha_1, \ldots, \alpha_n\} \subseteq K$ is a basis of K/F, and $\{\beta_1, \ldots, \beta_m\} \subseteq L$ is a basis of L/K. We claim that the mn elements $\alpha_i \beta_j$ for $1 \leq i \leq n$, $1 \leq j \leq m$ form a basis for L/F.

1. First, we show that $\{\alpha_i \beta_j\}$ spans L as an F-vector space. Let $\gamma \in L$. We know that

$$\gamma = b_1 \beta_1 + \cdots b_m \beta_m$$

for some $b_1, \ldots, b_m \in K$. This is because $\{b_j\}$ spans L as a K-vector space.

Then for each b_j, there exists $a_{1j}, \ldots, a_{nj} \in F$ such that

$$b_j = a_{1j} \alpha_1 + \cdots + a_{nj} \alpha_n,$$

since $\{a_i\}$ spans K as an F-vector space.

Combining the above two displayed equations gives

$$\gamma = \sum_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}} a_{ij} (\alpha_i \beta_j).$$

Thus, every $\gamma \in L$ can be written as a linear combination of the elements $\alpha_i \beta_j$ with coefficients in F.

2. Now, we show that the $\alpha_i \beta_j$ are linearly independent. Suppose that

$$\gamma = \sum_{\substack{1 \leq i \leq n \\ 1 \leq j \leq m}} a_{ij} (\alpha_i \beta_j) = 0,$$

for $a_{ij} \in F$. We combine terms to obtain

$$\sum_{1 \leq j \leq m} \left(\sum_{1 \leq i \leq n} a_{ij} \alpha_i \right) \beta_j = 0.$$

By linear independence of the β_j, we have, for all j,

$$\sum_{1 \le i \le n} a_{ij}\alpha_i = 0.$$

Then by linear independence of the α_i, we have $a_{ij} = 0$ for all i and j. This shows that the elements $\alpha_i\beta_j$ are linearly independent. Thus, they form a basis.

□

Corollary 19.14. *Let K/F be a finite extension of degree n, and let $\alpha \in K$. Then α is algebraic over F and its degree divides n.*

Proof. We have $F \subseteq F(\alpha) \subseteq K$. Then

$$n = [K : F] = [K : F(\alpha)][F(\alpha) : F],$$

so $F(\alpha)/F$ is a finite extension with degree dividing n. Thus, by Proposition 19.12, α is algebaric over F with degree dividing n. □

Example 19.15. Let

$$\omega = \frac{-1 + i\sqrt{3}}{2}$$

be a third root of unity, i.e., the 3 roots of the polynomial $x^3 - 1 = (x-1)(x^2 + x + 1)$ are 1, ω, and ω^2.

Let $\alpha = \sqrt[3]{2}$. Then the roots of the polynomial $x^3 - 2$ are

$$\alpha_1 = \alpha,\ \alpha_2 = \omega\alpha,\ \alpha_3 = \omega^2\alpha.$$

Then $\mathbb{Q}(\alpha_1, \alpha_2) = \mathbb{Q}(\alpha_1, \omega)$.

Note that α_1 has degree 3 over $\mathbb{Q}$, and ω has degree 2 over $\mathbb{Q}(\alpha_1)$. Thus, by the multiplicative property of degree for the extensions

$$\mathbb{Q} \subseteq \mathbb{Q}(\alpha_1) \subseteq \mathbb{Q}(\alpha_1, \omega),$$

we have

$$[\mathbb{Q}(\alpha_1, \alpha_2) : \mathbb{Q}] = 6.$$

20 Apr 6: Adjoining roots of polynomials, finite fields

20.1 Algebraic elements [15.3]

Corollary 20.1. *Let K/F be a field extension, and suppose $\alpha, \beta \in K$ are algebraic over F. Then $\alpha + \beta$ and $\alpha\beta$ are also algebraic over F.*

Proof. Recall that $\alpha \in K$ is algebraic over F if and only the degree $[F(\alpha) : F]$ is finite. Thus, it suffices to show that $F(\alpha+\beta)$ and $F(\alpha\beta)$ are finite extensions of F.

Consider the field extensions

$$F \subseteq F(\alpha) \subseteq F(\alpha, \beta).$$

Since α and β are algebraic over F, both $[F(\alpha) : F]$ and $[F(\alpha, \beta) : F(\alpha)]$ are finite, so by the multiplicative property of degree, $[F(\alpha, \beta) : F]$ is also finite.

Since $\alpha + \beta$ and $\alpha\beta$ are contained in $F(\alpha, \beta)$, the extensions $F(\alpha + \beta)$ and $F(\alpha\beta)$ are also finite over F. □

20.2 Irreducible polynomials [15.4]

More explicitly, we can show that $\alpha + \beta$ is algebraic by finding a linear dependence among the elements

$$1, \alpha + \beta, (\alpha + \beta)^2, \ldots$$

If α has degree over F and β has degree m over F, then we can write each power of $(\alpha+\beta)$ as a linear combination of the products $\alpha^i\beta^j$ for $0 \le i < n$ and $0 \le j < m$.

Example 20.2. The powers of $\gamma = \sqrt{2} + \sqrt{3}$ are

$$1, \quad \gamma = \sqrt{2} + \sqrt{3}, \quad \gamma^2 = 5 + 2\sqrt{6}, \quad \gamma^3 = 11\sqrt{2} + 9\sqrt{3}, \quad \gamma^4 = 49 + 20\sqrt{6}.$$

The elements $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$ form a basis of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Thus, we can see that the powers of γ satisfy the linear dependence

$$\gamma^4 - 10\gamma^2 + 1 = 0,$$

while $1, \gamma, \gamma^2, \gamma^3$ are linearly independent. Thus, the irreducible polynomial of γ is

$$f(x) = x^4 - 10x^2 + 1.$$

This polynomial factors as

$$f(x) = (x - (\sqrt{2} + \sqrt{3}))(x - (\sqrt{2} - \sqrt{3}))(x - (-\sqrt{2} + \sqrt{3}))(x - (-\sqrt{2} - \sqrt{3})).$$

In general, the monomials $\alpha^i\beta^j$ span $F(\alpha, \beta)$, but they do not necessarily form a basis of $F(\alpha, \beta)$. The degree of $F(\alpha, \beta)$ might be less than mn, as we saw for the extension $\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2}\omega)$ from last lecture, which has degree 6 (not 9) over $\mathbb{Q}$.

20.3 Adjoining roots [15.6]

Lemma 20.3. *Let $f \in F[x]$ be an* irreducible *polynomial. Then $F[x]/(f)$ is a field extension of F in which f has a root.*

Proof. Since f is irreducible, (f) is maximal, so $F[x]/(f)$ is a field. It contains a copy of F consisting of (the residues of) the constant polynomials in $F[x]$. The residue $\overline{x} \in F[x]/(f)$ is a root of f:

$$f(\overline{x}) = \overline{f(x)} = 0 \in F[x]/(f). \qquad \square$$

Definition 20.4. Let K/F be a field extension, and let $f \in F[x]$. We say that f *splits in* K if it factors into linear factors in K.

Example 20.5.

1. Every polynomial $f \in \mathbb{C}[x]$ splits in $\mathbb{C}$.
2. The polynomial $x^3 - 2$ has a root in $\mathbb{Q}(\sqrt[3]{2})$, but it does not split in $\mathbb{Q}(\sqrt[3]{2})$, since its other two roots are $\sqrt[3]{2}\omega$ and $\sqrt[3]{2}\omega^2$, which are not contained in $\mathbb{Q}(\sqrt[3]{2})$.
3. The polynomial $x^4 - 1$ splits in $\mathbb{Q}(i)$, since

$$x^4 - 1 = (x-1)(x-i)(x+1)(x+i).$$

Proposition 20.6. *Let F be a field, and let $f \in F[x]$ be any polynomial. There exists a field extension K of F in which f splits.*

Proof. We prove this by induction on $\deg f$. For linear polynomials f, the proposition is clear because we can just take $K = F$.

Suppose $\deg f > 1$, and let g be an irreducible factor of f. By Lemma 20.3 applied to g, there exists a field extension K_1/F such that g (and thus f) has a root $\alpha \in K_1$.

Then $f(x) = (x - \alpha)f_1(x)$ for some polynomial $f_1(x) \in F_1[x]$ with $\deg f_1 = \deg f - 1$. By induction on the degree, there exists a field extension K/F_1 such that $f_1(x)$ splits in K. Then f also splits in K by construction of f_1. $\square$

20.4 Multiple roots [15.6]

We first note the following facts about polynomial factorizations in extensions.

Proposition 20.7 (Proposition 15.6.4 of Artin)**.** *Let K/F be a field extension, and let $f, g \in F[x]$.*

1. *The greatest common divisor of f and g is the same, whether computed in $F[x]$ or $K[x]$.*
2. *If f and g have a common root in K, then they are not relatively prime in $F[x]$.*

Proof sketch. This essentially follows from the fact that the (monic) greatest common divisor of f and g is the monic polynomial d of minimal degree such that $d = pf + qg$ for some $p, q \in F[x]$. $\square$

Definition 20.8. Let K/F be a field extension, and let $f \in F[x]$. An element $\alpha \in K$ is called a *multiple root* of f if $(x - \alpha^2)$ divides f.

Proposition 20.9. *Let $f \in F[x]$. Then f has a multiple root in some field extension of F if and only if f and its derivative f' are not relatively prime.*

Proof. Let K be any extension of F. If $\alpha \in K$ is a root of f, then $f(x) = (x - \alpha)g(x)$ for some $g \in F[x]$. Note that α being a multiple root of f is the same as α being a root of g.

We compute the derivative

$$f'(x) = (x - \alpha)g'(x) + g(x),$$

so $f'(\alpha) = g(\alpha)$. Thus α is a root of g if and only if it is also a root of f'.

We conclude that f has a multiple root in some extension K if and only if f and f' have a common root in K, which is equivalent to f and f' having a common factor in $F[x]$ by Proposition 20.7 $\square$

Corollary 20.10. *Let f be an* irreducible polynomial *in $F[x]$. Then f has no multiple root in any field extension of F unless $f' = 0$.*

Proof. If f is irreducible, then f and f' must be relatively prime unless $f' = 0$, so this follows from the previous proposition. $\square$

20.5 Finite fields [15.7]

Recall that for any prime p, the ring $\mathbb{Z}/p\mathbb{Z}$ is a field, denoted $\mathbb{F}_p$ with exactly p elements.

Recall also that if $f \in \mathbb{F}_p$ is an irreducible polynomial of degree r, then we can construct another field $F = \mathbb{F}_p[x]/(f)$, which has degree r over $\mathbb{F}_p$. Since F is an r-dimensional vector space over $\mathbb{F}_p$, it has exactly p^r elements.

Example 20.11. The quadratic $x^2 + x + 1$ is an irreducible polynomial over $\mathbb{F}_2$. Thus,

$$\mathbb{F}_2[x]/(x^2 + x + 1)$$

is a field with 4 elements. It is **not** isomorphic to $\mathbb{Z}/4\mathbb{Z}$, which is not a field!

It turns out that all fields with exactly 4 elements are isomorphic to this one. More generally, there is exactly one finite field with q elements for every prime power $q = p^r$, and this is the complete classification of fields with finitely many elements.

Lemma 20.12. *If F is a field with finitely many elements, then it has positive characteristic p. Furthermore, F is an extension of $\mathbb{F}_p$, and it has p^r elements, where $r = [F : \mathbb{F}_p]$.*

Proof. A field must have either prime or zero characteristic, and it is clear that finite fields must have positive characteristic (e.g. by Lagrange's theorem applied to the additive group F).

If F has characteristic p, then the elements

$$0, 1, \ldots, p-1$$

form a subfield of F isomorphic to $\mathbb{F}_p$. Then F is an r-dimensional $\mathbb{F}_p$-vector space, so its elements can be identified with r-dimensional column vectors with entries in $\mathbb{F}_p$, of which there are p^r. □

Theorem 20.13. *Let $q = p^r$, where p is a prime and $r \geq 1$ is an integer.*

1. *Let F be a field with q elements. The elements of F are roots of the polynomial $x^q - x$.*
2. *Let F be a field with q elements. The multiplicative group $F^\times$ of nonzero elements of F is isomorphic to $\mathbb{Z}/(q-1)\mathbb{Z}$.*
3. *There is a unique field with exactly q elements. We denote it by $\mathbb{F}_q$.*
4. *$\mathbb{F}_{p^r}$ contains a subfield isomorphic to $\mathbb{F}_{p^k}$ if and only if k divides r.*
5. *The irreducible factors of the polynomial $x^q - x$ in $\mathbb{F}_p[x]$ are precisely the irreducible polynomials in $\mathbb{F}_p[x]$ of degree d dividing r.*

Example 20.14. Here are some examples of the factorization of $x^q - x$ over $\mathbb{F}_p$, for $p = 2$.

1. $r = 2$, $q = 4$: The factors of $x^4 - x$ are the irreducible degree 1 and degree 2 polynomials in $\mathbb{F}_2[x]$. We have

   $$x^4 - x = x(x-1)(x^2 + x + 1).$$

   The only irreducible polynomial of degree 2 is x^2+x+1. The factorization contains x and $x - 1$ because $\mathbb{F}_4$ contains $\mathbb{F}_2$.

2. $r = 3$, $q = 8$: The factors of $x^8 - x$ are the irreducible degree 1 and degree 3 polynomials in $\mathbb{F}_3[x]$. We have

   $$x^8 - x = x(x-1)(x^3 + x + 1)(x^3 + x^2 + 1).$$

   The irreducible quadratic polynomail $x^2 + x + 1$ does not appear in this factorization. This corresponds to the fact that $\mathbb{F}_8$ does not contain $\mathbb{F}_4$, because $2 = [\mathbb{F}_4 : \mathbb{F}_2]$ does not divide $3 = [\mathbb{F}_8 : \mathbb{F}_2]$.

   Note that there are two different irreducible cubics $f_1 = x^3 + x + 1$ and $f_2 = x^3 + x^2 + 1$. According to the theorem, the fields $\mathbb{F}_2[x]/(f_1)$ and $\mathbb{F}_2[x]/(f_2)$ are isomorphic.

Proof of Theorem 20.13.

1. By Lagrange's theorem applied to the abelian group $F^\times$ of nonzero elements of F under multiplication, which has $q-1$ elements, we have $\alpha^{q-1} = 1$ for all $\alpha \neq 0$.

   Thus, $\alpha^q - \alpha = 0$ for all $\alpha \in F$.

2. By the structure theorem for finite abelian groups,

$$F^\times \cong \mathbb{Z}/s_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/s_m\mathbb{Z}$$

   for some positive integers $s_1|s_2|\cdots|s_m$. We would like to show that $m = 1$ and $s_1 = q-1$.

   The key observation is that by the divisibility condition, every element of the direct sum has order dividing s_m. In other words, every element $\alpha \in F^\times$ satisfies $\alpha^{s_m} = 1$.

   But the degree s_m polynomial $x^{s_m} - 1$ can have at most s_m roots, while $F^\times$ has $q-1$ elements. Thus, we must have $s_m = q-1$.

3. (i) (Existence). Let $K/\mathbb{F}_p$ be a field extension of in which $f = x^q - x$ splits completely. Note that

$$f = q \cdot x^{q-1} - 1 = -1,$$

   which is coprime to f, so f has no multiple roots. Let F be the *set* of all q roots of $x^q - x$ in K. We claim that F is a subfield of K.

   - "Freshman's dream" works in characteristic p. If $a, b \in F$, then

$$(a+b)^p = a^p + b^p.$$

   Thus, if $\alpha, \beta \in F$, then $\alpha^q = \alpha$ and $\beta^q = \beta$, so

$$(\alpha+\beta)^q = \alpha+\beta, \quad (\alpha\beta)^q = \alpha\beta, \quad (\alpha^{-1})^q = \alpha^{-1}.$$

   This shows that $\alpha+\beta$, $\alpha\beta$, and α^{-1} are all in F as well. In addition $-1 \in F$, so F is a field.

   (ii) (Uniqueness). Now assume F and F' is both fields with exactly q elements. We want to show that they are isomorphic.

   Let $\alpha \in F$ be a generator of the multiplicative group $F^\times$, which exists since $F^\times$ is isomorphic to $\mathbb{Z}/(q-1)\mathbb{Z}$. Then $F = \mathbb{F}_p(\alpha)$. Let f be the irreducible polynomial of α over $\mathbb{F}_p$. Then F is isomorphic to $\mathbb{F}_p[x]/(f)$.

   Since α is a root of $x^q - x$, we have that f divides $x^q - x$. By part 1. of the theorem, $x^q - x$ splits completely in F', so all of its factors also split completely in F'. Thus, f has a root $\alpha' \in F'$, and $\mathbb{F}_p(\alpha')$ is isomorphic to $\mathbb{F}_p[x]/(f)$.

   However, $\mathbb{F}_p[x]/(f) \simeq F$ has degree r over $\mathbb{F}_p$, so in fact

$$F' = \mathbb{F}_p(\alpha') \simeq \mathbb{F}_p[x]/(f)$$

   as well.

4. Since $[\mathbb{F}_{p^r} : \mathbb{F}_p] = r$ and $[\mathbb{F}_{p^k} : \mathbb{F}_p] = k$, by the multiplicative property of degree, if $\mathbb{F}_{p^k}$ is a subfield of $\mathbb{F}_{p^r}$, we must have $k|r$. We omit the proof of the other direction.

5. Omitted. (You can find the proof on page 462 of Artin.)

$\square$

21 Apr 8: Splitting fields

21.1 Galois theory [16.4]

Example 21.1. The roots of quadratic polynomials come in pairs:

1. If $1+2i$ is a root of a polynomial $f \in \mathbb{R}[x]$, then $1-2i$ is also a root of f.
2. If $3+5\sqrt{2}$ is a root of a polynomial $f \in \mathbb{Q}[x]$, then $3-5\sqrt{2}$ is also a root of f.

More generally, the roots of cubics come in triples, the roots of degree n polynomials come in n-tuples... But the relationships between the roots can be more complicated.

Definition 21.2. Let K/F and K'/F be field extensions. An *F-isomorphism* between these two extensions is a field homomorphism $\sigma\colon K \to K'$ such that $\sigma(x) = x$ for all $x \in F$.

Definition 21.3. An *automorphism* of K/F is an F-isomorphism from K/F to itself. The automorphisms of K/F form a group, which we denote by $\mathrm{Aut}(K/F)$.

Automorphisms of field extensions are how we think about the "relationships between roots" of a polynomial.

Observation 21.4. Let K/F be an extension, and let $f \in F[x]$. Let $\sigma\colon K \to K$ be an automorphism of K/F. Suppose $\alpha \in K$ is a root of f. Then $\sigma(\alpha)$ is also a root of f.

Therefore, automorphisms of K/F must permute the roots of f in K.

Example 21.5.

1. $\mathrm{Aut}(\mathbb{C}/\mathbb{R})$ is a group of order 2 consisting of the identity and complex conjugation $z \mapsto \overline{z}$.
2. $\mathrm{Aut}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$ is also a group of order 2, consisting of the identity and the field homomorphism
$$\sigma(a+b\sqrt{2}) = a - b\sqrt{2}, \quad a, b, \in \mathbb{Q}.$$

Example 21.6.

1. The roots of $x^3 - 2$ are
$$\alpha_1 = \alpha,\ \alpha_2 = \alpha\omega,\ \alpha_3 = \alpha\omega^2$$
for $\alpha = \sqrt[3]{2}$ and $\omega = e^{2\pi i/3}$ a third root of unity.

   Let $K = \mathbb{Q}(\alpha_1, \alpha_2)$ be the field we get by adjoining all of the roots. An automorphism
$$\sigma\colon \mathbb{Q}(\alpha_1, \alpha_2) \to \mathbb{Q}(\alpha_1, \alpha_2)$$

is determined by the values of $\sigma(\alpha_1) = \alpha_i$ and $\sigma(\alpha_2) = \alpha_j$. It turns out that we can pick any $i \neq j$ and this gives an automorphism.

In other words, σ is determined by the permutation it induces on the set of roots $\{\alpha_1, \alpha_2, \alpha_3\}$, and any permutation occurs. So $\mathrm{Aut}(K/\mathbb{Q}) \simeq S_3$.

2. The polynomial $x^4 + x^3 + x^2 + x + 1$ is irreducible over $\mathbb{Q}$. Its roots are

$$\zeta, \zeta^2, \zeta^3, \zeta^4,$$

where $\zeta = e^{2\pi i/5}$ is a fifth root of unity.

Let $K = \mathbb{Q}(\zeta)$. Then an automorphism σ is determined by the value of $\sigma(\zeta)$.

In particular, $\sigma(\zeta)$ determines $\sigma(\zeta^2)$, $\sigma(\zeta^3)$, $\sigma(\zeta^4)$, so we cannot permute the roots however we like. Even though there are 4 roots, far less than $4! = 24$ permutations can occur.

There are 4 automorphisms $\sigma(\zeta) = \zeta^k$ for $k = 1, 2, 3, 4$. They form a group isomorphic to $(\mathbb{Z}/5\mathbb{Z})^\times \simeq \mathbb{Z}/4\mathbb{Z}$.

Example 21.7. The group $\mathrm{Aut}(\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q})$ contains only the identity map. This is because if

$$\sigma \colon \mathbb{Q}(\sqrt[3]{2}) \to \mathbb{Q}(\sqrt[3]{2})$$

is a field homomorphism, then $\sigma(\sqrt[3]{2})$ must also be a cube root of 2. The only cube root of 2 in $\mathbb{Q}(\sqrt[3]{2})$ (which is a subfield of $\mathbb{R}$) is the real cube root $\sqrt[3]{2}$.

21.2 Primitive element theorem [15.8]

Theorem 21.8. *Let F be a field of characteristic* 0*. Every finite extension K/F has a* primitive element*, i.e., an element $\alpha \in K$ such that $K = F(\alpha)$.*

Example 21.9. A primitive element of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ would be $\sqrt{2} + \sqrt{3}$.

21.3 Splitting fields [16.3, 16.4]

Definition 21.10. Let $f \in F[x]$ be any polynomial (not necessarily ireducible). A *splitting field* for f over K is a field extension K/F such that f splits completely in K, say

$$f = (x - \alpha_1) \cdots (x - \alpha_n), \quad \alpha_i \in K,$$

and $K = F(\alpha_1, \ldots, \alpha_n)$.

Example 21.11. $\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2}\omega)$ is a splitting field for $x^3 - 2$ over $\mathbb{Q}$.

Remark 21.12. If we are looking for a splitting field of a polynomial f inside some ambient field, e.g., $\mathbb{C}$, it is very straightforward to find one. As above, we just adjoin all of the roots of f, and in fact, this is the only splitting field.

The following lemma and proposition are concerned with an abstract setting, in which we do not initially have a field like $\mathbb{C}$ in which we can find all roots.

Lemma 21.13. *Every polynomial $f \in F[x]$ has a splitting field.*

Proof. We showed last time that there exists a field K' in which f splits completely. Then we can take the subfield $K = F(\alpha_1, \ldots, \alpha_n) \subseteq K'$ where $\alpha_1, \ldots, \alpha_n$ are the roots of f in K'. □

From now on, we assume that our fields have characteristic 0.

Proposition 21.14. *Let $f \in F[x]$. Any two splitting fields of f are isomorphic.*

Example 21.15. The fields $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}[x]/(x^2 - 2)$ are both splitting fields for $x^2 - 2$. They are not the same field, but they are isomorphic.

Proof of Proposition 21.14. Let K_1 and K_2 be two splitting fields of f.

1. Since $K_1 = F(\alpha_1, \ldots, \alpha_n)$ where the α_i are the roots of f in K_1, it is a finite extension of F. Thus, it has a primitive element $\gamma \in K_1$ such that $K_1 = F(\gamma)$.

2. Let g be the irreducible polynomial of γ over F. Let L be an extension of K_2 which contains a root γ' of g. Then $F(\gamma)$ is isomorphic to $F(\gamma') \subseteq L$ via the isomorphism $\gamma \mapsto \gamma'$.

3. Since $F(\gamma) = K_1$ is a splitting field of f, $F(\gamma')$ is also a splitting field of f.

4. However, this means that the roots $\alpha_1, \ldots, \alpha_n$ of f in L are contained in $F(\gamma')$ and generate $F(\gamma')$. But the same is true for K_2, since K_2 is a splitting field of f which is contained in L, so $K_2 = F(\gamma') \simeq K_1$.

□

21.4 Symmetric polynomials [16.1]

Let $R[u_1, \ldots, u_n]$ be a polynomial ring in the n variables $u_1, \ldots, u_n$. The symmetric group S_n acts on $f \in R[u_1, \ldots, u_n]$ by permuting the variables:

$$\sigma(f(u_1, \ldots, u_n)) = f(u_{\sigma(1)}, \ldots, u_{\sigma(n)}).$$

Example 21.16. If $f = u_1 u_2^3 + u_3^2$, and $\sigma = (1\ 2)$, then

$$\sigma(f) = u_2 u_1^3 + u_3^2.$$

If $\tau = (1\ 3\ 2)$, then

$$\tau(f) = u_3 u_1^3 + u_2^2.$$

Definition 21.17. A polynomial $f \in R[u_1, \ldots, u_n]$ is *symmetric* if it is fixed by *every* permutation in S_n.

Example 21.18. The polynomial

$$f = u_1 u_2^3 + u_1 u_3^3 + u_2 u_1^3 + u_2 u_3^3 + u_3 u_1^3 + u_3 u_2^3$$

is symmetric.

Definition 21.19. The *elementary symmetric functions* are the symmetric polynomials

$$s_1 = \sum_i u_i = u_1 + u_2 + \cdots + u_n$$
$$s_2 = \sum_{i<j} u_i u_j = u_1 u_2 + u_1 u_3 + \cdots$$
$$s_3 = \sum_{i<j<k} u_i u_j u_k = u_1 u_2 u_3 + \cdots$$

Theorem 21.20. *Any symmetric polynomial $f \in R[u_1, \ldots, u_n]$ can be expressed uniquely as a polynomial in the symmetric functions $s_1, \ldots, s_n$.*

Example 21.21. The symmetric polynomial $u_1^2 + u_2^2 + u_3^2$ is equal to

$$(u_1 + u_2 + u_3)^2 - 2(u_1 u_2 + u_1 u_3 + u_2 u_3) = s_1^2 - 2s_2.$$

21.5 Discriminant [16.2]

Let $f \in F[x]$, and let K/F be an extension of F in which f factors as

$$f = (x - \alpha_1) \cdots (x - \alpha_n), \quad \alpha_i \in K.$$

Lemma 21.22. *Suppose f is given by*

$$f = x^n - a_1 x^{n-1} + a_2 x^{n-2} - \cdots \pm a_n.$$

Then $a_i = s_i(\alpha_1, \ldots, \alpha_n)$.

Definition 21.23. The *discriminant* of f is defined to be

$$\Delta(f) = \prod_{i<j} (\alpha_i - \alpha_j)^2.$$

Note that the product

$$\prod_{i<j} (u_i - u_j)^2$$

is a symmetric polynomial in $u_1, \ldots, u_n$. Thus, it can be expressed as a polynomial in the symmetric functions $s_i(u_1, \ldots, u_n)$. But by the previous lemma, when we evaluate the s_i at $(\alpha_1, \ldots, \alpha_n)$, we get the coefficients of f.

- The discriminant of f can be expressed as a polynomial in the coefficients of f. It can be defined this way even when f does not factor into linear factors.

Example 21.24. For quadratic polynomials, we have

$$(u_1 - u_2)^2 = (u_1 + u_2)^2 - 4u_1 u_2 = s_1^2 - 4s_2.$$

Thus, for $f = x^2 - a_1 x + a_2$, we have

$$\Delta(f) = (\alpha_1 - \alpha_2)^2 = (\alpha_1 + \alpha_2)^2 - 4\alpha_1 \alpha_2 = a_1^2 - 4a_2.$$

Example 21.25. There is a formula for the discriminant of a cubic polynomial of the form

$$f = x^3 + px + q.$$

It is

$$\Delta(f) = -4p^3 - 27q^2.$$

22 Apr 13: Galois extensions

We assume that all of our fields have characteristic 0.

Definition 22.1. A *Galois extension* is a finite extension K/F such that $|\operatorname{Aut}(K/F)| = [K : F]$. If K/F is Galois, we write $\operatorname{Gal}(K/F)$ instead of $\operatorname{Aut}(K/F)$.

We saw last time that

$$\mathbb{C}/\mathbb{R}, \quad \mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2}\omega)/\mathbb{Q}, \quad \mathbb{Q}(\zeta)/\mathbb{Q}$$

are Galois extensions (where $\omega = e^{2\pi i/3}$ and $\zeta = e^{2\pi i/5}$), while $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not a Galois extension.

22.1 Fixed fields [16.5]

Definition 22.2. Let K be a field, and let H be a group of automorphisms of K. The *fixed field* of H is defined by

$$K^H := \{\alpha \in K : \sigma(\alpha) = \alpha \quad \forall \sigma \in H\}.$$

It is the subfield of K consisting of all elements of K which are fixed by *every* element of H.

- By definition, we always have $H \subseteq \operatorname{Aut}(K/K^H)$. In this subsection, we prove that $H = \operatorname{Aut}(K/K^H)$ and K/K^H is Galois.

Example 22.3. Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Then K has an automorphism σ given by

$$\sqrt{2} \mapsto -\sqrt{2}, \quad \sqrt{3} \mapsto \sqrt{3}.$$

Let $H = \{1, \sigma\}$. Then the fixed field of H is

$$K^H = \mathbb{Q}(\sqrt{3}).$$

Theorem 22.4. *Let H be a finite group of automorphisms of a field K, and let $\beta \in K$. Let*

$$\{\beta = \beta_1, \ldots, \beta_r\}$$

be the orbit of β under the action of H. Then β is algebraic over K^H, and its irreducible polynomial is

$$g(x) = (x - \beta_1) \cdots (x - \beta_r).$$

Proof. Note that any $\sigma \in H$ permutes the set $\{\beta_1, \ldots, \beta_r\}$. Since the coefficients of g are symmetric functions in the β_i, they are fixed by σ for all $\sigma \in H$. Thus, the coefficients of g lie in K^H, so $g \in K^H[x]$.

It is clear that β is a root of g. On the other hand, if β is a root of any polynomial $f \in K^H[x]$, then $\beta_1, \ldots, \beta_r$ are also roots of f. Thus, $g(x)$ is the irreducible polynomial of β. $\square$

An *algebraic extension* is an extension K/F such that every element of K is algebraic over F.

Lemma 22.5. *Suppose K/F is an algebraic extension which is not finite. Then there exist elements of K of arbitrarily large degree over F.*

Theorem 22.6 (Fixed field theorem)**.** *Let H be a finite group of automorphisms of a field K, and let $F = K^H$. Then $[K : F] = |H|$.*

Proof. By the above lemma and theorem, $[K : F]$ is finite, so we can use the primitive element theorem. Suppose $K = F(\gamma)$.

Then $[K : F]$ is equal to the degree of γ over F, which by Theorem 22.4, is equal to the size of the orbit of γ under H.

Since every element of H fixes $F = K^H$, any element which also fixes γ must be the identity, since $K = F(\gamma)$. So the stabilizer of γ in H is $\{1\}$, so the orbit has size $|H|$. □

Lemma 22.7.

1. *Let K/F be a finite field extension, and let $G = \mathrm{Aut}(K/F)$. Then $|G|$ divides $[K : F]$.*
2. *Let H be a finite group of automorphsims of a field K, and let $F = K^H$. Then K/F is Galois, and $\mathrm{Gal}(K/F) = H$.*

Proof.

1. We have
$$F \subseteq K^G \subseteq K,$$
and $|G| = [K : K^G]$, which divides $[K : F]$.
2. As mentioned before, $H \subseteq \mathrm{Aut}(K/F)$, so $|H|$ divides $|\mathrm{Aut}(K/F)|$.

   From 1. and the Fixed Field Theorem, we have that $|\mathrm{Aut}(K/F)|$ divides $[K : F] = |H|$.

   Thus, $\mathrm{Aut}(K/F) = H$, so K/F is Galois by definition, since
$$|\mathrm{Aut}(K/F)| = |H| = [K : F].$$

□

22.2 Splitting fields again [16.3, 16.6]

In this subsection and the next, we prove that "Galois extensions" and "splitting fields" are the same thing.

Lemma 22.8. *Let γ_1 be a primitive element of a finite extension K/F, and let f be the irreducible polynomial of γ_1. If $\gamma_1, \ldots, \gamma_r$ are the roots of f contained in K, then there are exactly r automorphisms of K/F, given by*
$$\sigma(\gamma_1) = \gamma_i, \quad \textit{for} \quad 1 \le i \le r.$$

Theorem 22.9. *Let K/F be the splitting field of some polynomial $f \in F[x]$. Then any irreducible polynomial $g \in F[x]$ which has a root in K splits completely in K.*

Proof sketch. Let $\alpha_1, \ldots, \alpha_n$ be the roots of f in K, so $K = F(\alpha_1, \ldots, \alpha_n)$.

Suppose β is a root of g. Write $\beta = p(\alpha_1, \ldots, \alpha_n)$ for some polynomial $p \in F[u_1, \ldots, u_n]$. We claim that the other roots of g can be found among

$$p(\alpha_{\sigma(1)}, \ldots, \alpha_{\sigma(n)}), \quad \sigma \in S_n.$$

This is shown by noting that the polynomial

$$\prod_{\sigma \in S_n} (x - p(\alpha_{\sigma(1)}, \ldots, \alpha_{\sigma(n)}))$$

has coefficients which are symmetric polynomials in the α_i, so they are in F, since the elementary symmetric functions evaluated at $(\alpha_1, \ldots, \alpha_n)$ give the coefficients of f. □

Corollary 22.10. *Let K/F be a splitting field. Then it is a Galois extension.*

Proof. Let γ be a primitive element of K/F, and let f be its irreducible polynomial. Then f has degree $n = [K : F]$, and by the above theorem, f has n roots $\gamma_1, \ldots, \gamma_n$ in K. Then by Lemma 22.8, there are exactly n automorphisms of K/F as well, so K/F is Galois. □

22.3 Equivalent definitions of Galois extension [16.6]

Theorem 22.11. *Let K/F be a finite extension of fields, and let $G = \mathrm{Aut}(K/F)$. The following are equivalent.*

1. *K/F is Galois.*
2. *$K^G = F$.*
3. *K is the splitting field of some polynomial over F.*

Proof. We already proved that 2. implies 1. Furthermore, if K/F is Galois, then $[K : K^G] = |G| = [K : F]$, so $K^G = F$, so 1. implies 2.

We also have shown that 3. implies 1. Now suppose K/F is Galois, with a primitive element γ that has irreducible polynomial f, of degree $n = [K : F]$. By Lemma 22.8, f has $|G| = [K : F]$ roots in K, which means that f splits completely in K. Thus, K is the splitting field of f over F. □

Example 22.12. Let $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, $F = \mathbb{Q}$. K is the splitting field over $\mathbb{Q}$ of the polynomial $(x^2 - 2)(x^2 - 3)$, so this is a Galois extension of degree 4.

The Galois group $\mathrm{Gal}(K/F)$ consists of the four automorphisms $\{1, \sigma, \tau, \sigma\tau\}$, where

$$\sigma\colon \sqrt{2} \mapsto -\sqrt{2},\ \sqrt{3} \mapsto \sqrt{3}, \quad \tau\colon \sqrt{2} \mapsto \sqrt{2},\ \sqrt{3} \mapsto -\sqrt{3},$$

and

$$\sigma\tau\colon \sqrt{2} \mapsto -\sqrt{2},\ \sqrt{3} \mapsto -\sqrt{3}$$

is their composition. This group is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Example 22.13. Let $K = \mathbb{Q}(\sqrt[3]{2})$, $F = \mathbb{Q}$. Then $G = \mathrm{Aut}(K/F)$ is the trivial group $\{1\}$, so the fixed field $K^G = \mathbb{Q}(\sqrt[3]{2})$ is not equal to F.

22.4 Intermediate fields [16.6]

Let K/F be a field extension. An *intermediate field* is a field

$$F \subseteq L \subseteq K.$$

Note that if L is an intermediate field, then

$$\mathrm{Aut}(K/L) \subseteq \mathrm{Aut}(K/F),$$

because if an automorphism σ of K is the identity map on L, then it is also the identity map on F.

Corollary 22.14. *Suppose K/F is a Galois extension, and L is an intermediate field. Then K/L is also a Galois extension.*

Proof. We know that K/F is the splitting field over F of some polynomial $f \in F[x]$. But then clearly K is also the splitting field of f over L, so K/L is also a splitting field, so it is also Galois. $\square$

22.5 Apr 15: Main theorem, cubics

22.6 Main theorem [16.7]

Theorem 22.15. *Let K/F be a Galois extension. Let $G = \mathrm{Gal}(K/F)$. There is a bijection*

$$\{\text{subgroups of } G\} \leftrightarrow \{\text{intermediate fields of } K/F\}$$

given by

$$\begin{aligned} H &\mapsto K^H \\ \mathrm{Gal}(K/L) &\leftarrow\!\shortmid L. \end{aligned}$$

Furthermore, L/F is Galois if and only if H is a normal subgroup of G. If H is normal, then $\mathrm{Gal}(L/F)$ is isomorphic to G/H.

Corollary 22.16. *If H corresponds to L, then $[K : L] = |H|$ and $[L : F] = [G : H]$.*

Example 22.17. For any Galois extension K/F, the subgroup $\{1\}$ corresponds to K, and the subgroup $\mathrm{Gal}(K/F)$ corresponds to F.

In general, if $H \subseteq H'$ and L and L' are the corresponding intermediate fields, then $L' \subseteq L$. (The correspondence is *inclusion-reversing.*)

Example 22.18. Returning to the example $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ and $F = \mathbb{Q}$ from before, $G = \mathrm{Gal}(K/F)$ has 5 subgroups:

$$\{1\},\ \langle\sigma\rangle,\ \langle\tau\rangle,\ \langle\sigma\tau\rangle,\ G.$$

In order, the corresponding intermediate fields are

$$K, \quad \mathbb{Q}(\sqrt{3}), \quad \mathbb{Q}(\sqrt{2}), \quad \mathbb{Q}(\sqrt{6}), \quad F.$$

The Galois group is abelian; all subgroups are normal. As we can see, all the intermediate fields are Galois over $\mathbb{Q}$.

Example 22.19. Let $K = \mathbb{Q}(\alpha_1, \alpha_2)$, $F = \mathbb{Q}$, where

$$\alpha_1 = \alpha,\ \alpha_2 = \alpha\omega,\ \alpha_3 = \alpha\omega^2$$

for $\alpha = \sqrt[3]{2}$ and $\omega = e^{2\pi i/3}$. Recall that we identified $\mathrm{Gal}(K/F)$ with the symmetric group S_3 via its action on these roots.

- The subgroup of $\mathrm{Gal}(K/F)$ corresponding to $\mathbb{Q}(\sqrt[3]{2})$ is the cyclic subgroup of S_3 generated by $(2\ 3)$.
- The intermediate field corresponding to the subgroup A_3 is the quadratic extension $\mathbb{Q}(\omega)$.

22.7 Proof of main theorem [16.7]

Proof of correspondence. We want to show that the maps $H \mapsto K^H$ and $L \mapsto \mathrm{Gal}(K/L)$ are inverses.

1. Let H be a subgroup of $\mathrm{Gal}(K/F)$. We showed last time that
$$\mathrm{Gal}(K/K^H) = H.$$
2. Let L be an intermediate field, and let $H = \mathrm{Gal}(K/L)$. Since K/L is Galois,
$$K^H = L.$$

□

Proof that normal subgroups correspond to Galois extensions. Let L be any intermediate field. Let ϵ_1 be a primitive element of the extension L/F, and let g be its irreducible polynomial over F.

Note that g splits completely in K since K/F is Galois. Let
$$\epsilon_1, \ldots, \epsilon_r$$
be the roots of g in K.

- L/F is a splitting field if and only if $\epsilon_1, \ldots, \epsilon_r \in L$.

Let $G = \mathrm{Gal}(K/F)$ and $H = \mathrm{Gal}(K/L)$. Note that G acts transitively on the set $\{\epsilon_1, \ldots, \epsilon_r\}$. We consider the stabilizers of the elements ϵ_i.

- The stabilizer of ϵ_1 is equal to H. This is because $\sigma \in G$ fixes ϵ_1, if and only if it fixes $F(\epsilon_1) = L$, which is true if and only if $\sigma \in H$.

By the same reasoning:

- The stabilizer of ϵ_i is equal to H if and only if $\epsilon_i \in L$. Indeed, if ϵ_i is fixed by H, then $\epsilon_i \in L$ by definition. On the other hand, if $\epsilon_i \in L$, then it is also a primitive element, so the same argument as above applies.

However, by properties of stabilizers:

- If $\epsilon_i = \sigma(\epsilon_1)$, the stabilizer of ϵ_i is equal to $\sigma H \sigma^{-1}$.

Thus, we see that
$$\sigma(\epsilon_1) \in L \iff H = \sigma H \sigma^{-1}$$
for all $\sigma \in G$.

Thus, if L/F is Galois, then H must be normal, and by transitivity of the action, if H is normal, then L/F is Galois. □

Proof that the Galois group is the quotient. Now suppose that H is a normal subgroup of G, and $L = K^H$. Then we have a group homomorphism

$$\varphi\colon \operatorname{Gal}(K/F) \to \operatorname{Gal}(L/F), \quad \sigma \mapsto \sigma|_L,$$

defined by restricting automorphisms of K to L. The kernel of φ is exactly $H = \operatorname{Gal}(K/L)$.

By the first isomorphism theorem, G/H is isomorphic to $\operatorname{im}(\varphi)$. On the other hand, since L/F is Galois, we have

$$|\operatorname{Gal}(L/F)| = [L : F] = [K : F]/[K : L] = |G/H| = |\operatorname{im}(\varphi)|,$$

so φ is surjective, so $\operatorname{Gal}(L/F)$ is isomorphic to G/H. □

22.8 Cubics [16.8]

In this subsection, we determine the Galois groups of the splitting fields of cubics.

Theorem 22.20. *Let K be the splitting field of an (irreducible) polynomial g of degree r over F. Then* $\operatorname{Gal}(K/F)$ *acts faithfully on the roots of g, so we can view* $\operatorname{Gal}(K/F)$ *as a (transitive) subgroup of the symmetric group S_r.*

Let $f(x) = x^3 - a_1x^2 + a_2x - a_3$ be an irreducible cubic in $F[x]$ with splitting field K. Suppose $\alpha_1, \alpha_2, \alpha_3$ are the roots of f in K. We have

$$F \subseteq F(\alpha_1) \subseteq F(\alpha_1, \alpha_2) \subseteq K.$$

In fact, since $\alpha_1 + \alpha_2 + \alpha_3 = a_1 \in F$, we always have $K = F(\alpha_1, \alpha_2)$. We also always have

$$[F(\alpha_1) : F] = 3,$$

since f is irreducible over F.

However, f has a linear factor $(x - \alpha_1)$ over the field $F(\alpha_1)$, so it factors as

$$f(x) = (x - \alpha_1)g(x)$$

for some quadratic polynomial g with coefficients in $F(\alpha_1)$. Furthermore, α_2 must be a root of g, so it has degree at most 2 over $F(\alpha_1)$, so

$$[F(\alpha_1, \alpha_2) : F(\alpha_1)] = 2 \text{ or } 1.$$

Thus, the splitting field K of a cubic has degree either 3 or 6 over F. The case $[K : F] = 3$ happens when $\alpha_2, \alpha_3 \in F(\alpha_1)$.

Example 22.21. For $f = x^3 - 2$ over $\mathbb{Q}$, we can conclude that $[K : \mathbb{Q}] = 6$ because K contains the degree 3 element $\sqrt[3]{2}$ as well as the degree 2 element ω.

Alternatively, we could note that the two complex roots of f are not contained in $\mathbb{Q}(\sqrt[3]{2})$, which is a subfield of $\mathbb{R}$.

In either case, since the Galois group has order 6, $\operatorname{Gal}(K/F)$ is the whole group S_3.

Example 22.22. We can show that $f = x^3 + 3x + 1$ is a strictly increasing function of x (by checking that its derivative is always positive) for real x, so f has one real root and two complex roots. The splitting field of this cubic over $\mathbb{Q}$ also has Galois group S_3.

Example 22.23. Let $f = x^3 - 3x + 1$ (again over $\mathbb{Q}$). Let α_1 be a root of f. Let $\alpha_2 = \alpha_1^2 - 2$. We compute

$$\begin{aligned} f(\alpha_2) &= (\alpha_1^2 - 2)^3 - 3(\alpha_1^2 - 2) + 1 \\ &= \alpha_1^6 - 6\alpha_1^4 + 9\alpha_1^2 - 1 \\ &= (\alpha_1^3 - 3\alpha_1 + 1)(\alpha_1^3 - 3\alpha_1 - 1) = 0. \end{aligned}$$

Thus, α_2 is also a root of f, and in this case, since $\alpha_2 \in F(\alpha_1)$, so $[K : F] = 3$. We now note that:

- The only transitive subgroups of S_3 are S_3 and A_3.

Thus, $\mathrm{Gal}(K/\mathbb{Q}) \simeq A_3$ in this case.

23 Apr 20: Quartics, cyclotomic extensions

23.1 Cubics cont'd [16.8]

More generally, the key to determining whether or the splitting field of a cubic f has Galois group S_3 or A_3 is by considering the special element

$$\delta = (\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)(\alpha_2 - \alpha_3).$$

This is an element of K, and it is equal to the square root of the discriminant of f. Recall that the discriminant is always an element of F.

Theorem 23.1. *Let K be the splitting field of an irreducible cubic $f \in F[x]$ with discrimant D. Then* $\mathrm{Gal}(K/F)$ *is*

- *S_3 if D is not a square in F.*
- *A_3 if D is a square in F.*

Proof. For any $\sigma \in S_n$, we have

$$\sigma(\delta) = \mathrm{sign}(\sigma) \cdot \delta.$$

Thus, δ is fixed by exactly the even permutations.

If $\delta \in F$, then it must be fixed by every element of $\mathrm{Gal}(K/F)$, so $\mathrm{Gal}(K/F) = A_3$. Conversely, if $\mathrm{Gal}(K/F) = A_3$, then the Galois group fixes δ, so $\delta \in F$. $\square$

Example 23.2.

1. The discriminant of $x^3 + 3x + 1$ is $-4 \cdot 27 - 27 = -135$, which is not a square in $\mathbb{Q}$.
2. The discriminant of $x^3 - 3x + 1$ is $4 \cdot 27 - 27 = 81$, which is a square in $\mathbb{Q}$.

Proposition 23.3. *Let K be the splitting field of an irreducible degree n polynomial $f \in F[x]$ with discriminant D. Then* $\mathrm{Gal}(K/F)$ *is contained in A_n if and only if D is a square in F.*

23.2 Quartics [16.9]

Let f be an irreducible quartic in $F[x]$, and let K be its splitting field. We know that $\mathrm{Gal}(K/F)$ is a transitive subgroup of S_4 (meaning that its action on the set $\{1, 2, 3, 4\}$ is transitive). There are 5 transitive subgroups of S_4:

$$S_4, A_4, D_4, C_4, D_2.$$

(Recall that D_n is the dihedral group of order $2n$, and C_n is the cyclic group of order n). The orders of these groups are 24, 12, 8, 4, and 4, respectively.

Today, we study quartics which are the irreducible polynomials of numbers of the form $\sqrt{a + \sqrt{b}}$.

Example 23.4. Let $\alpha = \sqrt{4+\sqrt{5}}$ and $F = \mathbb{Q}$. Its irreducible polynomial is

$$f = \left(x - \sqrt{4+\sqrt{5}}\right)\left(x - \sqrt{4-\sqrt{5}}\right)\left(x + \sqrt{4+\sqrt{5}}\right)\left(x + \sqrt{4-\sqrt{5}}\right),$$

so $f = x^4 - 8x^2 + 11$. (We omit the verification that this is irreducible.)

Let $\alpha' = \sqrt{4-\sqrt{5}}$, and let

$$\alpha_1 = \alpha,\ \alpha_2 = \alpha',\ \alpha_3 = -\alpha,\ \alpha_4 = -\alpha'.$$

The splitting field K of F is equal to $F(\alpha, \alpha')$.

We know that $\mathrm{Gal}(K/F)$ has order equal to $[K : F]$, so we are first interested in the degree of this extension. Note that we have the chain of extensions

$$F \subseteq F(\alpha) \subseteq F(\alpha, \alpha') = K.$$

The first extension always has $[F(\alpha) : F] = 4$ since α is the root of an irreducible quartic. Note that α' is the root of the quadratic

$$x^2 - (4 - \sqrt{5}),$$

which has coefficients in $F(\alpha)$ since $4 - \sqrt{5} \in \mathbb{Q}(\sqrt{5}) \subseteq \mathbb{Q}(\sqrt{4+\sqrt{5}})$. Thus, $[F(\alpha, \alpha') : F(\alpha)]$ has degree at most 2.

So $[K : F]$ is either 8 or 4, depending on whether or not $\alpha' \in F(\alpha)$. To determine which of these is the case, we consider the element $\alpha\alpha' = \sqrt{11}$.

Note that $F(\alpha)$ and $F(\alpha')$ are both quadratic extensions of $F' \coloneqq \mathbb{Q}(\sqrt{5})$, with $\alpha = \sqrt{a}, \alpha' = \sqrt{b}$ for $a, b \in F'$. It can be checked manually that

$$F'(\sqrt{a}) = F'(\sqrt{b}) \iff \sqrt{a}/\sqrt{b} \in F',$$

which is equivalent to $\sqrt{ab} = \alpha\alpha' \in F'$. But $\sqrt{11} \notin \mathbb{Q}(\sqrt{5})$, so $\alpha' \notin F(\alpha)$.

Thus, $[K : F] = 8$, and the only transitive subgroup of S_4 which has order 8 is D_4, so $\mathrm{Gal}(K/F)$ is isomorphic to D_4.

Remark 23.5. As a subgroup of the symmetric group of $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$, the Galois group is the subgroup generated by

$$\sigma = (1\ 2\ 3\ 4) \text{ and } \tau = (2\ 4).$$

In fact, this subgroup contains all the permutations which satisfy $\alpha_3 \mapsto -\alpha_i$ for $\alpha_1 \mapsto \alpha_i$.

Example 23.6. Let $\alpha = \sqrt{2+\sqrt{2}}$ and $F = \mathbb{Q}$. In this case, $f = x^4 - 4x^2 + 2$. We can define $\alpha' = \sqrt{2-\sqrt{2}}$ and $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ in the same way as before.

In this case, we have $\alpha\alpha' = \sqrt{2} \in \mathbb{Q}(\sqrt{2})$, so $\alpha' \in F(\alpha)$. Thus, $[K : F] = 4$, and $\mathrm{Gal}(K/F)$ is C_4 or D_2. We now have to distinguish between the two cases.

Since the Galois group is a transitive subgroup of the permutation group of $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\}$, there exists some element $\sigma \in \mathrm{Gal}(K/F)$ such that

$$\sigma(\alpha) = \alpha'.$$

Squaring both sides gives

$$\sigma(2+\sqrt{2}) = 2-\sqrt{2},$$

so $\sigma(\sqrt{2}) = -\sqrt{2}$. This means $\sigma(\alpha\alpha') = -\alpha\alpha'$, and since $\sigma(\alpha) = \alpha'$, we have

$$\sigma(\alpha') = -\alpha.$$

So σ does not have order 2 (in fact $\sigma = (1\ 2\ 3\ 4)$ as above), and $\mathrm{Gal}(K/F) \simeq C_4$. It is the cyclic group generated by σ.

Example 23.7. Let $\alpha = \sqrt{4+\sqrt{7}}$ and $F = \mathbb{Q}$. In this case, $f = x^4 - 8x^2 + 9$. In addition, we now have $\alpha\alpha' = 3 \in F$. Again, $[K : F] = 4$, and the Galois group is either C_4 or D_2.

If $\sigma \in \mathrm{Gal}(K/F)$ satisfies $\sigma(\alpha) = \alpha'$, then $\sigma(\alpha\alpha') = \alpha\alpha'$, so $\sigma(\alpha') = \alpha$. Similarly, we can see that every element of $\mathrm{Gal}(K/F)$ has order 2, so $\mathrm{Gal}(K/F) \simeq D_2$.

It is the subgroup generated by σ^2 and τ with σ and τ as in Remark 23.5.

Remark 23.8. In the case when the Galois group is D_2, we can always write α as a sum of un-nested square roots. In the example above,

$$\sqrt{4+\sqrt{7}} = \frac{\sqrt{14}}{2} + \frac{\sqrt{2}}{2}.$$

Indeed, $K = \mathbb{Q}(\sqrt{2}, \sqrt{7})$, which we have already seen has Galois group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

23.3 Cyclotomic fields [16.10]

For any integer n, let $\zeta_n = e^{2\pi i/n}$. It is a root of the polynomial $x^n - 1$.

Proposition 23.9. *Let p be a prime. The irreducible polynomial of $\zeta = \zeta_p$ is*

$$f = x^{p-1} + \cdots + 1.$$

The roots of f are $\zeta, \zeta^2, \ldots, \zeta^{p-1}$.

Proposition 23.10. *Let p be a prime, let $\zeta = \zeta_p$, and let $K = \mathbb{Q}(\zeta)$. Then $\mathrm{Gal}(K/\mathbb{Q})$ is isomorphic to the cyclic group $(\mathbb{Z}/p\mathbb{Z})^\times$.*

Proof. For $1 \le i \le p-1$, let σ_i denote automorphism of K given by

$$\sigma_i(\zeta) = \zeta^i.$$

This is a field homomorphism because ζ and ζ^i have the same irreducible polynomial!

Since K is generated by ζ, it is clear that $\mathrm{Gal}(K/F) = \{\sigma_1, \ldots, \sigma_{p-1}\}$. The map

$$\mathrm{Gal}(K/F) \to (\mathbb{Z}/p\mathbb{Z})^\times, \quad \sigma_i \mapsto i$$

is a group homomorphism since

$$\sigma_i \circ \sigma_j(\zeta) = (\zeta^j)^i = \zeta^{ij}$$

so $\sigma_i \circ \sigma_j = \sigma_{ij}$. Finally, recall we showed that $(\mathbb{Z}/p\mathbb{Z})^\times = \mathbb{F}_p^\times$ is a cyclic group of order $p-1$. $\square$

Theorem 23.11. *In general, the irreducible polynomial of ζ_n is*

$$f = \prod_{\substack{1 \le k \le n-1 \\ \gcd(k,n)=1}} (x - \zeta_n^i).$$

The Galois group of $\mathbb{Q}(\zeta_n)$ over $\mathbb{Q}$ is isomorphic to $(\mathbb{Z}/n\mathbb{Z})^\times$.

23.4 Quadratic subextensions of $\mathbb{Q}(\zeta_p)$ [16.10]

For $p \neq 2$ a prime, note that $\mathbb{Z}/(p-1)\mathbb{Z}$ has exactly one subgroup of index 2, the subgroup $\{0, 2, \ldots, p-3\}$. By the Main Theorem, this means that there is exactly one quadratic extension of $\mathbb{Q}$ contained inside $\mathbb{Q}(\zeta_p)$.

Theorem 23.12. *Let p be an odd prime, and let L be the quadratic extension of $\mathbb{Q}$ contained in $\mathbb{Q}(\zeta_p)$. If $p \equiv 1 \pmod 4$, then $L = \mathbb{Q}(\sqrt{p})$, and if $p \equiv 3 \pmod 4$, then $L = \mathbb{Q}(\sqrt{-p})$.*

Example 23.13. Let $\zeta = \zeta_{17}$. Consider the automorphism σ_3 of $K = \mathbb{Q}(\zeta)$ given by

$$\sigma_3(\zeta) = \zeta^3.$$

We choose this one because 3 is a generator of $(\mathbb{Z}/17\mathbb{Z})^\times$. The powers $3^0, \ldots, 3^{15} \pmod{17}$, in order, are

$$1, 3, -8, -7, -4, 5, -2, -6, -1, -3, 8, 7, 4, -5, 2, 6.$$

So the index 2 subgroup of $G = \mathrm{Gal}(K/\mathbb{Q})$ is

$$H = \{1, \sigma_{-8}, \sigma_{-4}, \sigma_{-2}, \sigma_{-1}, \sigma_8, \sigma_4, \sigma_2\}.$$

Let

$$\alpha_1 = \zeta + \zeta^{-8} + \zeta^{-4} + \zeta^{-2} + \zeta^{-1} + \zeta^8 + \zeta^4 + \zeta^2,$$
$$\alpha_2 = \zeta^3 + \zeta^{-7} + \zeta^5 + \zeta^{-6} + \zeta^{-3} + \zeta^7 + \zeta^{-5} + \zeta^6.$$

Then α_1 and α_2 are both fixed by H, and they form a G-orbit, so they are the roots of the quadratic

$$x^2 - (\alpha_1 + \alpha_2)x + \alpha_1\alpha_2.$$

It is easy to see that $\alpha_1 + \alpha_2 = -1$, since the sum of all of the p-th roots of unity is 0.

On the other hand, to compute $\alpha_1\alpha_2$, we note the following lemma.

Lemma 23.14. *Let p be a prime and $\zeta = \zeta_p$. If $\alpha = c_1\zeta + c_2\zeta^2 + \cdots + c_{p-1}\zeta^{p-1}$ is a rational number, then $c_1 = \cdots = c_{p-1} = c$, and $\alpha = -c$.*

Then $\alpha_1\alpha_2$ is a sum of 64 powers of ζ, and since it is rational, the number of times each nonzero power occurs in the sum is the same. Indeed, none of the powers are zero, so

$$\alpha_1\alpha_2 = 4(\zeta^1 + \cdots + \zeta^{16}) = -4.$$

Then the irreducible polynomial of α_1 is

$$x^2 + x - 4,$$

which has discriminant 17, so it generates the extension $\mathbb{Q}(\sqrt{17})$.

Example 23.15. Let $\zeta = \zeta_{11}$. Now we take the generator 2 of $(\mathbb{Z}/11\mathbb{Z})^\times$. In this case,

$$H = \{1, \sigma_4, \sigma_5, \sigma_{-2}, \sigma_3\}.$$

We find the quadratic with roots

$$\alpha_1 = \zeta + \zeta^4 + \zeta^5 + \zeta^{-2} + \zeta^3,$$

$$\alpha_2 = \zeta^2 + \zeta^{-3} + \zeta^{-1} + \zeta^{-4} + \zeta^{-5}.$$

The product $\alpha_1\alpha_2$ is a sum of 25 powers of ζ, with ζ^0 occuring exactly 5 times, so

$$\alpha_1\alpha_2 = 5 + 2(\zeta + \cdots \zeta^{10}) = 3,$$

so the irreducible polynomial of α_1 is

$$x^2 + x + 3.$$

This has discriminant -11, so it generates the extension $\mathbb{Q}(\sqrt{-11})$.

What happens in general is that the product $\alpha_1\alpha_2$ has ζ^0 terms if and only if $-1 = a^k$ for k odd, where a is a generator of $(\mathbb{Z}/p\mathbb{Z})^\times$. But we always have $k = (p-1)/2$ because -1 is the unique order 2 element of $(\mathbb{Z}/p\mathbb{Z})^\times$, so the parity of k is determined by $p \pmod 4$.

24 Apr 22: Quintic equations

Today, we prove that there is no "quintic formula."

24.1 Kummer extensions [16.11]

Theorem 24.1. *Let p be a prime, and let F be a subfield of $\mathbb{C}$ which contains the p-th root of unity. Let K/F be a Galois extension of degree p. Then $K = F(\beta)$ for some β such that $\beta^p \in F$.*

Proof. The Galois group $\mathrm{Gal}(K/F)$ is cyclic of order p. Let σ be a generator of $\mathrm{Gal}(K/F)$. Then $\sigma^p = 1$. We view σ is a linear operator on the F-vector space K.

Theorem 24.2. *If A is a matrix such that $A^r = I$ for some $r > 0$, then A is diagonalizable over $\mathbb{C}$.*

Since $\sigma^p = 1$, all eigenvalues of σ are p-th roots of unity. In particular, all of the eigenvalues of σ are in F. Since σ is diagonalizable and not the identity matrix, it has an eigenvalue $\lambda \neq 1$.

Let $\beta \in K$ be an element such that $\sigma(\beta) = \lambda \cdot \beta$. Then

$$\sigma(\beta^p) = \lambda^p \beta^p = \beta^p,$$

so $\beta^p \in F$. Finally, $\sigma(\beta) \neq \beta$ since $\lambda \neq 1$, so $\beta \notin F$. Thus, $K = F(\beta)$ since β has degree p over F. □

Proposition 24.3. *Let F be a subfield of $\mathbb{C}$ which contains all p-th roots of unity. Let $b \in F$. The polynomial $x^p - b$ is either irreducible over F, or it splits completely.*

So if you adjoin a p-th root to F which contains all p-th roots of unity, you always get a degree p Galois extension.

24.2 Definition of solvability [16.12]

By a "quintic formula" we mean a formula which gives the roots of a degree 5 polynomial in terms of its coefficients using only radicals $\sqrt[n]{}$ and $+, -, \times, \div$. Formally, we make the following definition.

Definition 24.4. Let α be a complex number, and let F be a subfield of $\mathbb{C}$. The following conditions on α are equivalent, and if either one is satisfied, then we say that α is *solvable* over F.

1. There exists a chain of extensions $F = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_r = K$ in $\mathbb{C}$ such that $\alpha \in K$, and for $1 \leq j \leq r$, $F_j = F_{j-1}(\beta)$ for some β such that a power of β is in F_{j-1}.

2. There exists a chain of extensions $F = F_0 \subseteq F_1 \subseteq \cdots \subseteq F_s = K$ in $\mathbb{C}$ such that $\alpha \in K$, and for $1 \leq j \leq s$, the extension F_j/F_{j-1} is Galois of prime degree.

We first note that in 2., we can replace "Galois of prime degree" with "abelian extension" — an *abelian extension* is a Galois extension whose Galois group is abelian.

Lemma 24.5. *Let K/F be an abelian extension. There exists a chain of intermediate fields $F = F_0 \subseteq F_1 \subseteq \cdots F_m = K$ such that for $1 \leq j \leq m$, the extension F_j/F_{j-1} is Galois of prime degree.*

Proof. Let $G = \mathrm{Gal}(K/F)$. Since G is abelian, all of its subgroups are normal. If G does not already have prime order, then it has a proper normal subgroup H, corresponding to some intermediate field $F \subseteq L \subseteq K$ such that L/F and K/L are both abelian extensions. We repeat this process until we get the desired chain. □

Proof sketch for Definition 24.4. □

24.3 Quintic equations [16.12]

Theorem 24.6. *Let $f \in F[x]$ be an irreducible quintic whose splitting field K has Galois group $G = A_5$ or S_5 over F. Then the roots of f are not solvable over F.*

Proof. First, we reduce to the case when $G = A_5$. To do this, we simply replace F by the intermediate field F' corresponding to the subgroup $A_5 \subseteq S_5$. If α is not solvable over F', then it is not solvable over F.

Assume for the sake of contradiction, that α satisfies 2. of Definition 24.4, so we have a chain $F_0 \subseteq \cdots \subseteq F_s$ such that $F = F_0$ and $\alpha \in F_s$ (note that F_s here does not be contained in K, the splitting field of f).

$$\begin{tikzcd}
 & K_1 \arrow[dl, dash, "H"'] \arrow[dr, dash, "G_1"] & \\
K \arrow[dr, dash, "A_5"'] & & F_1 \arrow[dl, dash, "C_p"] \\
 & F &
\end{tikzcd}$$

Let K_1 be the *compositum* of K and F_1, the smallest subfield of $\mathbb{C}$ containing both K and F_1. It is also Galois over F: one way of seeing this is by noting that F_1 is the splitting field of some degree p polynomial $g \in F[x]$, with roots $\beta_1, \ldots, \beta_p \in F_1$. Then

$$K_1 = F(\alpha_1, \ldots, \alpha_5, \beta_1, \ldots, \beta_p)$$

is the splitting field of fg, where $\alpha_1, \ldots, \alpha_5$ are the roots of f.

Claim 24.7. *The Galois group $G_1 = \mathrm{Gal}(K_1/F_1)$ is also isomorphic to A_5.*

Proof. Let $G' = \mathrm{Gal}(K_1/F)$, and denote the Galois groups as in the diagram above. (C_p denotes the cyclic group of order p.) We have that H and G_1 are normal subgroups of G' such that

$$\varphi\colon G' \to G'/H \xrightarrow{\sim} A_5, \quad \psi\colon G' \to G'/G_1 \xrightarrow{\sim} C_p.$$

Furthermore,

$$H \cap G_1 = \{1\}.$$

This follows from the Main Theorem since K_1 is the smallest field containing K and F_1, and the correspondence is inclusion-reversing.

Thus, if we restriction ψ to H, we get an injective map $\psi\colon H \to C_p$. Since the only subgroups of C_p are $\{1\}$ and C_p, either

1. $H = \{1\}$. Then $G' \simeq A_5$, so it has no proper normal subgroups! Then it is impossible for G'/G_1 to be isomorphic to C_p, so this case is not possible.

2. $H = C_p$. Restricting φ now to G_1, we have an injection $G_1 \to A_5$. However, we have

$$|G'| = |H| \cdot |A_5|, \quad |G'| = |G_1| \cdot |C_p|,$$

so $|G_1| = |H| \cdot |A_5|/|C_p| = |A_5|$, so $G_1 = A_5$, as desired. □

The point of the claim is that we can replace F by F_1, and the quintic f still generates an A_5 extension.

More precisely, we can define $K_2, \ldots, K_s$ in the same way as we defined K_1 and by induction, $\mathrm{Gal}(K_s/F_s) = A_5$, and K_s is the splitting field of f over F_s. But we said that $\alpha \in F_s$, so this is a contradiction. □

24.4 Example of non-solvable quintic [16.12]

Lemma 24.8. *Every transitive subgroup G of S_5 contains a 5-cycle.*

Proof. A group acting with an orbit of size 5 has order $|G|$ divisible by 5, so it has an element of order 5. In S_5, the elements of order 5 are the 5-cycles. □

Lemma 24.9. *Let G be a subgroup of S_5 which contains a 5-cycle and a transposition. Then $G = S_5$.*

Corollary 24.10. *Let $f \in \mathbb{Q}[x]$ be an irreducible quintic which has exactly 3 real roots and 2 complex roots. Then the splitting field K of f has Galois group S_5 over $\mathbb{Q}$.*

Proof. We know that $G = \mathrm{Gal}(K/\mathbb{Q})$ contains a 5-cycle because it is transitive. It also contains the restriction of complex conjugation to K. (Since $K/\mathbb{Q}$ is a splitting field, complex conjugation sends K to K because the roots of polynomials in $\mathbb{Q}[x]$ come in complex conjugate pairs.) This automorphism acts on the roots by swapping the two complex roots. Thus, G also contains a transposition, so by the lemma, $G = S_5$. □

Example 24.11. Any small perturbation of the polynomial $x^5 - 16x = x(x^2 - 4)(x^2 + 4)$ has Galois group S_5. For example, $x^5 - 16x + 2$ is a quintic whose roots are not solvable over $\mathbb{Q}$.